

Bayesian analysis of agricultural field experiments

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Summary. The paper describes Bayesian analysis for agricultural field experiments, a topic that has received very little previous attention, despite a vast frequentist literature. Adoption of the Bayesian paradigm simplifies the interpretation of the results, especially in ranking and selection. Also, complex formulations can be analysed with comparative ease, by using Markov chain Monte Carlo methods. A key ingredient in the approach is the need for spatial representations of the unobserved fertility patterns. This is discussed in detail. Problems caused by outliers and by jumps in fertility are tackled via hierarchical-*t* formulations that may find use in other contexts. The paper includes three analyses of variety trials for yield and one example involving binary data; none is entirely straightforward. Some numerical comparisons with frequentist analyses are made.

Keywords: Agricultural field trials; Bayesian computation; Bayesian inference; Binary data; Combining information; Crop experiments; Intrinsic autoregressions; Markov chain Monte Carlo methods; Markov random fields; Prior distributions; Ranking and selection; Spatial statistics; Variety trials

1. Introduction

There is a vast literature on the design and analysis of agricultural field experiments, dating back to the early 1900s, almost all of which adopts a frequentist standpoint. In the present paper, we consider instead Bayesian formulations, based on Besag and Higdon (1993) and Besag *et al.* (1995), section 5. Our aim is not particularly to argue for the Bayesian paradigm but rather to explore its ramifications in a modern computing environment, to encourage discussion, to highlight some current weaknesses and hopefully to stimulate further research. The data sets are available at

<http://www.stat.duke.edu/~higdon/trials/data.html>

Although statistical concepts of design and analysis were developed by R. A. Fisher, F. Yates and many others primarily in the context of agricultural field experiments, modern courses and text-books usually concentrate on industrial and medical applications, in which spatial effects are unlikely to be a central issue. Field trials are rather different, for it has always been recognized that generally a crop will perform much more similarly on two experimental units (plots) that are close together than on plots that are further apart. Fisher (1928), p. 229, states

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'... the peculiarity of agricultural field experiments lies in the fact, verified in all careful uniformity trials, that the area of ground chosen may be assumed to be markedly heterogeneous, in that its fertility varies in a systematic, and often a complicated manner from point to point'.

Indirect assessments suggest that variation in fertility is typically of the same order as the effects of interest and cannot be neglected. There follow some brief comments.

As regards design, the ideal arrangement for a variety trial, say, would usually involve replication with compact blocks of small plots, each pair of varieties appearing close together as often as possible. Some compromises must be made, as, for example, in α -designs of Patterson and Williams (1976). In practice, severe restrictions can be imposed by expense, machinery, management practice and shortage of land and possibly seed. Plots are usually long and narrow. Unfortunately, many experiments employ inappropriate designs (Preece, 1982): thus, the El Batán trial in Table 1 in Section 3 uses a randomized complete-block (RCB) design, despite involving 50 varieties of wheat.

As regards statistical analysis, we assume that the goal is to compare a particular set of treatments (or varieties) and that this involves effects that act additively on fertility, perhaps after a suitable transformation of the data. Variation in fertility must generally be taken into account indirectly and the obvious approach is to devise a stochastic model, as a component of the overall formulation, as we discuss further below and in Section 2. However, recall that the original Fisher–Pitman–Welch solution is quite distinct, being conceived in terms of the distribution induced by the randomization of the treatment allocation, within the block constraints of the design. The basis for such an analysis remains valid for *any* pattern of fertility. In addition, standard least squares analysis, treating fertilities inappropriately as independent Gaussian errors, produces results that correspond closely to the randomization analysis, at least for fairly simple designs. This point is sometimes misrepresented in arguing in favour of overtly spatial analysis.

Thus, for the El Batán data, it is the accuracy rather than the validity of the RCB analysis that is at issue, though one may also question the practical relevance of the reference distribution. For a trial that suggests such pronounced within-block fertility patterns as in the raw yields in Fig. 1(a) in Section 3 it would seem clear that the standard analysis can be improved on but this is not obvious for well-designed trials, with small (incomplete) blocks within which constant fertility provides a plausible approximation.

Alternatively, the stochastic modelling approach, whether Bayesian or frequentist, must invoke a more realistic representation of fertility. We stress that there is no intention here to abandon sound design principles, but any blocking structure that does not have a physical presence is omitted from the model and from the subsequent analysis. Note that the fertilities can be interpreted as fixed or random effects: in the Bayesian formulation in this paper, they are fixed. Of course, it is crucial that the fertility model should represent a wide range of plausible patterns, unlike polynomial and other trend surfaces, and yet its flexibility must be limited, so as not to absorb genuine treatment effects by overfitting. Treatment replication, in some form, is essential in making this a realistic task.

As a consequence of an entirely model-based approach to the analysis, we believe it is highly desirable to provide some robustness to aberrations in the form of possible outliers and jumps in fertility (Papadakis, 1984). Gaussian models are especially vulnerable. Thus, a feature of our approach is that we propose formulations based on *t*-distributions, in which the degrees of freedom are guided by the data. The methodology requires some further development for two-dimensional fertility adjustment but we think that the problems are not insurmountable and also that the general approach has applications in quite different

contexts where model uncertainty is an issue (e.g. Draper *et al.* (1992), Besag *et al.* (1997) and Byers and Besag (1998)).

As we have indicated, our preference for the Bayesian paradigm is partly in a spirit of exploration, fuelled by the fact that complicated functionals of high dimensional posterior distributions can be calculated with comparative ease by Markov chain Monte Carlo (MCMC) methods. We shall have rather little to say about MCMC sampling itself and, for further details, refer to Smith and Roberts (1993), Besag and Green (1993), Tierney (1994), Besag *et al.* (1995), Green (1995) and the accompanying references and discussions. The analyses in this paper were all carried out using simple or block Gibbs samplers, except for the occasional inclusion of some Hastings steps. It would be of interest to investigate the extent to which corresponding frequentist analyses can be implemented. However, our reasons for preferring a Bayesian analysis also reflect the view that, particularly in spatial statistics, formulations are often representations rather than actual physical models (Besag (1974) used the term 'schemes' for the former).

One advantage of the Bayesian paradigm is that one is able to pose important practical questions that have no direct counterparts in frequentist inference. This is evident in ranking and selection, where the plant breeder wishes to compare the effects of different varieties, under particular experimental conditions. Whereas the frequentist resorts to an indirect approach based on multiple comparisons, whose interpretation is fraught with difficulties, the Bayesian is free to evaluate, for example, the posterior probability that any given variety has the optimal effect, or that the top or perhaps the top three effects are covered by a particular group of varieties. The answers to such questions may be crucial in planning future experiments and are readily available from the MCMC output. Simultaneous credible regions for sets of effects can also be determined (Besag *et al.*, 1995).

In organizing the paper, we decided to present the methodology first, followed by applications. However, we have tried to leave some flexibility for the reader to skip in and out of this sequence. In Section 2, we describe some Bayesian formulations for field trials, beginning with a basic one in which

- (a) the treatment or variety effects have no special structure,
- (b) the variation in fertility is essentially one dimensional and
- (c) Gaussian assumptions are appropriate, both in the likelihood and in the prior.

Each restriction is relaxed subsequently.

Sections 3–6 provide specific applications. Thus, Section 3 reanalyses the El Batán data, mentioned above, in which there is substantial variation and even jumps in fertility, as well as evidence of outliers. One-dimensional fertility adjustment is sufficient in this example but not in the other three. In Section 4, we consider a small experiment on eight varieties of strawberry, in which the proximity of a hedge creates a complication. We contrast Bayesian and frequentist formulations. In Section 5, we examine one aspect of a trial on seed counts from morning-glory plants, in which the treatment structure follows a factorial design. In particular, we focus on whether or not plants produce seeds, according to colour intensity and seed pool. The existence of pronounced fertility patterns suggests that a standard frequentist formulation would be inappropriate.

In Section 6, we analyse six trials on 64 varieties of corn, carried out in a single season in different counties of the coastal plain of North Carolina. We provide comparisons with frequentist results, including incomplete-block and geostatistical analyses, for the individual trials. We also construct a Bayesian hierarchical model that combines information between the trials, some of which are individually rather inconclusive. The combination of information

from *successive* trials in a plant breeding programme is perhaps less controversial and, although we do not have an example, the methodology and computations would be quite similar. Such combination is especially appealing in the Bayesian paradigm, with the posterior distribution of variety effects from previous trials providing prior information for the current one. Finally, Section 7 contains some discussion.

2. Bayesian formulation of field experiments

2.1. Introduction

For definiteness, we focus on a variety trial, in which it is intended to make yield comparisons between m varieties of a crop. We assume that measurements are effectively continuous, resulting in an n -vector y of yields over n rectangular plots. However, our formulation extends to more complicated treatment structures and to discrete observations, as in Section 5. We embed the lay-out of the plots in a rectangular array L , with q_1 rows and q_2 columns, so that $n \leq q_1 q_2$. This allows for conventional missing observations and for aberrations in an otherwise rectangular lay-out but also we may introduce conceptual ‘border’ plots in L , purely as a computational device, to offset edge effects that often frustrate spatial formulations. In practice, experiments may involve several different arrays, as in Sections 5 and 6, and these may become separate columns of plots, as in Section 3, but here we refer to a single array, unless otherwise stated. We index the plots in L by row i_1 and column i_2 , and allow integer pair subscripts $i = (i_1, i_2)$ for matrices and vectors, when appropriate. We order the elements of y in some convenient manner (e.g. lexicographically) and define an $n \times q$ incidence matrix F , where $q = (q_1, q_2)$, to provide identification between the two indexing systems.

Let τ and T denote the m -vector of centred variety effects and the associated $n \times m$ plot-by-variety incidence matrix; more generally, τ would represent treatment effects and T the corresponding design matrix. In our basic formulation, we suppose that

$$y|\tau, \psi, \lambda_y \sim N(F\psi + T\tau, I_n/\lambda_y), \quad (2.1)$$

where ψ is a q -vector representing a $q_1 \times q_2$ array of fertility effects, I_n is the $n \times n$ identity matrix and λ_y is the unknown precision of the data, which we assign a diffuse but proper $\Gamma(a, b)$ prior (i.e. a gamma distribution with mean a/b and variance a/b^2). In Section 2.4, we replace expression (2.1) by its analogue based on a t -distribution, whose degrees of freedom are unknown, with the general aim of accommodating outliers; see also Section 3. Continuous distributions can be replaced by discrete ones, if appropriate. Whatever the eventual choice, the formulation is completed by specifying prior distributions for τ and ψ , with ψ being used to absorb the overall effect (in each array, if there is more than one).

2.2. Prior distributions for variety or treatment effects

The simplest choice of prior for τ in a single trial is either a centred uniform distribution over the real line (effectively) or centred $N(0, I_m/\lambda_\tau)$, with $\lambda_\tau \sim \Gamma(c, d)$, sufficiently diffuse that substantial shrinkage towards zero occurs only when the data provide little evidence of differences between varieties. Since our fertility priors in Section 2.3 are invariant to the addition of a constant, the centring of variety effects need not be respected at each stage of the MCMC computations but can be left until the end. The Gaussian prior could be replaced by one with heavier tails, to allow for one or more discrepant effects, but we have not tried this. Other choices are appropriate if the varieties can be partitioned into more homogeneous groups or T has a particular structure or there is substantial knowledge about variety

differences, gleaned from previous trials. In the last case, it may be appropriate to carry out a combined analysis in which, in effect, the posterior from earlier experiments provides the prior for the current one. In this paper, our default prior is the simple Gaussian or uniform distribution: the latter is useful mainly in making numerical comparisons with standard frequentist analyses in which shrinkage does not occur. We comment further in individual applications.

When combining several different but closely related trials, as in Section 6, we permit discrepancies between the centred variety effects across trials. Thus, we adopt a hierarchical model, in which the effect τ_k of variety k is replaced by $\tau_k^t = \tau_k + \xi_k^t$, where ξ_k^t represents the differential effect of variety k in trial t . The ξ_k^t for variety k are then assigned an $N(0, 1/\lambda_{\tau_k})$ prior and the λ_{τ_k} are given separate or linked gamma distributions.

2.3. Prior distributions for fertility effects

2.3.1. Basic Gaussian representation

In specifying a basic prior distribution for the fertilities ψ , over a $q_1 \times q_2$ array, we restrict attention to a class of non-stationary Gaussian intrinsic autoregressions (Künsch, 1987; Besag *et al.*, 1991; Besag and Kooperberg, 1995). Let $i \sim j$ denote pairs of directly adjacent plots and let ∂i identify plots that are directly adjacent to i . For each pair $i \sim j$, define a symmetric coefficient ω_{ij} , which can be fixed or have a prior distribution of its own. Let W denote the $q \times q$ symmetric matrix with (i, j) element W_{ij} , where $W_{ii} = \omega_{i+}$, $W_{ij} = -\omega_{ij}$ for $i \sim j$, $W_{ij} = 0$ otherwise and $\omega_{i+} > 0$ is the sum over $j \in \partial i$ of the ω_{ij} . We shall ensure that W is positive semidefinite; clearly, W cannot be strictly definite because its row and column sums are 0. Then our (improper) prior density for ψ is

$$\pi(\psi|\omega, \lambda_\psi) \propto \lambda_\psi^{q_1 q_2 / 2} |W|^{1/2} \exp(-\frac{1}{2} \lambda_\psi \psi^T W \psi), \quad (2.2)$$

where λ_ψ is a scale parameter to which we give a $\Gamma(g, h)$ distribution, and $|W|$ denotes the product of the non-zero eigenvalues of W , which is a required factor if ω itself has a prior distribution.

More transparently, the full conditional distribution of ψ_i , given ω , λ_ψ and all other plot fertilities ψ_{-i} , is Gaussian with mean and variance

$$\begin{aligned} E(\psi_i|\psi_{-i}, \omega, \lambda_\psi) &= \sum_{j \in \partial i} \frac{\omega_{ij}}{\omega_{i+}} \psi_j, \\ \text{var}(\psi_i|\psi_{-i}, \omega, \lambda_\psi) &= \frac{1}{\lambda_\psi \omega_{i+}}, \end{aligned} \quad (2.3)$$

and depends only on the plot fertilities $\psi_{\partial i}$ directly adjacent to it. The identity

$$\psi^T W \psi = \sum_{i \sim j} \omega_{ij} (\psi_i - \psi_j)^2 \quad (2.4)$$

implies that expression (2.2) is a special case of a pairwise difference prior distribution (Besag, 1989; Besag *et al.*, 1995). We now discuss the choice of the ω_{ij} s.

It is convenient first to imagine a model on the doubly infinite rectangular array, so that edge effects for all plots $i = (i_1, i_2)$ can be temporarily ignored. Then we would choose

$$E(\psi_i|\psi_{-i}, \beta, \lambda_\psi) = \frac{1}{2} \beta (\psi_{i_1-1, i_2} + \psi_{i_1+1, i_2}) + \frac{1}{2} \bar{\beta} (\psi_{i_1, i_2-1} + \psi_{i_1, i_2+1}), \quad (2.5)$$

$$\text{var}(\psi_i|\psi_{-i}, \beta, \lambda_\psi) = 1/2 \lambda_\psi, \quad (2.6)$$

with $\beta + \bar{\beta} = 1$ and β fixed or given an approximately uniform prior on the unit interval. For any fixed $\beta \in (0, 1)$, the sole indeterminacy in this model is its overall level. Indeed, the intrinsic case can be interpreted as the limit of a stationary autoregression, in which there is no shrinkage towards a common mean level of fertility. We see no practical justification in an assumption of stationarity at the scale of a typical field experiment, though it is a feature of many of the spatial formulations in the literature. For some related comments regarding parameter estimates, see, for example, Besag (1974) and Besag and Kooperberg (1995).

We return now to the finite array. For interior plots, there is correspondence between the mean structures in equations (2.3) and (2.5) if $\omega_{ij} = \beta$ for $i \sim j$ in the same column and $\omega_{ij} = \bar{\beta}$ for $i \sim j$ in the same row. Unfortunately, this implies that, for example, $\text{var}(\psi_i - \psi_j)$ for adjacent plots i and j in the same row or in the same column changes radically between the centre and the edges of the array, at least when $\beta \approx \frac{1}{2}$. Nor are there any simple remedies: see the related discussion and numerical examples for stationary autoregressions in Besag and Kooperberg (1995). In principle, the ideal solution would employ the finite restriction of the infinite lattice autoregression to a $q_1 \times q_2$ array. However, the conditional mean for any edge plot then depends not only on directly adjacent ψ_j s but also on the ψ_j s on *all* other boundary plots. When $\beta = \frac{1}{2}$, the required coefficients can be determined from results in section 3.3 of Besag and Kooperberg (1995), though general formulae do not seem yet to exist. Note that the invariances diminish as β takes on more extreme values and we now pause to consider the important special cases $\beta = 0$ and $\beta = 1$.

When $\beta = 1$ in equation (2.5), we obtain a set of independent random walks along the columns of the array, each with an $N(0, 1/\lambda_\psi)$ increment distribution, and similarly along the rows if $\beta = 0$. These random walks are improper in that each has an arbitrary level. We adopt $\beta = 1$ for one-dimensional adjustment, when a trial is laid out as a single column of plots, or as several columns that can be treated independently, given λ_ψ ; see Section 3 and, for previous examples, Besag and Higdon (1993), Besag *et al.* (1995) and, in a frequentist setting, Besag and Kempton (1986). Note that the random walk interpretation implies that edge corrections do not arise in transferring to finite lay-outs. The indeterminacy in each column corresponds to the usual practice of fitting arbitrary block effects.

We return now to $\beta \in (0, 1)$, for which the lack of a simple independent increments interpretation creates the complication of edge effects. We contain the problem by either of the following procedures. In the first, we augment the minimal rectangle containing the actual lay-out by a few layers of virtual plots and then, for this $q_1 \times q_2$ array, adopt the specification described below equations (2.5) and (2.6). Thus, the ω_{ij} s for the minimal array match those for the finite restriction of the infinite lattice process on all interior plots and converge quickly to the correct values on its boundary as the number of layers increases. In the analysis itself, the fertilities on virtual plots are treated in the same manner as on plots with genuinely missing data, a procedure that can be implemented rigorously via MCMC sampling, as in Sections 5 and 6.

In practice, two or three extra layers seem adequate. As a numerical example, relevant to Section 6, consider an 18×11 lay-out and suppose that $\beta = \frac{1}{2}$. Then, without virtual plots, the ratio of the maximum to the minimum of $\text{var}(\psi_i - \psi_j)$ for $i \sim j$ is 1.39. However, with the inclusion of borders of widths 2 and 3 respectively, the ratio drops to 1.02 and 1.01 for the original pairs of plots; and these results are rather insensitive to the dimensions of the lay-out. The device of adding layers can also be applied to non-Gaussian formulations for ψ but, in the Gaussian case, there are simplifications in the evaluation of the eigenvalues of W that can be used to speed up the computations and hence introduce a continuous prior for β ; see Section 2.5.

The second procedure retains the original array but chooses the non-zero ω_{ij} s in equations (2.3) such that $\text{var}(\psi_i - \psi_j)$ is constant for all adjacent plots within columns, and similarly within rows, i.e., for any given β in equation (2.5), the values of these two variances are matched with those in the infinite lattice process, using a modification of Dempster's algorithm in Besag and Kooperberg (1995). In this case, we adopt a uniform prior on the 12 values $\beta = 0.01, 0.05, 0.15, \dots, 0.95, 0.99$ and compute the corresponding W s off line. The value of β is not critical and there is no real disadvantage in restricting it to a small finite set. For an example, see Section 4.

2.3.2. Locally quadratic representation

When $\beta = 0.5$, equation (2.5) is equivalent to *locally planar* interpolation of fertilities by the method of ordinary least squares. An alternative is to adopt *locally quadratic* interpolation, based on the eight plots closest to i , in which case the weights attached to directly and diagonally adjacent plots are $\frac{1}{2}$ and $-\frac{1}{4}$ respectively. Curiously, the restriction of the infinite lattice process to a $q_1 \times q_2$ array is now straightforward, with

$$\pi(\psi|\lambda_\psi) \propto \exp \left\{ -\frac{1}{2} \lambda_\psi \sum_{i_1=1}^{q_1-1} \sum_{i_2=1}^{q_2-1} (\psi_{i_1, i_2} - \psi_{i_1+1, i_2} - \psi_{i_1, i_2+1} + \psi_{i_1+1, i_2+1})^2 \right\}, \quad (2.7)$$

another example of a pairwise difference prior distribution, with $i \sim j$ in expression (2.2) now also including diagonal adjacencies. The ω_{ij} s follow immediately from expression (2.7) and are given in section 3.4 of Besag and Kooperberg (1995).

Clearly, expression (2.7) is invariant to the addition of arbitrary row and column effects. Such a strong alignment with the plot axes would be fatal in most spatial applications but here it can be defended because management practices may induce row and column patterns. The resulting analysis has close ties with the frequentist methods developed by Cullis and Gleeson (1991) and investigated empirically by Kempton *et al.* (1994). The prior is an intrinsic limit of the *separable* stationary autoregressions of Martin (1979, 1990). Edge effects are absent because expression (2.7) has an interpretation in terms of independent two-dimensional differences and it might be of interest to experiment with non-Gaussian distributions for these increments, analogous to the one-dimensional formulation in Section 2.4. Our own view is that the above invariance to row and column effects is undesirable, at least in a Bayesian context. For some further comments, see Section 7. Note that the invariance does not occur in adopting locally quadratic interpolation based on the nearest 12, rather than eight, plots (Besag and Kooperberg, 1995) but edge effects then re-enter the formulation and the representation is perhaps too intricate for routine use in a typical experiment.

Furthermore, although locally quadratic interpolation is very appealing in principle, because it does not attenuate peaks and troughs, we suspect that there is a danger of over-fitting the fertility pattern, to the detriment of genuine variety effects. Nevertheless, Section 4 includes a sample analysis. Note that the adoption of independent second differences for one-dimensional fertility adjustment is discussed in a non-Bayesian context by Green *et al.* (1985).

2.3.3. Other Gaussian representations

An appealing alternative approach is to represent fertility by a process in continuous space and to integrate over each plot to obtain corresponding average values ψ_i , as proposed in the pioneering work of Whittle (1954) and Matérn (1960). Of course, the main attraction of continuum models is that they are self-consistent as plot size varies. We first give the flavour

by following a very simple recipe, which is essentially one dimensional and assumes that the underlying fertility process, when averaged across a single column of plots, is a Brownian motion with arbitrary level. Then standard properties of the integrated process, together with lemma 3.1 of Besag and Kooperberg (1995), imply that, at the plot level, ψ has a Gaussian pairwise difference distribution (2.4) with easily computed but fully rich ω_{ij} s. Direct implementation involves costly matrix manipulations but these can be avoided by the introduction of auxiliary variables $\gamma_0, \dots, \gamma_n$, representing the fertilities at the end points of the n plots. The γ_i s then follow a Gaussian random walk with precision λ' , say, conditional on which the ψ_i s are independent and Gaussian, with means $\frac{1}{2}(\gamma_{i-1} + \gamma_i)$ and precisions $12\lambda'$.

Limited experience with this formulation suggests that it is effectively equivalent to that based on Gaussian random walks. For example, this holds for the data in Section 3. Incidentally, we have also carried out closely related experiments in which the plot size for the random walk fertility model is halved and the yields in each pair of subplots are treated as missing values but summing to the observed plot yield.

Although corresponding two-dimensional continuum models would be of considerable interest, the computational problems seem quite challenging and perhaps prohibitive if one allows for possible discontinuities in fertility. An alternative is to adopt the 'geostatistical' approach, which generally involves an assumption of stationarity and the direct estimation of a parametric covariance structure using the variogram. For discussion of this approach, see, for example, Brewer and Mead (1986), Zimmerman and Harville (1991) and Brownie *et al.* (1993). These papers adopt a frequentist standpoint and only the last includes an analysis of experimental data; see Section 6 for an additional example and some comparisons. Bayesian versions of geostatistical formulations have attracted recent attention. A fully Bayesian approach, as in Diggle *et al.* (1998), takes account of uncertainty in the estimated variogram and can be implemented via MCMC methods. However, in agricultural applications with moderately large data sets, such as in Section 5, the computational task of inverting large non-sparse matrices becomes very demanding for routine use. In comparison, variogram estimation is unreliable in small data sets. There is clear scope for future developments in devising and implementing continuum models but there seems to be no compelling *practical* advantage over the aesthetically less appealing but more flexible discrete representations.

2.4. Hierarchical- t formulations for yields and fertilities

We remarked in Section 1 that, in a model-based approach, it is desirable to accommodate possible outliers and jumps in the level of fertility, a point first addressed by Papadakis (1984) from a frequentist viewpoint; see also Besag and Seheult (1989). Taplin and Raftery (1994) discuss the problem from a Bayesian perspective but their formulation is based on stationary time series models for fertility.

Here, we develop the approach first suggested in Besag and Higdon (1993). For outliers, we replace the Gaussian distribution in expression (2.1) by one with heavier tails, such as Laplace or Student's t_{ν_y} . In the latter case, we allow ν_y itself to have a prior distribution and describe this as a *hierarchical- t* formulation. In practice, we have usually specified the prior to be uniform on 1, 2, 4, 8, 16, 32 and 64 degrees of freedom, though of course any other choice could be made.

As regards fertility, we restrict attention to one-dimensional adjustment, for which solutions are more readily available. In particular, we adopt an approach that is complementary to the one above and allows a non-Gaussian distribution for the increments of the random walk(s), with the intention that jumps in fertility are then less prone to be smeared out. If the choice is

a t_{ν_ψ} -distribution, we again assign a prior to the degrees of freedom. Thus, the hierarchical- t formulation for yields and fertilities incorporates an allowance for *model uncertainty*. Besag *et al.* (1995) describe this very briefly and mention an example in which there is close agreement with the basic formulation but, in Section 3, we analyse a more awkward data set. Some computational issues are discussed at the end of Section 2.5.

2.5. Posterior distribution

If we combine the basic Gaussian likelihood (2.1), fertility formulation (2.2) and variety prior with the remaining priors, we obtain the posterior density

$$\begin{aligned} \pi(\tau, \psi, \beta, \lambda|y) &\propto \lambda_y^{n/2} \exp\{-\frac{1}{2}\lambda_y(y - F\psi - T\tau)^T(y - F\psi - T\tau)\} \\ &\times \lambda_\tau^{m/2} \exp(-\frac{1}{2}\lambda_\tau\tau^T\tau) \lambda_\psi^{q_1q_2/2} |W|^{1/2} \exp(-\frac{1}{2}\lambda_\psi\psi^TW\psi) \\ &\times \pi(\beta) \lambda_y^{a-1} \exp(-b\lambda_y) \lambda_\tau^{c-1} \exp(-d\lambda_\tau) \lambda_\psi^{g-1} \exp(-h\lambda_\psi). \end{aligned} \quad (2.8)$$

If the variety prior is uniform, the terms involving λ_τ are absent. Below, we briefly re-examine the role of W .

First, consider a single column of q_1 plots, with fertility represented by a Gaussian random walk, so that W is now an ordinary $q_1 \times q_1$ matrix. Then $\pi(\beta)$ and $|W|$ in expression (2.8) are irrelevant and the only non-zero off-diagonal elements in W are $W_{i,i\pm 1} = -1$, with W_{ii} ensuring zero row and column sums; we denote this matrix by W_{q_1} . If there are q_2 separate columns, or for one-dimensional adjustment along the columns of a $q_1 \times q_2$ array, then W is effectively block diagonal, i.e. $W = W_{q_1} \otimes I_{q_2}$, where $\otimes$ denotes the Kronecker matrix product. In all these cases, the full conditionals for blocked and individual components are given in Besag *et al.* (1995), p. 23, and a Gibbs sampler is quite straightforward, with draws from Gaussian and gamma distributions; we usually employ Cholesky decomposition to update τ and ψ .

For two-dimensional adjustment, we have described alternative procedures for dealing with edge effects in equation (2.5) and we now add a further remark on the first method, relating to the updates of β . With the above notation, it follows that the $q \times q$ matrix W for the extended array can be written

$$W = \beta W_{q_1} \otimes I_{q_2} + \bar{\beta} I_{q_1} \otimes W_{q_2},$$

and that the q_1q_2 eigenvalues of W are the elements of $\beta\rho_{q_1} \oplus \bar{\beta}\rho_{q_2}$, where ρ_{q_1} and ρ_{q_2} are the vectors of eigenvalues of W_{q_1} and W_{q_2} , and $\oplus$ denotes the Kronecker matrix sum. We calculate ρ_{q_1} and ρ_{q_2} off line via the QL -decomposition (Press *et al.* (1988), p. 337) of the tridiagonal matrices W_{q_1} and W_{q_2} . Hence $|W|$, the product of the non-zero eigenvalues of W , is immediately available for any β . We use Metropolis steps for the updates.

Finally, we comment on computation in hierarchical- t formulations for y and ψ . Although non-Gaussian components in the prior and in the likelihood generally lead to complications in the posterior that are best handled via Hastings rather than Gibbs steps, t -distributions provide an exception. Thus, in considering y , an equivalent but simplified posterior distribution is obtained if I_n in expression (2.1) is replaced by D^{-1} , where D is an $n \times n$ diagonal matrix, whose diagonal elements δ_s are independent $\Gamma(\frac{1}{2}\nu_y, \frac{1}{2}\nu_y)$ variates. Then the full conditionals for τ and ψ are once more Gaussian and those for the δ_s s are independent gammas, so a simple Gibbs sampler is again available. Moreover, a significant computational improvement can be made by blocking δ with ν_y , as below. Note that y_s is suggested as an outlier if the posterior mean of δ_s is markedly less than its nominal value of 1.

For the hierarchical- t random walks used in one-dimensional fertility adjustment, an analogous procedure can be devised. Suppose that the trial is embedded in a $q_1 \times q_2$ array, with fertility represented by a random walk within each column. Let W in the Gaussian formulation (2.8) have non-zero off-diagonal elements ω_{ij} , corresponding to within-column plot adjacencies $i \sim j$, but now assign these independent $\Gamma(\frac{1}{2}\nu_\psi, \frac{1}{2}\nu_\psi)$ distributions. Then this gives the required t_{ν_ψ} increments. Blocking of ω and ν_ψ again provides a significant improvement computationally; also ω_{ij} s with unusually small posterior means suggest the locations of fertility jumps.

These representations lead to an equivalent posterior density for the hierarchical- t formulation, incorporating both outliers and fertility jumps,

$$\begin{aligned} \pi(\tau, \psi, \delta, \omega, \lambda, \nu|y) &\propto \lambda_y^{n/2} |D|^{1/2} \exp\{-\frac{1}{2}\lambda_y(y - F\psi - T\tau)^T D(y - F\psi - T\tau)\} \\ &\times \lambda_\tau^{m/2} \exp(-\frac{1}{2}\lambda_\tau \tau^T \tau) \lambda_\psi^{q_1 q_2/2} |W|^{1/2} \exp(-\frac{1}{2}\lambda_\psi \psi^T W\psi) \\ &\times \prod_{s=1}^n \frac{(\frac{1}{2}\nu_y)^{\nu_y/2}}{\Gamma(\frac{1}{2}\nu_y)} \delta_s^{\nu_y/2-1} \exp(-\frac{1}{2}\nu_y \delta_s) \prod_{i \sim j} \frac{(\frac{1}{2}\nu_\psi)^{\nu_\psi/2}}{\Gamma(\frac{1}{2}\nu_\psi)} \omega_{ij}^{\nu_\psi/2-1} \exp(-\frac{1}{2}\nu_\psi \omega_{ij}) \\ &\times \lambda_y^{a-1} \exp(-b\lambda_y) \lambda_\tau^{c-1} \exp(-d\lambda_\tau) \lambda_\psi^{g-1} \exp(-h\lambda_\psi) \pi(\nu_y) \pi(\nu_\psi), \end{aligned} \quad (2.9)$$

the final two terms representing priors for the degrees of freedom. Note that here $|D| = \Pi_s \delta_s$ and $|W| = \Pi_{i \sim j} \omega_{ij}$. Now define

$$Q_\tau = \lambda_y T^T D T + \lambda_\tau I_m$$

and

$$Q_\psi = \lambda_y F^T D F + \lambda_\psi W$$

and denote the fertility and variety effects on plot s by $\psi_{[s]}$ and $\tau_{[s]}$ respectively, with obvious modification if T has treatment structure. Then the full conditionals for τ , ψ , λ_y , λ_τ and λ_ψ are

$$\begin{aligned} \tau | \dots &\sim N\{\lambda_y Q_\tau^{-1} T^T D(y - F\psi), Q_\tau^{-1}\}, \\ \psi | \dots &\sim N\{\lambda_y Q_\psi^{-1} F^T D(y - T\tau), Q_\psi^{-1}\}, \\ \lambda_y | \dots &\sim \Gamma\{a + \frac{1}{2}n, b + \frac{1}{2}(y - F\psi - T\tau)^T D(y - F\psi - T\tau)\}, \\ \lambda_\tau | \dots &\sim \Gamma(c + \frac{1}{2}m, d + \frac{1}{2}\tau^T \tau), \\ \lambda_\psi | \dots &\sim \Gamma(g + \frac{1}{2}q_1 q_2, h + \frac{1}{2}\psi^T W\psi). \end{aligned}$$

We can block update δ and ν_y by sampling successively from the distributions

$$\begin{aligned} \pi(\nu_y | \tau, \psi, \omega, \lambda, \nu_\psi, y) &\propto \pi(\nu_y) \prod_{s=1}^n \frac{\Gamma(\frac{1}{2}(\nu_y + 1))}{\Gamma(\frac{1}{2}\nu_y)} \nu_y^{-1/2} \left\{ 1 + \frac{\lambda_y (y_s - \psi_{[s]} - \tau_{[s]})^2}{\nu_y} \right\}^{-(\nu_y+1)/2}, \\ \delta_s | \tau, \psi, \omega, \lambda, \nu, y &\sim \Gamma\{\frac{1}{2}\nu_y + \frac{1}{2}, \frac{1}{2}\nu_y + \frac{1}{2}\lambda_y (y_s - \psi_{[s]} - \tau_{[s]})^2\}, \quad s = 1, \dots, n, \end{aligned}$$

the first of which has been marginalized over δ . Note that here the order of updating ν_y and δ is crucial. Similarly, ω and ν_ψ can be block updated by drawing successively from

$$\pi(\nu_\psi | \tau, \psi, \lambda, \nu_y, y) \propto \pi(\nu_\psi) \prod_{i \sim j} \frac{\Gamma\{\frac{1}{2}(\nu_\psi + 1)\}}{\Gamma(\frac{1}{2}\nu_\psi)} \nu_\psi^{-1/2} \left\{ 1 + \frac{\lambda_\psi(\psi_i - \psi_j)^2}{\nu_\psi} \right\}^{-(\nu_\psi+1)/2},$$

$$\omega_{ij} | \tau, \psi, \delta, \lambda, \nu, y \sim \Gamma\{\frac{1}{2}\nu_\psi + \frac{1}{2}, \frac{1}{2}\nu_f + \frac{1}{2}\lambda_\psi(\psi_i - \psi_j)^2\}, \quad i \sim j,$$

the first of which has been marginalized over ω , so that order is again important. We use this procedure in Section 3.

3. A randomized complete-block variety trial on wheat

In this section, we reanalyse an RCB experiment on wheat, carried out at El Batán, Mexico, by the International Center for Improvement of Maize and Wheat. There are three separate single-column replicates, each containing 50 varieties, and so we apply one-dimensional fertility adjustment. The lay-out and the corresponding yields are given in Table 1, with the yields standardized to have variance 1, as is our usual practice before attributing prior distributions (Besag and Higdon, 1993; Besag *et al.*, 1995). The raw yields are also shown in Fig. 1(a). This is an awkward data set, previously discussed from a non-Bayesian perspective by Wilkinson (1984) and Besag and Seheult (1989). A simple RCB decomposition of the data suggests a strong spatial pattern in the residuals, so that naïve variety estimates have little worth.

We note here that the basic Bayesian formulation, with Gaussian assumptions for fertility and yield but a vague prior applied to varieties (see equation (5.3) of Besag *et al.* (1995), with $\lambda_\tau = 0$), produces variety effects that agree closely with those from a superficially similar frequentist extended first-differences analysis (see Besag and Kempton (1986)). The same holds for the standard deviations and corresponding standard errors for variety differences. In places, the variety rankings differ substantially from those of the RCB analysis. However, there is also evidence that the data contain aberrations, in the form of outliers and jumps in fertility. Thus, we turn to the hierarchical-*t* formulation of Section 2.4. We also prefer a Gaussian prior for the variety effects.

3.1. Results

Fig. 1 shows two different additive decompositions of the yields. The upper one corresponds to the basic Bayesian formulation, with Gaussian components for the likelihood and the variety and fertility priors. This agrees closely with the decomposition when Gaussian fertility increments are replaced by Laplace ones, as in Besag and Higdon (1993), or by Brownian motion, as described in Section 2.3.3. The lower decomposition is that obtained from the hierarchical-*t* formulation, for which Table 2 provides the approximate bivariate posterior distribution of ν_y and ν_ψ , with probability spread over quite a wide range of models. Fig. 2 shows 90% credible intervals under the two alternatives. Note that the credible intervals in the hierarchical-*t* formulation are sometimes shorter than their counterparts, despite the additional flexibility in the modelling. We return to this point at the end of the section.

Table 3 compares the rankings and the posterior means for the top 15 varieties under the Gaussian and hierarchical-*t* analyses. The means are generally shrunk towards 0 in comparison with corresponding values under the uniform prior. This must always happen but here the effect is quite pronounced, in agreement with the wide credible intervals (with or without shrinkage). Table 3 also includes the posterior probability that each of the varieties is best and that each lies among the top five. For example, variety 4 has posterior probability about 0.5 of being best and probability in excess of 0.9 of being among the best five. Such

Table 1. El Batán variety trial on wheat: lay-out and transformed yields in three single-column replicates

<i>Variety</i>	<i>Yield</i>		<i>Variety</i>	<i>Yield</i>		<i>Variety</i>	<i>Yield</i>
1	3.38		18	3.49		14	3.01
2	3.59		20	4.70		16	3.91
3	4.33		44	3.29		28	2.31
4	4.54		26	3.66		32	4.33
5	2.59		21	2.82		6	3.15
6	2.41		15	3.47		11	4.17
7	2.64		17	4.49		31	3.63
8	4.00		39	3.43		43	3.08
9	4.35		41	2.87		1	4.93
10	3.98		32	3.43		36	2.22
11	2.78		8	2.45		18	2.68
12	2.96		12	3.66		46	2.13
13	2.27		24	2.68		7	2.36
14	3.10		23	1.90		29	2.59
15	2.50		47	2.78		2	3.43
16	2.64		29	1.90		21	2.31
17	3.19		37	2.27		45	1.90
18	3.01		25	2.59		23	1.85
19	2.64		50	2.96		17	3.15
20	3.43		46	1.81		25	2.31
21	1.85		33	1.85		47	2.18
22	2.41		34	1.99		34	2.18
23	1.81		42	2.96		3	2.36
24	1.81		49	1.76		24	1.76
25	1.85		6	2.04		20	1.67
26	1.85		13	2.64		27	1.43
27	1.81		3	2.45		22	2.45
28	1.34		27	2.18		42	2.45
29	1.76		28	1.67		4	3.19
30	2.04		4	2.96		19	1.99
31	2.22		9	2.92		30	1.43
32	2.13		5	2.45		5	1.85
33	1.85		7	1.81		15	1.71
34	2.18		35	2.13		49	1.81
35	2.04		19	1.62		44	1.81
36	1.06		10	2.04		50	1.94
37	2.27		11	2.82		10	1.76
38	2.45		1	1.90		26	1.76
39	2.82		36	1.11		9	2.41
40	3.19		31	1.48		13	1.85
41	3.56		2	2.27		35	1.81
42	3.29		38	1.43		38	1.48
43	2.04		14	2.08		41	1.57
44	2.59		43	2.08		40	1.81
45	1.57		40	2.59		37	1.60
46	1.34		48	1.53		33	1.43
47	1.71		22	2.18		12	2.50
48	1.85		45	1.76		8	1.81
49	1.81		30	1.34		48	1.99
50	3.10		16	2.13		39	1.67

calculations are trivial from the MCMC output. One can also decide which varieties need to be carried over to a future experiment in order to have a prescribed probability of including the optimal one.

Finally, we resolve some differences between the Gaussian and the hierarchical-*t* analyses. In particular, varieties 1, 10, 11 and 20 are highlighted in Figs 1 and 2. Variety 1 has a

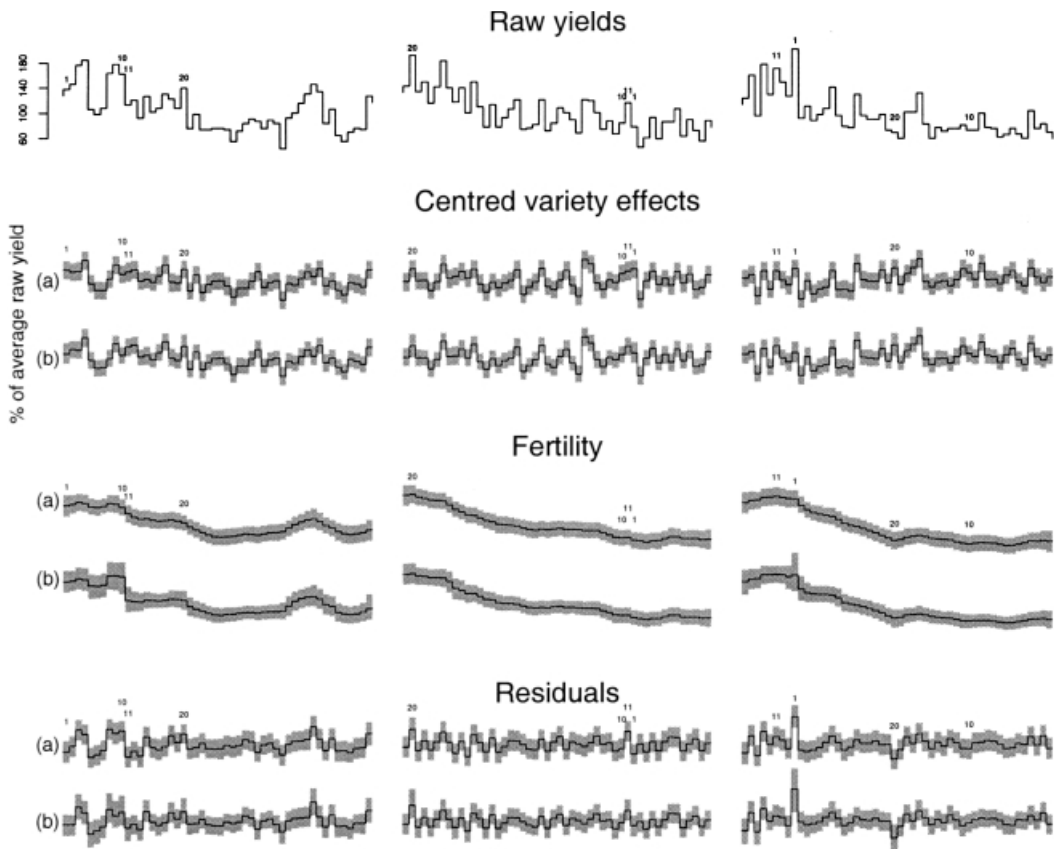


Fig. 1. El Batán variety trial: yields and additive decompositions into variety, fertility and residual effects under (a) Gaussian and (b) hierarchical- t formulations, with the same scale throughout: shaded regions provide pointwise 90% credible intervals; locations of varieties 1, 10, 11 and 20 in each replicate are identified

Table 2. El Batán variety trial: bivariate and marginal posterior distributions of ν_y and ν_ψ in the hierarchical- t formulation

ν_y	Probabilities for the following values of ν_ψ :						
	1	2	4	8	16	32	64
1	.	.	.	.	.	.	.
2	0.03	0.04	0.01	.	.	.	0.08
4	0.13	0.14	0.03	0.01	0.01	.	0.32
8	0.10	0.10	0.01	0.01	.	.	0.23
16	0.07	0.05	0.01	.	.	.	0.14
32	0.06	0.04	0.01	.	.	.	0.12
64	0.06	0.04	0.01	.	.	.	0.11
	0.45	0.41	0.07	0.03	0.02	0.01	0.01

suspiciously large yield in the final replicate and hence is penalized by the t -analysis. This analysis also suggests a fall in fertility in the first replicate between varieties 10 and 11, which therefore demotes the former and promotes the latter. Variety 20 has one of the lowest yields in the final replicate but otherwise performs very well. Its mean is enhanced by the t -analysis

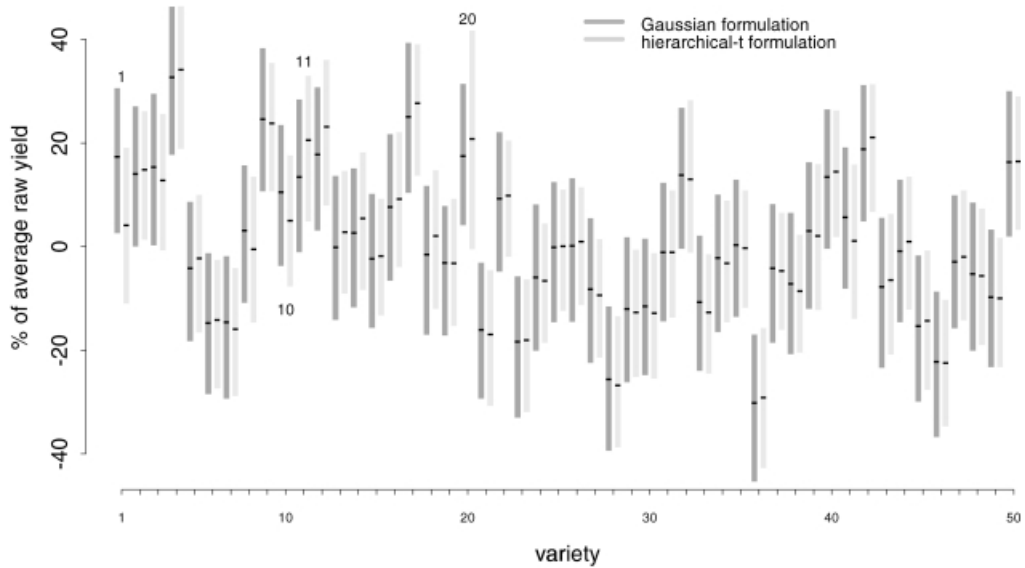


Fig. 2. El Batán trial: 90% credible intervals for the centred variety effects under the Gaussian and hierarchical-*t* formulations

Table 3. El Batán variety trial: rankings of varieties and posterior probabilities under Gaussian and hierarchical-*t* formulations†

Var	Rank		$E(\tau_i y)$		$P(\tau_i \text{ is best})$		$P(\tau_i \in \text{top } 5)$	
	$t_\infty - t_\infty$	$t_\eta - t_\eta$	$t_\infty - t_\infty$	$t_\eta - t_\eta$	$t_\infty - t_\infty$	$t_\eta - t_\eta$	$t_\infty - t_\infty$	$t_\eta - t_\eta$
4	1	1	32	32	48	48	90	91
17	2	2	25	26	15	16	65	73
9	3	3	24	23	14	7	65	55
42	4	5	18	19	3	3	37	39
20	5	6	17	19	3	8	31	41
12	6	4	17	22	3	7	28	53
1	7	15	17	7	3	1	28	8
50	8	8	16	16	3	2	27	24
3	9	12	15	13	1	1	21	14
2	10	9	14	14	0	1	17	17
11	11	7	13	18	1	3	18	32
32	12	10	13	13	1	1	15	17
40	13	11	13	13	1	1	15	14
10	14	16	10	4	0	0	8	2
22	15	13	9	10	0	0	6	6

†Var, variety number; rank, rank by posterior mean; $E(\tau_i|y)$, posterior mean in per cent of average raw yield; $P(\tau_i \text{ is best})$, posterior probability times 100 that variety *i* is the best yielding; $P(\tau_i \in \text{top } 5)$, posterior probability times 100 that variety *i* is among the five best yielding varieties; $t_\infty - t_\infty$, basic Gaussian formulation; $t_\eta - t_\eta$, hierarchical-*t* formulation.

but the most notable change is the increase in the width of its credible interval, which is an indication of model uncertainty. In a Gaussian formulation, the existence of a few anomalies spreads uncertainty among all effects, whereas the *t*-formulation anticipates occasional aberrations. This explains why there is not a general increase in the widths of the credible intervals, despite the additional flexibility.

Table 4. Variety trial on strawberries: lay-out and yields in pounds per plot

Lay-out

Block I				Block II			
G	V	Rl	F	Re	M	E	P
E	P	M	Re	G	V	F	Rl
V	F	Rl	G	P	E	Re	M
E	Re	M	P	G	F	V	Rl
Block III				Block IV			

H
e
d
g
e

Yields

Block I				Block II			
5.8	6.3	4.9	6.5	4.5	5.2	6.5	3.8
6.9	7.6	7.9	5.6	7.0	5.5	4.0	2.7
7.6	6.4	5.0	6.9	7.4	5.3	5.2	3.0
7.5	7.0	6.1	7.2	6.5	5.6	5.8	1.4
Block III				Block IV			

H
e
d
g
e

Table 5. Strawberry trial: centred variety effects under different formulations

	<i>Effects for the following varieties:</i>							
	<i>V</i>	<i>Rl</i>	<i>F</i>	<i>G</i>	<i>Re</i>	<i>E</i>	<i>M</i>	<i>P</i>
Naïve	0.53	−2.27	−0.14	0.78	−0.19	0.78	−0.22	0.73
Quadratic trend	0.50	−1.64	−0.21	0.26	−0.42	0.82	−0.08	0.77
2-dimensional 1st order	0.53	−1.77	−0.23	0.43	−0.41	0.74	−0.07	0.77
2-dimensional locally quadratic	0.37	−1.32	−0.35	0.23	−0.64	0.67	0.21	0.84

4. A variety trial on strawberries

In this section, we compare several different analyses of a small data set involving eight varieties of strawberries. The data are reproduced in Table 4; see also Mead (1988), chapter 10. As is evident, shading from a hedge severely delayed crop production in its vicinity, so a simple analysis is inappropriate, whether or not one allows for the apparent within-block structure. One of us first encountered these data in 1968 as an undergraduate in a course given by Professor J. A. Nelder at the University of Birmingham. Students were instructed to fit a quadratic trend across the columns and to correct the mean yields of the varieties by an analysis of covariance. It was suggested that the block contrast (I + III)−(II + IV) should be incorporated in the error term.

The first two rows of Table 5 provide estimates of centred variety effects for the complete-block and the quadratic trend covariance analyses. The latter, for which the implicit fertility map is shown in Table 6, adjusts for the particularly favourable locations of variety G and the unfavourable ones of P, M and especially Rl. Varieties E and P now rank first and second

Table 6. Strawberry trial: fertility map for the quadratic trend analysis

Block I				Block II			
6.2	6.5	6.6	6.4	6.1	5.4	4.6	3.5
6.2	6.5	6.6	6.4	6.1	5.4	4.6	3.5
6.5	6.8	6.8	6.7	6.2	5.6	4.7	3.6
6.5	6.8	6.8	6.7	6.2	5.6	4.7	3.6
Block III				Block IV			

H
e
d
g
e

Table 7. Strawberry trial: fertility map for the first-order two-dimensional Bayesian analysis

Block I				Block II			
6.0	6.2	6.7	6.5	5.6	5.2	5.2	3.7
6.3	6.6	7.1	6.3	6.1	5.1	4.8	4.0
6.6	6.7	6.7	6.4	6.2	5.1	5.1	3.7
6.7	6.9	6.5	6.4	6.0	5.5	5.0	3.7
Block III				Block IV			

H
e
d
g
e

but, with an average standard error of 0.56 for variety differences, the results remain rather inconclusive, apart from confirming the poor performance of R1. The downturn at the left-hand end of the fertility map is perhaps a little suspicious: it is conceivable that this is an artefact of fitting a deterministic polynomial trend.

We turn now to a Bayesian analysis. If a Gaussian prior is used for variety effects, there is very substantial shrinkage in the posterior means. This is consistent with the large standard errors noted above but it confuses numerical comparisons and so here we focus on a uniform prior. For the fertility effects, it is natural, given the lay-out of the trial, to adopt a two-dimensional fertility model and to ignore the blocking structure. With the first-order prior (2.5), the posterior means are as shown in the third row of Table 5, so that, apart from the marginal reversal of E and P, the ranking of varieties agrees with the covariance analysis. The posterior standard deviations for variety differences range between 0.54 and 0.68. The posterior probability that P has maximal effect is 0.38; for either P or E this probability is 0.72, and for P, E or V it is 0.87. Individual 80% credible intervals for P, E and V are (0.29, 1.26), (0.23, 1.26) and (0.04, 1.01) respectively, with corresponding simultaneous intervals (0.09, 1.45), (0.02, 1.48) and (−0.16, 1.21), calculated as in section 6.3 of Besag *et al.* (1995), which ensures equivariance under strictly monotone transformations. None of these results suggests that the experiment is very conclusive.

The corresponding fertility map is shown in Table 6 and is similar to Table 7: the posterior distribution of the anisotropy parameter β implies heavy concentration on within-column adjustment. Since the Bayesian model does not involve a quadratic trend term, Table 7

supports the downturn in fertility at the left-hand end of the trial in the covariance analysis. Incidentally, the *locally* quadratic Bayesian fertility model produces the same ranking of varieties but, because it absorbs arbitrary row and column effects, the posterior means are a little different; see the final row of Table 5 and further comments below.

We now examine what would have happened using one-dimensional adjustment. If a Gaussian random walk is assumed for the fertilities within each row of the experiment, then the posterior means of the variety effects are indistinguishable from the naïve estimates. There is an almost linear decrease in fertility from left to right in each row but this is so slight (0.04–0.06) that the adjustment is negligible. All 80% credible intervals, except for R1, cover 0. The corresponding hierarchical-*t* analysis provides some compensation, so that P is rated best, with V, G and E tying for second, though P is the only variety whose lower 10% point is positive. Within-row adjustment cannot detect the consistency of the low yields in the final column.

Fitting random walks to the fertilities in each column of the trial tells a different story. The individual fits are rather flat but, of course, their levels are independent from column to column. Hence, the fit is very similar to that obtained from the locally quadratic fertility model. In either case, there is the danger of spurious overfitting to some columns, since the number of observations in each is so small. This danger does not occur with the first-order two-dimensional fertility model.

5. Binary data from an experiment on morning-glory plants

In this section, we describe an analysis of binary observations, though the complete data are in the form of counts. This is one aspect of an experiment concerning the maintenance of genetic variation in morning-glory (*Ipomoea purpurea*) plants: in particular, the relationship between floral pigmentation and fitness, as in Rausher and Fry (1993). First, we provide some background.

Morning-glory flowers display one of three pigment intensities, determined by the W-locus: dark (WW), white (ww) and light (Ww). In natural populations in south-eastern USA, the W-locus is generally polymorphic, so at least two of the pigment intensities are present, with the frequency of the white allele being typically less than 40% (Epperson and Clegg, 1988). However, the combined effects of factors, such as pollinator behaviour, out-crossing rates and the contribution to the pollen pool, would seem to bestow an advantage on the white allele (Rausher and Fry, 1993). In this experiment, which involved the planting of more than 3000 seedlings, it was hoped to detect other forces, which might explain the anomaly.

The method of seed production for the trial involved a combination of selfing and crossing, based on 12 Ww ‘great-grandparents’, each a cross between two homozygotes. Eventually, six ‘pools’ were formed, within each of which the seeds had the same pair of great-grandparents and were of known genotypes. These six pools were used in the experiment itself, so that seeds from the same pool were genetically more similar than those from different pools. The effects of primary interest include not only those due to the three pigment intensities and the six seed pools but also possible interactions between them.

The experimental plots, each containing a single seed, were spread over two separate but nearby fields; see Fig. 3. The left-hand field contained two blocks, in which 56 WW, 42 Ww and 16 ww seeds were planted from each pool, the relative frequencies reflecting those found in natural populations. The right-hand field had an additional part block in which there were 409 rather than 684 seeds. In total, 3145 seeds were planted: 1550 dark, 1158 light and 437 white. The high level of replication was employed because pigment intensity was expected to

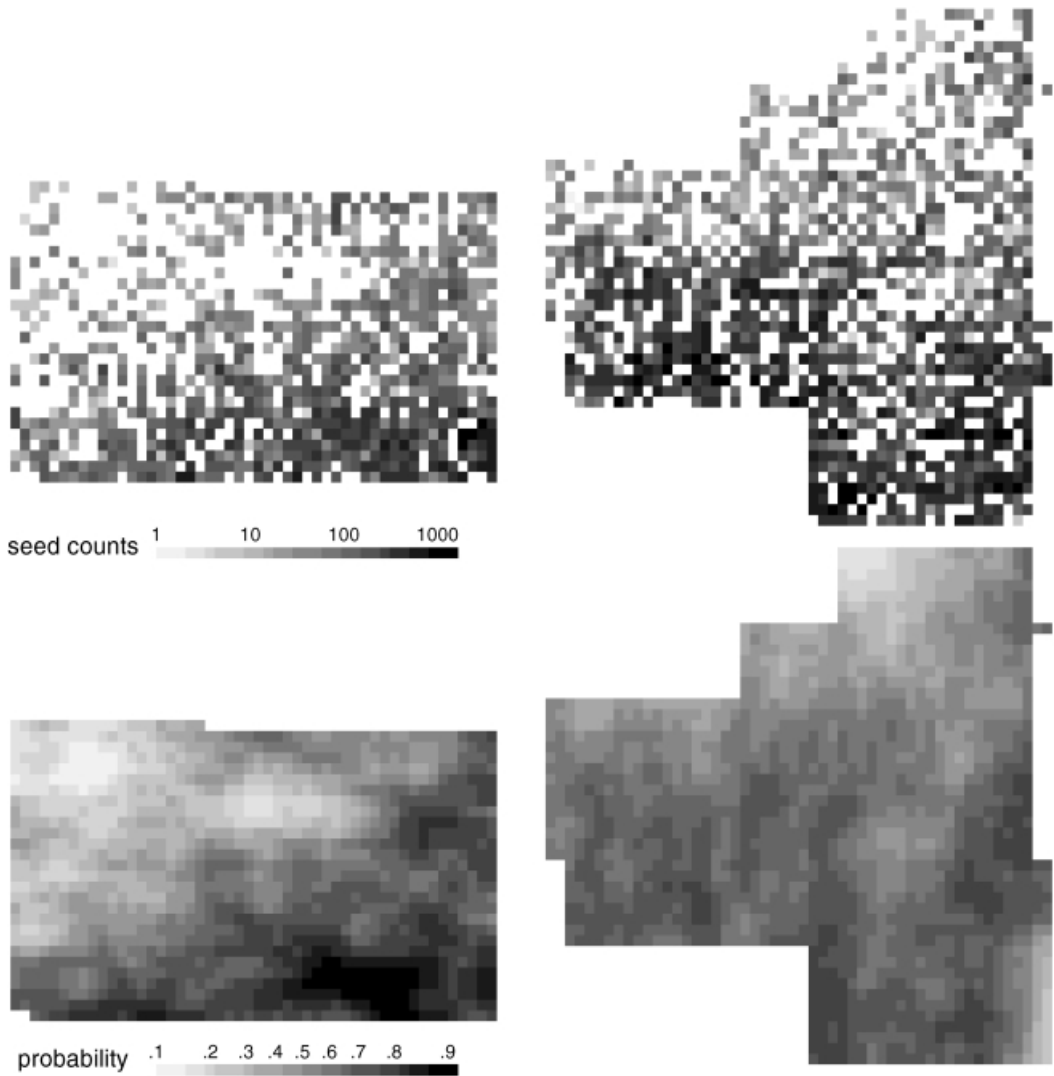


Fig. 3. Raw data (top) and posterior fertility map (bottom) in the morning-glory seed trial: for the raw data, each pixel represents the number of seeds collected from a single plant—the darker the pixel, the higher the count—white pixels identify plots in which no seed was produced; fertilities are shown for all locations where seeds were planted

have a relatively small effect. The number of seeds produced by each seedling s was recorded, with the outcomes represented in Fig. 3: the darker the pixel, the higher the count. There are 1277 0s, indicated by white pixels. The non-zero counts range between 1 and 1665, with a median of 98 and mean of 197. There is compelling visual evidence of spatial variation in fertility. Indeed, the experimenters noted that the left-hand field suffered from accumulation of excess water in the upper left region, caused by a slight gradient, whereas the right-hand field contained soil of relatively poor quality. Here, we consider only the presence or absence of seed for each s , writing $y_s = 1$ or $y_s = 0$, correspondingly. An overall analysis of all counts would be of more interest but, although a prototype exists, the two aspects of the experiment

require separate parameterizations and it is therefore defensible first to address the presence–absence data in their own right. Note that an analysis of the positive counts, with the 0s discarded as ‘missing data’, would not be legitimate.

Let τ denote the 18 centred effects associated with the various combinations of pigment intensities and seed pools and as usual write ψ for the plot fertilities. Let p_s denote the probability of producing seed in plot s , given τ and ψ . Then, our model postulates that the corresponding vector p is of the form

$$p = \frac{\exp(F\psi + T\tau)}{1 + \exp(F\psi + T\tau)},$$

in the same notation as equation (2.1), where exponentiation and division on the right-hand side occur element by element. The likelihood follows on assuming that plants produce seeds independently, given ψ and τ . Originally, we also allowed p to depend on a vector of unmeasured plot effects but these proved to be negligible and were subsequently omitted.

We take a vague prior for τ , subject to the zero-sum constraint. We comment subsequently on this choice. For an example of constrained Gaussian priors in a factorial experiment on yields, see Besag and Higdon (1993), section 3. For ψ , we adopt separate two-dimensional fertility priors in the two fields, each defined over a rectangular array that allows a conceptual border of width at least two plots, to eliminate edge effects; see Section 2. We permit different hyperparameters in the two fields for the reasons described above. In the computations, we use Hastings steps to accommodate the non-Gaussian likelihood.

We note that morning-glory plants are susceptible to competition through overtopping but this is not an issue at the spacing in this experiment. In other applications, competition might be addressed through an autologistic component (Besag, 1974) in which the interaction parameters for nearest neighbours are constrained to be non-positive. Incidentally, we consider the direct approach to general interaction between binary variables via autologistic models, exemplified in section 7.1 of Besag (1974) and used by some researchers more recently, to be outdated; see also the discussion and rejoinder to Besag (1974). Of course, any serious investigation of competition would require a range of different plant spacings.

5.1. Results

The posterior fertility means in the lower panel of Fig. 3 show a very strong spatial pattern, as expected, with the probability of seed production ranging from about 0.1 to 0.9 over the layout. The corresponding precisions λ_ψ are comparable in the two fields but the anisotropy parameters β differ considerably, with means 0.74 (left) and 0.39 (right). Fig. 4 displays posterior means and 90% credible intervals for the centred effects of all 18 combinations of intensities and pools. The widths of the intervals reflect the wide disparity in the number of seedlings planted for each colour intensity. There is no evidence of main effects, either in Fig. 4 or more formally. At first sight, there may be the hint of an interaction effect but this is likely to be an artefact of the above disparity. If random samples of sizes n and $3n$ are taken from two independent $N(0, \kappa)$ populations, there is probability $\frac{2}{3}$ that the n -sample mean will be the larger in magnitude. Thus, we find no evidence that colour intensity or seed pool have any effect in whether plants produce seeds. Our conclusion would only be strengthened if the uniform treatment prior were to be replaced by a Gaussian distribution. However, there remains the possibility of an interaction with fertility.

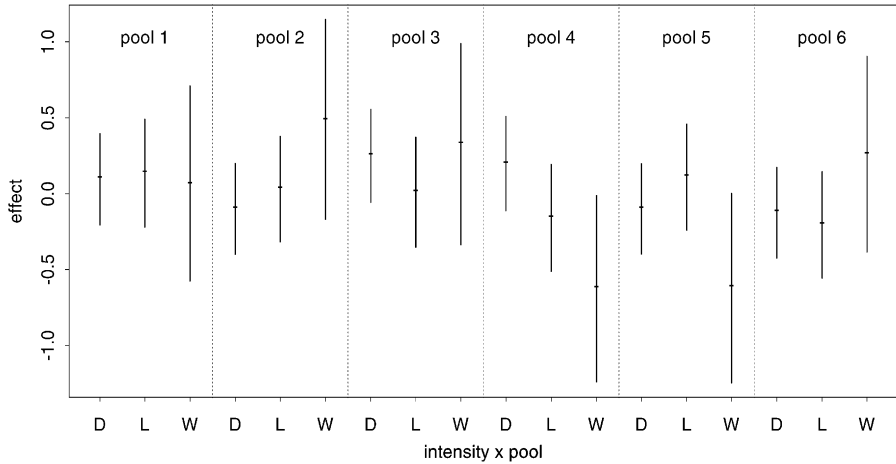


Fig. 4. Pointwise 90% credible intervals for the intensity $\times$ pool interaction for the seed trial

6. Six variety trials on corn

Our final application focuses on a set of six trials, each containing the same 64 varieties of corn. The trials were conducted in a single growing season in six different counties in the coastal plain of North Carolina. Although the soil and general weather patterns are similar throughout the region, there is some effect on yield due to variation in rainfall. Each trial conformed to the practices of the farmers in the immediate locality: in particular, no irrigation was used and there were slight differences in physical row spacing, harvesting equipment and fertilizer application. The information distributed about such trials includes the statement ‘Growers are cautioned against making varietal selection decisions based on an individual location in any one year’ (Bowman, 1995).

Each trial involved three replicates, laid out in an incomplete-block design, within an 18×11 lattice of rectangular plots, whose length-to-width ratio was about 2:1, as shown in Fig. 5. Each experiment had 24 blocks of eight plots, with a boustrophedon lay-out along the rows. The yields of the remaining six plots in each trial were not recorded and are treated as missing values. We again add two layers of virtual plots to neutralize edge effects, as described in Section 2.3.

6.1. Comparison of separate analyses

We first present separate Bayesian analyses of the six trials and make some comparisons with frequentist results. We adopt the basic two-dimensional Gaussian formulation but with uniform priors for the variety effects, again for comparability. The raw yields and the posterior means of the fertility effects are shown in Fig. 5. The posterior mean for the asymmetry parameter β varies between 0.3 and 0.8, except in county 4, where the mode is close to 0, corresponding to independent column adjustment. Note that, with a Gaussian prior for τ , the variety effects for counties 2, 4 and 6 show substantial shrinkage towards 0.

We also carried out two frequentist analyses for each trial. The first was a standard incomplete-block analysis, with recovery of interblock information, using restricted maximum likelihood (REML) estimation. The second was a geostatistical analysis, with fertility ψ represented by a stationary Gaussian process having lag (r_1, r_2) autocovariance,

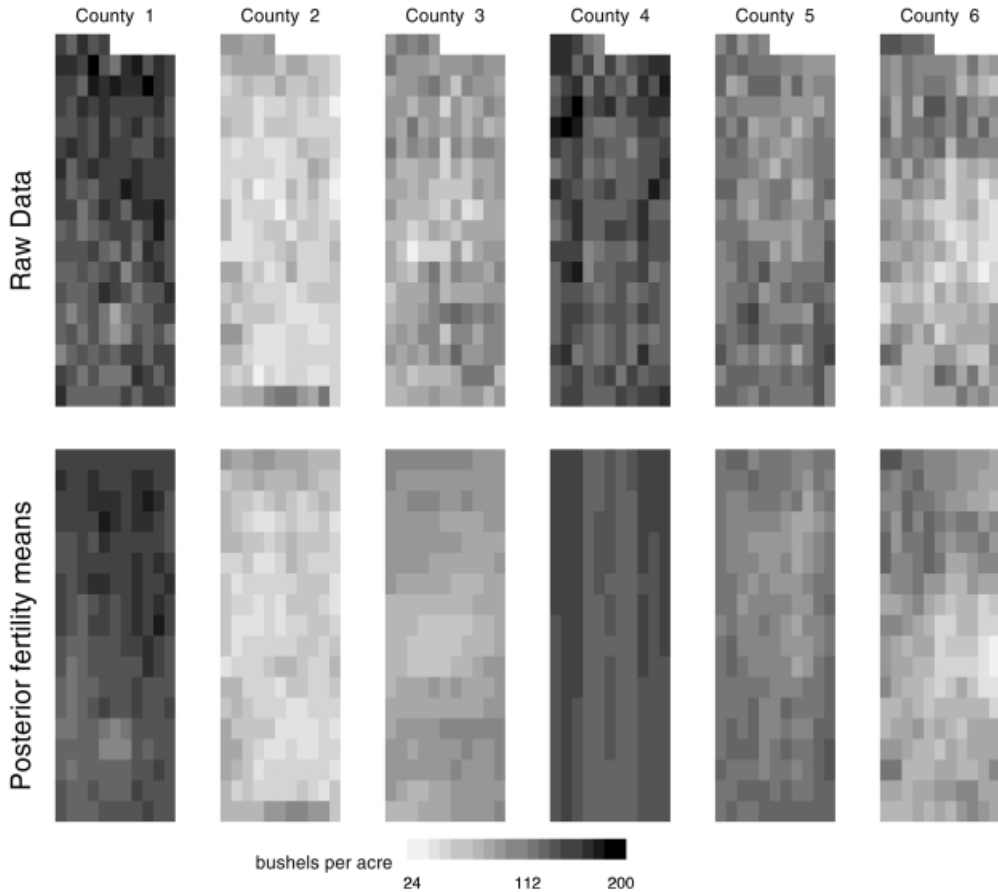


Fig. 5. Raw yields and fertility maps under separate Bayesian analyses for the corn trials

$$\theta \exp \{-(\alpha_1 |r_1|)^{\delta_1} - (\alpha_2 |r_2|)^{\delta_2}\}, \quad (6.1)$$

where $\theta, \alpha_1, \alpha_2 > 0$ and $0 < \delta_1, \delta_2 \leq 2$. This is one possible generalization of the isotropic covariance function in Brewer and Mead (1986); see also Diggle *et al.* (1998). The δ s affect the smoothness of ψ along rows and columns. The five parameters were again estimated by REML.

The variety effects for the Bayesian and geostatistical analyses are very similar but those for the incomplete-block analysis differ substantially in places. Thus, Fig. 6 plots the variety estimates from the Bayesian analysis against those from the two frequentist analyses for each of the six trials. Fig. 6 also shows twice the average posterior standard deviation (Bayesian) and twice the average standard errors (frequentist) of the centred variety effects for each trial. The standard errors for the incomplete-block analyses are noticeably larger and this may reflect the strong row orientation of the blocks.

In view of the numerical results, it is not surprising that the estimated fertility patterns for the two types of spatial analysis are mostly rather similar. However, county 4 provides an exception. In the geostatistical analysis, the estimated range of dependence is very small, so fertility corresponds closely to yield and is rather noisy. In the Bayesian analysis, the fertility

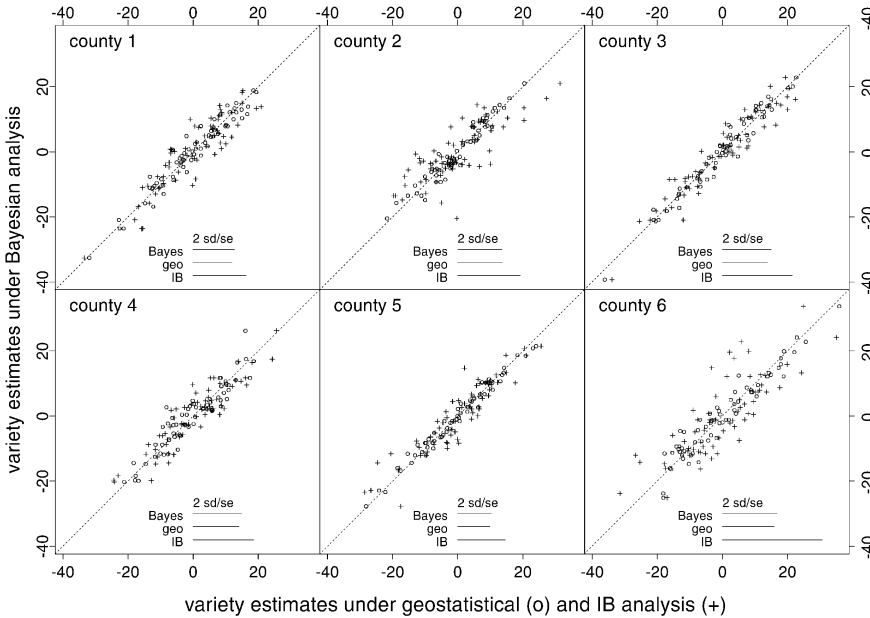


Fig. 6. Variety estimates under the Bayesian formulation (y-axis), plotted against the corresponding estimates from the incomplete-block (IB) analysis (x-axis; +) and estimates from the geostatistical (geo) analysis (x-axis; ○) for the corn trials

map shows little variation within columns. Finally, note that the geostatistical formulation does not allow for uncertainty in estimating the parameters in expression (6.1), though this could be corrected in a corresponding fully Bayesian analysis (Diggle *et al.*, 1998).

6.2. Combining information across trials

Although it is of course possible to amalgamate the results of the individual analyses, it is natural in a Bayesian setting to adopt a hierarchical variety prior, as described in Section 2.2, and hence to produce a combined analysis. Thus, we accommodate a different overall level in each trial and allow variety $\times$ trial interactions by replacing τ_k by $\tau'_k = \tau_k + \xi_k^t$ in trial t , where the ξ_k^t are given independent $N(0, 1/\lambda_{\tau_k})$ priors and the λ_{τ_k} have independent gamma distributions. If the data support large values of the precisions λ_{τ_k} , then the τ'_k are shrunk towards a common value τ_k .

In the combined analysis, the posterior distributions of the fertilities ψ and the asymmetry parameters β agree closely with those obtained previously. However, since the combined analysis adjusts each variety effect τ'_k towards a common combined value τ_k , the posterior distribution for each λ_ψ is shifted slightly to the left and that for each λ_ψ slightly to the right. Results for the top 10 varieties, according to the posterior means of the τ_k s, are given in Fig. 7, together with the associated 90% credible intervals. The previous posterior means for each of these varieties are also shown, with county number as the plotting symbol, and are joined by dotted lines to the corresponding estimates under the combined analysis. The heavy shrinkage confirms that the individual trials are not very informative. As an example, Fig. 8 compares the variety estimates for counties 1 and 5, with Fig. 8(a) and Fig. 8(b) showing the results from separate and combined analyses respectively. Finally, as we might expect, there is

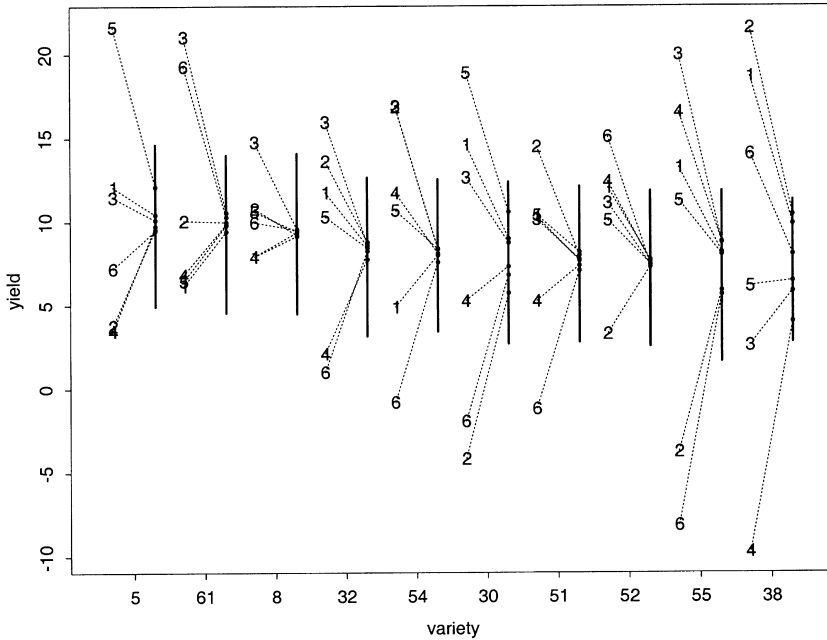


Fig. 7. Individual and combined variety estimates for the top 10 varieties: individual variety estimates for each county are connected (from left to right) by dotted lines to their corresponding estimates under the combined analysis; the magnitudes of the separate trial estimates are plotted using the county in which the trial was conducted as the plotting symbol ($\circ$, 90% credible intervals for the combined common estimate τ_k for the varieties)

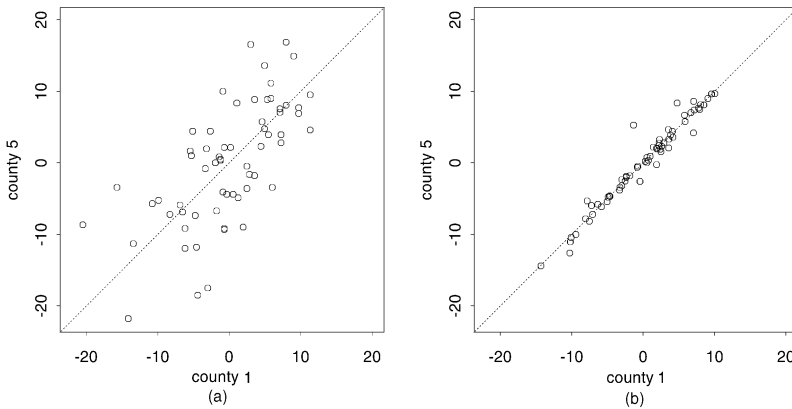


Fig. 8. Variety estimates from county 1 plotted against those from county 5: (a) separate analyses; (b) combined analysis, showing shrinkage that results from the hierarchical prior

a substantial improvement in consistency between the individual and combined results when Gaussian, rather than uniform, variety priors are used, because of the additional shrinkage.

7. Discussion

In this paper, we have developed Bayesian methodology for the analysis of agricultural field experiments, a topic that has received little attention previously despite a vast frequentist

literature. The most important peculiarity of field experiments is the presence of fertility patterns, occurring naturally and through management practices. The accuracy with which treatment or variety effects are estimated depends crucially on the means by which these unmeasured patterns are accommodated in the analysis. In the Bayesian paradigm, it is necessary to specify a corresponding prior distribution, which must capture the main features of the spatial variation and yet must not be too intricate, to guard against overfitting which might smear out the effects of primary interest. We contend that the most important requirement of the prior distribution is that fertilities in plots that are close together should generally be encouraged to be rather similar but that this should not compromise overall flexibility, which militates against an assumption of spatial stationarity; see, for example, Wilkinson *et al.* (1983).

Thus, we have adopted a four-neighbour Gaussian intrinsic autoregression as our base-line representation of two-dimensional variation in fertility. This particular choice involves two unknown parameters, controlling scale and lateral asymmetry, and has dual interpretations, both as a limiting version of a first-order stationary Gaussian (conditional) autoregression (Besag, 1974) and as locally planar stochastic interpolation. Also, as the asymmetry parameter $\beta \in (0, 1)$ tends to each of its extremal values, we obtain random walks along the rows or along the columns of the trial and these we employ for one-dimensional fertility adjustment. Such random walks epitomize the non-stationarity in our prior distributions and were first adopted by Besag and Kempton (1986) in frequentist spatial analysis.

Empirical investigations of lattice-based fertility models are described by Cullis and Gleeson (1991), Kempton *et al.* (1994) and Grondona *et al.* (1996) but with a restriction to separable autoregressive integrated moving average processes, which excludes our specific proposal. The papers all support parsimony, with which we concur, but differ in their detailed conclusions. Thus, the first two favour differencing as a component and hence non-stationarity but the third promotes the (stationary) $\text{AR}(1) \times \text{AR}(1)$ or doubly geometric process, studied extensively by Martin (1979, 1990): note that, somewhat confusingly, the authors refer to this as '2d $\text{AR}(1)$ ', although the conditional expectation of the fertility in any interior plot, given all others, depends on the nearest eight rather than the adjacent four fertilities. Recall that, viewed as a conditional autoregression, the $\text{AR}(1) \times \text{AR}(1)$ process has an intrinsic limit which corresponds to independence of two-dimensional differences and the locally quadratic representation, discussed in Section 2.3.2 and illustrated in Section 4. There, we considered such differencing unsuitable in a prior distribution because it absorbs arbitrary row and column effects and is therefore highly degenerate. Gilmour *et al.* (1997) reached an analogous frequentist conclusion that double differencing is inefficient because it loses too many degrees of freedom. Hence, they resurrected the stationary $\text{AR}(1) \times \text{AR}(1)$ process and resorted to cubic smoothing splines to remove large scale variation. However, the vast majority of intrinsic autoregressions do not suffer from severe degeneracy and this includes the one that we propose, with its single (desirable) deficiency in rank producing non-stationarity, as in one-dimensional differencing. We do not think that intricate modelling of fertility patterns of the type employed by Gilmour *et al.* (1998) is necessary; and it is certainly not desirable in routine use, quite apart from the dangers of overfitting fertility effects to limited data.

Our paper also extends the Gaussian priors and likelihoods of our basic formulation by introducing hierarchical-*t* distributions to cater for outlying observations, aberrant treatment or variety effects and, in one-dimensional adjustment, sudden jumps in fertility. We contend that simple row or column adjustment is often sufficient in practice, though the direction is not always the one suggested by plot dimensions, as in the example in section 6 of Cullis and Gleeson (1991). Incidentally, one-dimensional adjustment may be preferred if boustrophedon

harvesting has a significant effect on yield, as in the South Australian wheat trial discussed by Gilmour *et al.* (1997), which adds further complications to their sequence of models. A simple hierarchical- t analysis of the data suggests perhaps three breaks in fertility but no outliers and produces an average posterior standard deviation of variety differences that is marginally less than the corresponding standard errors in Gilmour *et al.* (1997). Of course, this may be fortuitous but again we doubt the merits of very intricate formulations.

We have not discussed discontinuities in two-dimensional adjustment, though one could adapt edge detection methods from Bayesian image analysis. This is computationally feasible but, once more, we suspect that it is overelaborate for routine use. Among many other topics given scant attention, we note the rather arbitrary choice of hyperpriors and the inelegant rescaling of the data, though neither has produced any practical problems in the analysis of many different data sets, so far as we know. The paper does not consider formal procedures for choosing between different formulations. This reflects a personal viewpoint with which others may disagree. However, our approach already incorporates some model uncertainty, both in the basic formulation (2.8), where values of β close to its extremes produce essentially one-dimensional adjustment, and in the hierarchical- t representation, through variation in the degrees of freedom. In the former case, one might prefer to include the extremes themselves, as well as the two-dimensional formulation, along lines first suggested by Dr A. H. Seheult some 10 years ago. This would then involve models of differing dimensions and presumably the reversible jumps MCMC technology of Green (1995). We have also left others to comment on the important issues of Bayesian design, though these will not be influential until Bayesian analyses, in some form, are accepted by practitioners.

We have described several different applications. These involve discrete as well as continuous measurements and both small and large experiments, with numbers of observations ranging from a few tens to several thousands. None of the examples is straightforward and, indeed, a referee has suggested that it would have been preferable to concentrate on 'simple' data sets. We do not agree: in a model-based formulation, it is highly desirable to establish that the complications, that may occur in practice and that may not be evident in the data, can be dealt with in a fairly routine fashion. We hope that we have gone some of the way towards achieving this goal. No doubt, our analyses could be improved on but we hope that the examples will help in persuading others that Bayesian analysis is worth pursuing in an area that remains perhaps the staunchest frequentist stronghold.

Indeed, we have maintained that Bayesian inference has some important practical advantages in analysing field experiments. For example, the results are easier to interpret, particularly in ranking and selection and when communicating with non-statisticians; the results from previous experiments can be incorporated in a rather natural manner into the prior for treatments or varieties in a subsequent trial, though we have only illustrated the analogous problem of combining information from several different trials; and there seems more freedom in using MCMC methods to analyse reasonably realistic formulations and to address model uncertainty.

Acknowledgements

JB's interest in field experiments was first stimulated by John Nelder and, in the applicability of spatial methods, by Maurice Bartlett, though their views no doubt differ substantially from those expressed in the paper. Subsequent influences include Rob Kempton, Richard Martin, Roger Mead, Desmond Patterson, J. S. Papadakis, Clifford Pearce, Graham Wilkinson and, at the University of Durham, Peter Green, Chris Jennison and, from a Bayesian viewpoint

especially, Allan Seheult. More recently, the authors are grateful for contributions from Simon Byers, Sharon Browning, Steven MacEachern, Christopher Triggs, the referees and the Research Section Committee. They also thank Mark Rausher and Daryl Bowman for providing the data in Sections 5 and 6 respectively. This work was supported in part by grants DMS-91-14402, DMS-92-16158, DMS-9505114 and DMS-9704425 from the National Science Foundation.

References

- Besag, J. (1974) Spatial interaction and the statistical analysis of lattice systems (with discussion). *J. R. Statist. Soc. B*, **36**, 192–236.
- (1989) Towards Bayesian image analysis. *J. Appl. Statist.*, **16**, 395–407.
- Besag, J. and Green, P. J. (1993) Spatial statistics and Bayesian computation (with discussion). *J. R. Statist. Soc. B*, **55**, 25–37, 53–102.
- Besag, J., Green, P. J., Higdon, D. M. and Mengersen, K. L. (1995) Bayesian computation and stochastic systems (with discussion). *Statist. Sci.*, **10**, 3–66.
- Besag, J. and Higdon, D. M. (1993) Bayesian inference for agricultural field experiments. *Bull. Int. Statist. Inst.*, **55**, book 1, 121–136.
- Besag, J. and Kempton, R. A. (1986) Statistical analysis of field experiments using neighbouring plots. *Biometrics*, **42**, 231–251.
- Besag, J. and Kooperberg, C. L. (1995) On conditional and intrinsic autoregressions. *Biometrika*, **82**, 733–746.
- Besag, J., Mengersen, K. L. and Higdon, D. M. (1997) Combining information in logistic regression. *Report*. Institute of Statistics and Decision Sciences, Duke University, Durham.
- Besag, J. and Seheult, A. (1989) Discussion on Leave-*k*-out diagnostics for time series (by A. G. Bruce and R. D. Martin). *J. R. Statist. Soc. B*, **51**, 405–406.
- Besag, J., York, J. and Mollié, A. (1991) Bayesian image restoration, with two applications in spatial statistics (with discussion). *Ann. Inst. Statist. Math.*, **43**, 1–59.
- Bowman, D. T. (1995) Measured crop performance: corn and silage 1995. *Crop Science Research Report 154*. North Carolina State University, Raleigh.
- Brewer, A. C. and Mead, R. (1986) Continuous second order models of spatial variation with application to the efficiency of field crop experiments (with discussion). *J. R. Statist. Soc. A*, **149**, 314–348.
- Brownie, C., Bowman, D. T. and Burton, J. W. (1993) Estimating spatial variation in analysis of data from yield trials: a comparison of methods. *Agron. J.*, **85**, 1244–1253.
- Byers, S. D. and Besag, J. (1998) Combining information in the adjustment of physical constants. *Report*. Department of Statistics, University of Washington, Seattle.
- Cullis, B. R. and Gleeson, A. C. (1991) Spatial analysis of field experiments—an extension to two dimensions. *Biometrics*, **47**, 1449–1460.
- Diggle, P. J., Tawn, J. A. and Moyeed, R. A. (1998) Model-based geostatistics (with discussion). *Appl. Statist.*, **47**, 299–350.
- Draper, D., Graver, Jr, D. P., Goel, P. K., Greenhouse, J. B., Hedges, L. V., Morris, C. N., Tucker, J. R. and Waternaux, C. M. (1992) *Combining Information: Statistical Issues and Opportunities for Research*. Washington DC: National Academy Press.
- Epperson, B. K. and Clegg, M. T. (1988) Genetics of flower color polymorphism in the common morning glory (*Ipomoea purpurea*). *J. Hered.*, **79**, 64–68.
- Fisher, R. A. (1928) *Statistical Methods for Research Workers*. Edinburgh: Oliver and Boyd.
- Gilmour, A. R., Cullis, B. R. and Verbyla, A. P. (1997) Accounting for natural and extraneous variation in the analysis of field experiments. *J. Agric. Biol. Environ. Statist.*, **2**, 269–293.
- Green, P. J. (1995) Reversible jump Markov chain Monte Carlo computation and Bayesian model determination. *Biometrika*, **82**, 711–732.
- Green, P., Jennison, C. and Seheult, A. (1985) Analysis of field experiments by least squares smoothing. *J. R. Statist. Soc. B*, **47**, 299–315.
- Grondona, M. O., Crossa, J., Fox, P. N. and Pfeiffer, W. H. (1996) Analysis of variety yield trials using two-dimensional separable ARIMA processes. *Biometrics*, **52**, 763–770.
- Kempton, R. A., Seraphin, J. C. and Sword, A. M. (1994) Statistical analysis of two-dimensional variation in variety yield trials. *J. Agric. Sci.*, **122**, 335–342.
- Künsch, H. R. (1987) Intrinsic autoregressions and related models on the two-dimensional lattice. *Biometrika*, **74**, 517–524.
- Martin, R. J. (1979) A subclass of lattice processes applied to a problem in planar sampling. *Biometrika*, **66**, 209–217.
- (1990) The use of time-series models and methods in the analysis of agricultural field trials. *Communs Statist. Theory Meth.*, **19**, 55–81.
- Matérn, B. (1960) Spatial variation. *Medd. Stat. Skogsforsk.*, **49**, no. 5.

- Mead, R. (1988) *The Design of Experiments: Statistical Principles for Practical Applications*. Cambridge: Cambridge University Press.
- Papadakis, J. S. (1984) Advances in the analysis of field experiments. *Proc. Acad. Athens*, **59**, 326–342.
- Patterson, H. D. and Williams, E. R. (1976) A new class of resolvable incomplete block designs. *Biometrika*, **63**, 83–92.
- Preece, D. A. (1982) The design and analysis of experiments: what has gone wrong? *Util. Math. A*, **21**, 201–244.
- Press, W. H., Flannery, B. P., Teukolsky, S. A. and Vetterling, W. T. (1988) *Numerical Recipes in C: the Art of Scientific Computing*. New York: Cambridge University Press.
- Rausher, M. D. and Fry, J. D. (1993) Effects of a locus affecting floral pigmentation in *ipomoea purpurea* on female fitness components. *Genetics*, **134**, 1237–1247.
- Smith, A. F. M. and Roberts, G. O. (1993) Bayesian computation via the Gibbs sampler and related Markov chain Monte Carlo methods (with discussion). *J. R. Statist. Soc. B*, **55**, 3–23, 53–102.
- Taplin, R. H. and Raftery, A. E. (1994) Analysis of agricultural field trials in the presence of outliers and fertility jumps. *Biometrics*, **50**, 764–781.
- Tierney, L. (1994) Markov chains for exploring posterior distributions (with discussion). *Ann. Statist.*, **22**, 1701–1762.
- Whittle, P. (1954) On stationary processes in the plane. *Biometrika*, **41**, 434–449.
- Wilkinson, G. N. (1984) Nearest neighbour methodology for design and analysis of field experiments. In *Proc. 12th Int. Biometric Conf.*, pp. 64–79. Washington DC: Biometric Society.
- Wilkinson, G. N., Eckert, S. R., Hancock, T. W. and Mayo, O. (1983) Nearest neighbour (NN) analysis of field experiments (with discussion). *J. R. Statist. Soc. B*, **45**, 151–211.
- Zimmerman, D. L. and Harville, D. A. (1991) A random field approach to the analysis of field-plot experiments and other spatial experiments. *Biometrics*, **47**, 223–239.

Discussion on the paper by Besag and Higdon

Chris Glasbey (*Biostatistics and Statistics Scotland, Edinburgh*)

I welcome this paper, which proposes the analysis of agricultural field experiments by using both a spatial model and a Bayesian framework. Of course, it would also be possible to carry out a Bayesian analysis with the traditional blocking structure, or a frequentist analysis with spatial modelling, as others have done.

The major advances in the paper are in being able to model jumps in fertility and to handle uncertainties in treatment rankings. Perhaps, for expository purposes, the emphasis is on the analysis of single trials. However, I think that it is in the analysis of series of agricultural trials that Bayesian methods have most to offer, and there is little room for improvement over traditional methods in single trials. For example, spatial modelling is needed in the first two experiments only because they were badly designed.

In the strawberry trial, my preference would be to omit from the analysis the column of plots nearest the hedge, because their yields have been suppressed to half those of the rest of the experiment. I would be wary about assuming that treatment and location effects are additive, even though the data seem to be consistent with this assumption. Table 8 shows treatment means for a randomized block analysis with and without the last column of plots. In the latter case, the results are very similar to the Bayesian two-dimensional first-order analysis: the only change in the treatment ranking is that V and G have swapped. Standard errors of treatment differences similarly range between 0.54 and 0.70.

I would like to discuss possibilities for Bayesian analyses of series of agricultural trials.

- (a) Each cycle of a plant breeding programme involves taking several crosses to generate a large number of new genotypes. For example, 15000 new genotypes are produced in a potato programme. These seeds are grown into plants, and a subset is selected. Through a series of field and glasshouse trials, a progressively smaller number of genotypes is selected in each successive year, until finally a few are passed to commercial partners for multiplication and large scale trialing. Current selection methods use a combination of ‘culling’ (i.e. if a genotype does not meet a

Table 8. Treatment means

Analysis	V	Rl	F	G	Re	E	M	P
‘Naïve’ analysis	6.3	3.5	5.6	6.6	5.6	6.6	5.6	6.5
Last column omitted	6.3	4.4	5.6	6.6	5.6	6.6	6.3	7.2

minimum standard for a character it is rejected), selection indices and breeders' instincts. My colleague, Jim McNicol, has plans to develop a Bayesian decision framework to integrate all the data generated in a breeding programme, using utilities to quantify the value that breeders put on each of the characters.

- (b) In animal research and development programmes, financial and ethical pressures mean that experiments are carried out using only small numbers of subjects. Therefore, each experiment lacks power when analysed in isolation. However, the power can be increased by using Bayesian methodology to take into account information from similar previous experiments. In particular, David Elston (of Biomathematics and Statistics Scotland) is investigating the use of Bayesian methods to pool variances, to incorporate information about variability from past experiments.
- (c) The analysis of national variety trials continues to present statisticians with many challenges and opportunities. For example, Mike Talbot is currently developing methods for predicting cereal variety performance in particular localities, but I shall leave it to him to say more in his own contribution to the discussion on this paper.

Finally, I learned a new word from this paper: 'boustrophedon'. According to my dictionary, it means 'to switch between left to right and right to left in alternate rows', can apply equally to writing and harvesting and derives from the Greek for 'as an ox turns in ploughing'. In conclusion, I am pleased to propose a vote of thanks for this paper.

Donald A. Preece (*University of Kent, Canterbury*)

Experiments on agricultural crops may be on land at a research station, or on farmers' fields or possibly on terrain newly cleared for agriculture. The land may be alluvial soil, or savannah, or a terrace on a mountain-side. Statistical methodology that is beneficial for results from some sites may not be universally useful, but that should not deter us.

Not all error variability in data from agricultural trials is due to soil fertility or the physical and biological environment. R. A. Fisher gave 'local control' as the reason for blocking, but the blocks in a design also have an organizational, or operational, or management role, for picking up components of variability arising from how an experiment is run (Dyke (1974), pages 42–43, and Pearce (1983), pages 42–43). Operations such as hand-weeding or harvesting should usually be done block by block: either each block should be completed before the next is begun, or separate teams should be assigned to separate blocks. Havoc can be caused to a soft fruit experiment if parts of some blocks have their fruit picked on Friday afternoon and the experiment is finished next week, when the fruit may be riper or mouldier; picking-time differences may be confounded with fertility differences. Because of the operational role of blocks in good experimentation, parameters for blocks are needed in *all* initial models for well-run experiments with blocks.

A yield variate from a variety trial may arise from harvesting all plots at one go, or from harvesting each variety when that particular variety is deemed to be ripe. For some crops such as strawberries, each plot will be harvested on successive occasions to give a total yield over harvests. Fertility effects may change over a longish harvesting period, making detailed data modelling hard.

In a typical variety trial, several variates are recorded, probably including more than one yield variate. In a trial on a cereal crop, separate yields of grain and straw might be recorded; in a fruit or root trial, yield might be recorded separately for different grades of produce. Just as the blocking in a design may be helpful for one variate but not another, so the benefits of any new form of analysis may differ between variates; unexplainable differences might indicate overanalysis.

In a variate described as 'grain yield', the individual values may have been adjusted to a standard (percentage) moisture content. This will have involved taking a sample from each plot and drying it to obtain its *actual* moisture content. This suggests another way in which variability can be introduced into data.

Mention of moisture reminds me of a trial whose results suggested a huge fertility trend. In fact, the crop was grass, and harvesting had started at one end of the site, early in the day — when the grass was heavy with dew; as the hours went by, the sun scorched off the moisture, reducing the harvested weights.

Not all crop varieties are self-pollinating. In some trials, differences within and between varieties could arise from pollination problems.

These various practical matters are important for variety trials generally, and in particular for tonight's examples, which were also well chosen to illustrate other general points.

In Table 1, giving yields for the El Batán trial, the varieties appear in numerical order for block 1,

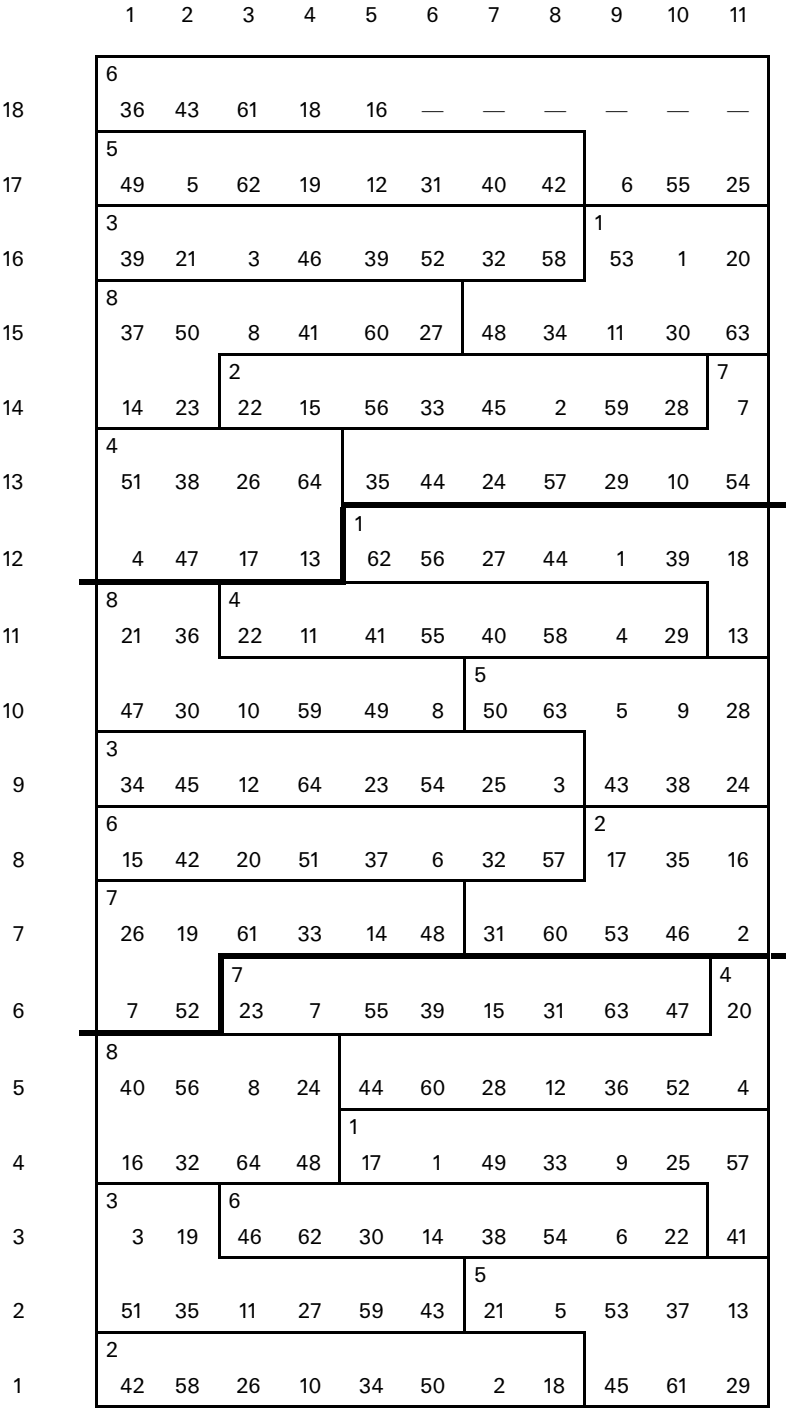


Fig. 9. Lay-out of the North Carolina corn trial in county 1: the design is a triple lattice (—, boundaries between the replicates; —, boundaries between blocks within replicates); the varieties are numbered as in the data set, as are the blocks within replicates

suggesting that this block may not have been randomized. Alas, the practice of not randomizing an initial block has sometimes been recommended and adopted. Then, if varieties are systematically numbered, say by parentage or according to a previous ranking, a varietal effect could be confounded with a fertility effect in the initial block.

The North Carolina trials seem to have design problems too. Seemingly, a central authority asked each of six locations to set up an 18×11 grid of plots to test 64 varieties. The mismatch of the numbers may have arisen from practical considerations, such as the width of machinery. But the lay-out of the first of the trials (Fig. 9) entitles us to doubt whether this is good design. The design is a triple lattice (Cochran and Cox (1957), section 10.22) in the sense of Yates (1937), p. 87, but it is also a jigsaw puzzle design, in the sense of Hamleys of Regent Street's Christmas offers, 1998 — 'complete' with the deliberately missing piece in one corner! In incomplete-block analyses of this and the other North Carolina trials, is recovery of interblock information beneficial?

In the second North Carolina trial, each of two varieties has its largest yield equal to roughly three times its smallest, with the large yields on end plots. Indeed several, but not all, end plots have yields so large as to be 'way out of line'. Perhaps fertilizer was spilt at the bottom end of the site, but one anomalous yield is at the top. Perhaps border areas of crop were harvested as if they belonged to plots — a mistake always to be watched out for. But, whatever the true explanation, the data for this trial must be treated with suspicion.

Other anomalies in the El Batán and North Carolina data reinforce a harsh lesson that I learned from handling data from hundreds of agricultural and horticultural trials: whenever possible, *work forward from the original plot yields in their original units*. Errors in copying and in arithmetic are all too frequent.

When it comes to analysis, the authors say that 'the frequentist resorts to . . . multiple comparisons'. I myself cannot see that multiple-comparison procedures can have the slightest use for *anyone* handling variety trials. Early in a testing programme, the only need is to decide which varieties go forward to the next stage; those that do so will be the 'best' on some criterion or criteria, except that we may wish to include, say, at least one variety of any particular parentage. At a later stage, decisions need to be multivariate; mere ranking on yield will be of little use if some varieties are more susceptible than others to disease.

Results of variety trials may lead to litigation, especially if different varieties come from different agrotechnology companies. This should not stop us from trying to make statistical and scientific advances, but the inability of law-courts to cope with deep quantitative arguments will limit the analysis procedures that can be adopted.

In studies of methods of analysis for variety trials, the *lengths* and *breadths* of plots, blocks and sites need to be reported, to give some indication of the circumstances for which the findings may be applicable. This point may come to have extra importance if the testing of genetically modified varieties leads to changes in the siting, design and lay-out of trials, with the aim of preventing the spread of new genetic material beyond the experimental area.

I am honoured to second the vote of thanks.

The vote of thanks was passed by acclamation.

Peter J. Diggle (*Lancaster University*)

It is a pleasure to comment on this very interesting paper. It presents a convincing case for a model-based approach to inference by its thoughtful occupation of the middle ground between, on the one hand, an overly cautious concern for the robustness imparted by a design-based approach and, on the other, an optimistic disregard for the potential fragility of overelaborate modelling. Indeed, I would argue that the major contribution of the paper is its advocacy of hierarchical modelling, in which dependence is modelled indirectly through a latent stochastic process, rather than in its Bayesian inferential stance. I sometimes think that the only real difference between Bayesian and non-Bayesian hierarchical modelling is whether random effects are labelled with Greek or Roman letters. Of course, the Bayesian paradigm implemented via Markov chain Monte Carlo methods gives a convenient solution to all manner of nasty inferential problems, making it often the only realistic option.

I was intrigued by the authors' remark that a full treatment of the morning-glory data requires 'separate parameterizations' of the binary and count responses. I sketch here a possible approach, inspired by the authors' comment that treating zero counts as missing would be invalid. Well, maybe it would and maybe it would not. Presumably the authors mean that treating a zero as a *non-informatively missing* value would be invalid.

Suppose that the viability of a plant and the number of seeds produced conditional on viability each reflect a combination of genetic and environmental effects, but in possibly different proportions. A model-based description of this might be obtained by postulating a bivariate latent process $W(s) = \{W_1(s), W_2(s)\}$ such that, conditional on $W(\cdot)$,

- (a) the probability $p(x)$ that a plant at location x is viable is given by

$$\text{logit}\{p(x)\} = \alpha_1(x) + W_1(x),$$

where $\alpha_1(x)$ incorporates relevant covariate information, and

- (b) the number of seeds produced by a plant at location x (considered as informatively missing if the observed number is 0) is Poisson distributed with mean $\mu(x)$, where

$$\log\{\mu(x)\} = \alpha_2(x) + W_2(x)$$

and $\alpha_2(x)$ similarly incorporates relevant covariate information.

By varying the strength of the correlation between $W_1(x)$ and $W_2(x)$ in this model, we can accommodate a range of assumptions about the extent to which viability and abundance of seed are interrelated. Fig. 3 suggests that, in the case of the morning-glory data, they are strongly interrelated.

A special case of this model, in the superficially unrelated setting of survival analysis with intermediate measured outcomes, is described in Diggle and Henderson (1998).

S. C. Pearce (*University of Kent, Canterbury*)

For too long the Bayesians and the practitioners in agricultural experimentation have ignored one another, so I greatly welcome this attempt to bring them together. For my part I am not an anti-Bayesian but in my work I am a non-Bayesian because I can rarely accept the suggested priors.

If we examine any field we shall indeed find trends, but also sporadic patches of shallow soil, impeded drainage, bird damage and much else. Our priors must allow for all of that. I have recently published a paper extending Papadakis's method. It is based on the idea that the only safe covariate must be strictly local (Pearce (1998), methods P2 and P2i) and pay no regard to what might be happening in other parts of the field, so I am interested to see results that are similar to mine appearing in Section 2.3.2. That is the only prior I would be happy to use, not because I suppose it to be more effective than others but because I could defend it by rational argument.

A similar point arises with the strawberry experiment. The hedge is thought to be diminishing the yields of plots adjacent to it. That presents no problems; the usual method is to set up a pseudovariate in which all plots are assigned 0 except those affected by the fault, which are assigned 1. Then the analysis of covariance is used to adjust all values to 0. If we do that, the plots next to the hedge have their data increased by 2.5. Also, the standard errors of treatment differences lie in the range (0.53, 0.59), which is marginally better than with the Bayesian analysis presented. I do not commend the method for that reason, however, but because it is open and objective. People can understand it. Those who did the experiment can justify themselves to their sponsors and to critical referees; they could not do so with priors that they did not understand. If I asked six people their priors and received six different answers, should I do six analyses and present six diverse conclusions to the experimenter for him or her to make a choice? Unless priors are objective and defensible they do not help.

Hans R. Künsch (*Eidgenössische Technische Hochschule, Zurich, and Isaac Newton Institute, Cambridge*)

I congratulate the authors on their interesting paper and would like to add some remarks on their prior for the fertility effects. First, intrinsic autoregressions in two dimensions are considerably more complicated than in one dimension and thus people might be sceptical about using them. For instance, in one dimension, a first-order intrinsic autoregression simply means that the first differences are white noise. But, in two dimensions, the increments in the two main directions have complicated auto-covariances and cross-covariances. However, it is obvious that they cannot be independent since the sum of increments over all loops must be 0 (a loop is a closed path on the lattice $\mathbf{Z}^2$ with edges between nearest neighbours). Julian Besag suggested to me many months ago that these constraints might be sufficient to explain the second moments of an intrinsic autoregression. This is indeed true: if (Z_1) and (Z_2) are two independent white noises on $\mathbf{Z}^2$, then, conditionally on the sums over all loops being 0, (Z_1, Z_2) has the same second moments as the increments of a first-order intrinsic autoregression. The proof

relies on the spectral theory for best linear predictors of multivariate stationary processes; see for example Priestley (1981), sections 9.2.1 and 10.3.

Second, if W is the precision matrix of a first-order autoregression on a rectangle as defined in equation (2.4), then the eigenvalues and eigenvectors are available in closed form; see Künsch (1994). So there is no need to rely on numerical routines as described in Section 2.5. In Künsch (1994) I also showed that replacing the squares in equation (2.4) by a Huber function can preserve sharp edges without increasing the computational difficulty too much.

Finally, I would like to confirm that it is extremely difficult to find simple models which give a good fit to uniformity trials. With stationary autoregressions we need large neighbourhoods but we still find many unexplained structures in the data, which suggest various kinds of non-stationary behaviour; see Künsch (1982). Using intrinsic autoregressions which are not too restrictive for the fertility effects therefore seems to be a good idea.

Steven G. Gilmour (*University of Reading*)

The authors have been careful to ensure that the methods recommended are sufficiently simple to be used in practice. I wonder whether they have made the methods less relevant to make them simpler to apply. This may be a matter of presentation, rather than any inherent limitation of the Bayesian methods.

In the El Batán variety trial, the results are summarized in Table 3 in terms of posterior expected yields and posterior probabilities. A more relevant summary would be provided by the posterior expected losses,

$$E_{\theta|Y}\{(\tau_{\max} - \tau_i)|y\},$$

where θ is the vector of all the parameters and $\tau_{\max}$ is the true, but unknown, maximal variety effect. It is more useful to know how far from the optimum our chosen variety is likely to be, rather than to know the probability that it is optimal. For example, the two best varieties may give almost identical yields. If the objective is not to pick the best variety, but to select a few for further experimentation, the 10th percentile of the posterior distribution of the loss, or something similar, may be more useful. Are these quantities also easily obtainable from the Markov chain Monte Carlo output?

It seems unnatural to do a fairly sophisticated modelling of underlying fertility trends, but not to model all the other things which may affect our responses. These would include management practices (which will very often imply discrete block effects even if the blocks do not have a physical presence), interference or competition effects, possible damage to particular parts of the field and so on. However, it may be impracticable to define prior distributions for all these.

I must comment on the designs of the experiments presented. The paper could, and I believe should, have been titled ‘Bayesian analysis of badly designed agricultural field experiments’. Presumably in well-designed experiments the benefits of the Bayesian analysis will be reduced and may disappear completely. However, even with the Bayesian analysis, a well-designed experiment will still perform better than a badly designed experiment. It would be interesting to quantify these statements.

What should a consulting statistician do when presented with data from such a badly designed experiment? We may derive the most out of *this experiment* by doing something like the Bayesian analysis proposed. However, we are likely to have a greater positive effect on our organization’s research output by gently persuading the experimenters that in future they should consult us *before* doing their experiments.

Carmen Fernández and Peter Green (*University of Bristol*)

We congratulate the authors for their insightful perspective on the use of spatially smoothing stochastic models for agricultural field experiments, and for demonstrating the host of opportunities for inference that the Bayesian setting provides.

In current research with Sylvia Richardson, we are developing stochastic models for geographical epidemiology, where spatial patterns are often an issue. Part of our work focuses on mixture modelling with spatially correlated allocation variables, and we wonder whether this could provide an interesting alternative to the approach in this paper. As an illustration, we re-examine the ‘El Batán’ data set. We model the response (yield) as

$$y_{ij} = \tau_{t_{ij}} + \beta_i + \mu_{z_{ij}} + \sigma\epsilon_{ij}, \quad i = 1, \dots, 3, \quad j = 1, \dots, 50,$$

where i indexes the column, j the plot within the column, t_{ij} the variety employed and $\tau_1, \dots, \tau_{50}$ and

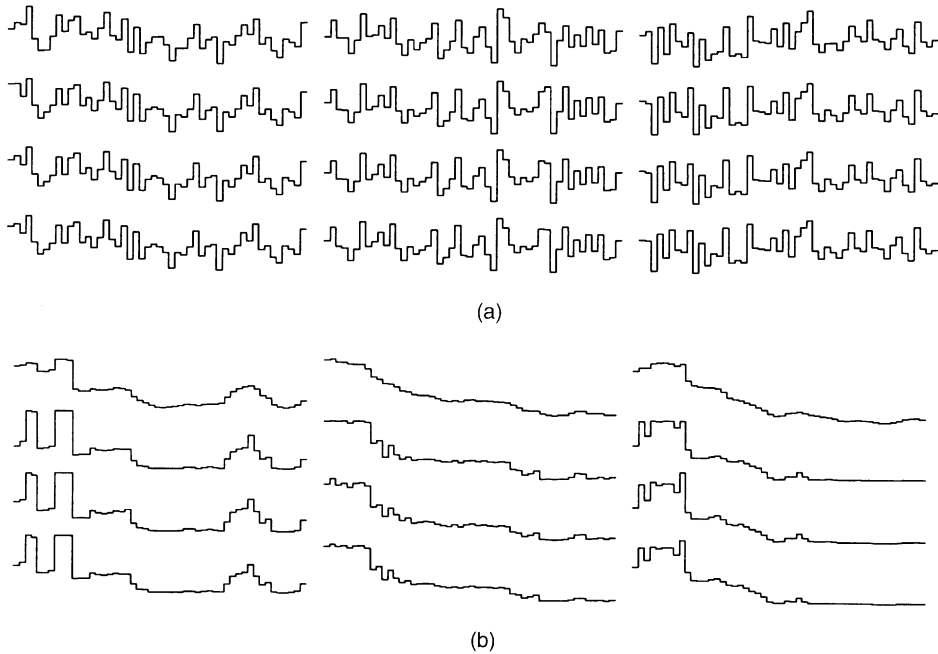


Fig. 10. Hierarchical- t (top traces) versus spatial mixture (three lower traces) formulations respectively using $(k = 3, h = 100)$, $(k = 6, h = 100)$ and $(k = 6, h = 1000)$: (a) posterior means for variety effects; (b) posterior means for fertility effects (hierarchical- t) and $E(\beta_i + \mu_{z_{ij}} | y)$ (spatial mixtures)

$\beta_1, \dots, \beta_3$ respectively denote the centred variety and column effects. The difference from the authors' formulation is that in place of a fertility surface plus noise we aim to explain the remaining variability in the data through a mixture of k normally distributed components, with means $\mu_1 < \dots < \mu_k$ and common variance σ^2 , which capture both plot error and the non-spatial component of fertility. A separation of the last two elements could be done through prior assumptions, but that will be avoided here. All the parameters are assigned weakly informative priors. The allocation variables $\{z_{ij}\}$ are assumed independent for different columns. Within each column, the use of a limiting case of a logistic autonormal multinomial allocation scheme, controlled by a smoothing parameter $h > 0$, ensures that the prior probability that two plots are allocated to the same component decreases as they become further apart. Although here we apply a one-dimensional adjustment, our methodology is directly applicable to arbitrary graphs. In addition, it allows borrowing strength from all plots allocated to the same component, regardless of their location.

Figs 10–12 display results from our analysis, using the authors' hierarchical- t formulation as a benchmark for comparison. Fertility patterns are smoother with the hierarchical- t model, although the posterior means for fertility effects are not entirely comparable with the quantities $E(\beta_i + \mu_{z_{ij}} | y)$ plotted for the spatial mixture specifications, as explained in the previous paragraph. Inference on variety effects is very similar in all cases.

Although this is not how we would interpret our model, it is formally equivalent to a block model where the allocations of plots to blocks (i.e. components) are not asserted *a priori* but are data dependent. Model averaging leads to the relatively smooth fertility patterns that we have displayed.

Neil Butler (*University of Reading*)

I would like to ask whether likelihood-based methods of statistical inference, Bayesian or frequentist, are necessarily the most effective way of analysing the model's non-location parameters (i.e. the neighbour and variance parameters). Certainly, in other areas of statistical research where stochastic

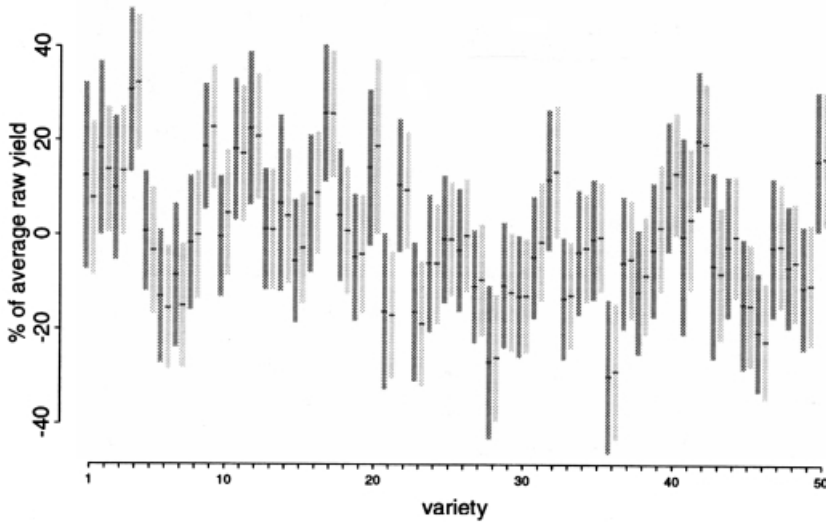


Fig. 11. Posterior 90% credible intervals for variety effects under hierarchical- t (□) and spatial mixture (■) formulations with $k = 6$ and $h = 100$

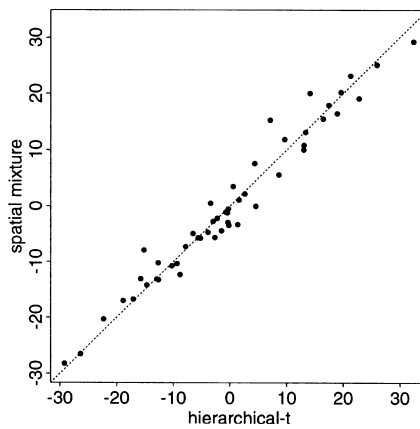


Fig. 12. Posterior means of variety effects under hierarchical- t and spatial mixture formulations with $k = 6$ and $h = 100$

processes are used, likelihood-based methods of inference for non-location parameters do not always perform well. This is particularly true when the inevitable misspecification of the ‘true’ model is exacerbated by a conflict between the likelihood and the objective of the statistical analysis. This was first demonstrated to me in the context of time series by my former doctoral supervisor, Jeff Harrison. There, likelihood, which can be shown to be inextricably linked with one-step-ahead forecasting, may conflict with the objective of producing good medium- and long-term forecasts. A remedy for this when making inferences about the time series parameters is to increase the weighting of the low frequency components of the data relative to the likelihood (see, for example, Haywood and Tunnicliffe Wilson (1997)). Although the objectives of a field trial are very different from time series forecasting, there may still be an argument for altering the relative weightings of the frequencies when making inferences about the non-location parameters, possibly by increasing the weighting of the higher frequencies as in generalized cross-validation (see Butler (1998)). It is also perhaps worth noting that one of the

authors has also expressed doubts about likelihood, or at least maximum likelihood estimation, in the context of image analysis (see Besag (1992)).

R. A. Bailey (*Queen Mary and Westfield College, London*)

I welcome this contribution to the problem of how best to design and analyse agricultural field experiments, not least because it gives me the opportunity to withdraw my unfounded criticism of Besag and Kempton (1986) in Bailey (1998).

The perennial problem is how to allow for spatial patterns of fertility. The allowance should be enough for spatial variation not to mask the treatment effects but not so much that estimators have large variances. Azaïs *et al.* (1998) advocate neighbour-balanced designs with appropriate randomization and frequentist spatial analysis.

The models proposed in Section 2 all seem reasonable, except that they deliberately exclude fixed block effects. In any large trial there are some management operations, such as harvesting, that cannot be accomplished by one person in one session. If there are large blocks in the trial then good management practice stops such operations at block boundaries, so that any difference between days or operators is absorbed into the fitted effects of the large blocks. I do not think it practical to omit such effects from the model.

I am glad that the experiments chosen to illustrate the analysis are real and messy, but I regret that they were so badly designed. The El Batán trial used complete blocks of 50 plots, even though 50 is evidently too large for the assumption of within-block homogeneity. Why not use blocks of size 10, using the dual of design T18 in Clatworthy (1973)? The different interpretations of the strawberry trial given by Mead (1988), section 10.0, and the present authors suggest an unacknowledged double blocking confounding one treatment contrast with half-rows (cf. Yates (1939), section 10). Neither system of blocks allows for the hedge. In the maize trial, blocks of consecutive eight plots in a boustrophedon lay-out which turns every 11 plots are absurd. Why not use a 16×12 lay-out with blocks of size 12? Again, the dual of a partially balanced design with two associate classes is suitable. The proponents of Bayesian methods do not make their case by comparing their methods with bad non-Bayesian methods.

The authors comment on when to promote Bayesian design. In my experience, practitioners are reluctant to implement either new designs or new methods of analysis. I commend the approach of Philippeau *et al.* (1996, 1997), who persuaded plant breeders to let them run three parallel trials one year, each with its own prescribed design and analysis, as a way of comparing (and introducing) complete design-and-analysis packages.

Richard L. Smith (*University of North Carolina, Chapel Hill, and Isaac Newton Institute, Cambridge*)

One of the questions raised by this fascinating paper is the appropriate form of spatial model for fertility effects. Past researchers (including Professor Besag) tended to favour stationary models but this paper makes a strong case for models which are intrinsically stationary, but not stationary. What kinds of diagnostics are available for determining the right spatial model?

One widely used tool for a spatial field Z is the variogram, defined as $2\gamma(h) = \text{var}\{Z(s+h) - Z(s)\}$ for vector locations s and h . This is defined for any intrinsically stationary process with finite second moments, and, for a stationary process $\gamma(h)$ tends to a constant as $|h| \rightarrow \infty$. The shape of this might, therefore, provide some clue to the right form of model.

For the North Carolina corn data, Figs 13(a)–13(f) show variogram plots for the fertility effect in three counties. Treatment effects were accounted for by simply subtracting the treatment means, and the interplot distances h were classified so that those within 45° of the north–south line were counted as north–south distances, the remainder as east–west distances. Figs 13(g)–13(l) show the same plots for three simulations of model (2.5)–(2.6).

Of the six real data variograms, Figs 13(b), 13(e) and 13(f) arguably show evidence of a continually increasing variogram, indicating non-stationarity. Figs 13(a) and 13(b) indicate a directionally dependent effect for county 1, but from Figs 13(c) and 13(d) there is no sign of any corresponding effect for county 4, which is very surprising in view of the results in the paper. Among the simulations, the visible evidence for non-stationarity is not at all clear, and for $\beta = 0.95$ in Figs 13(k) and 13(l) there is not much evidence for directional dependence either, despite the strong directionality in the model. Robust variograms (Cressie, 1993) were also calculated and produced very similar results.

If these results show anything, they probably support the use of a non-stationary model, but the

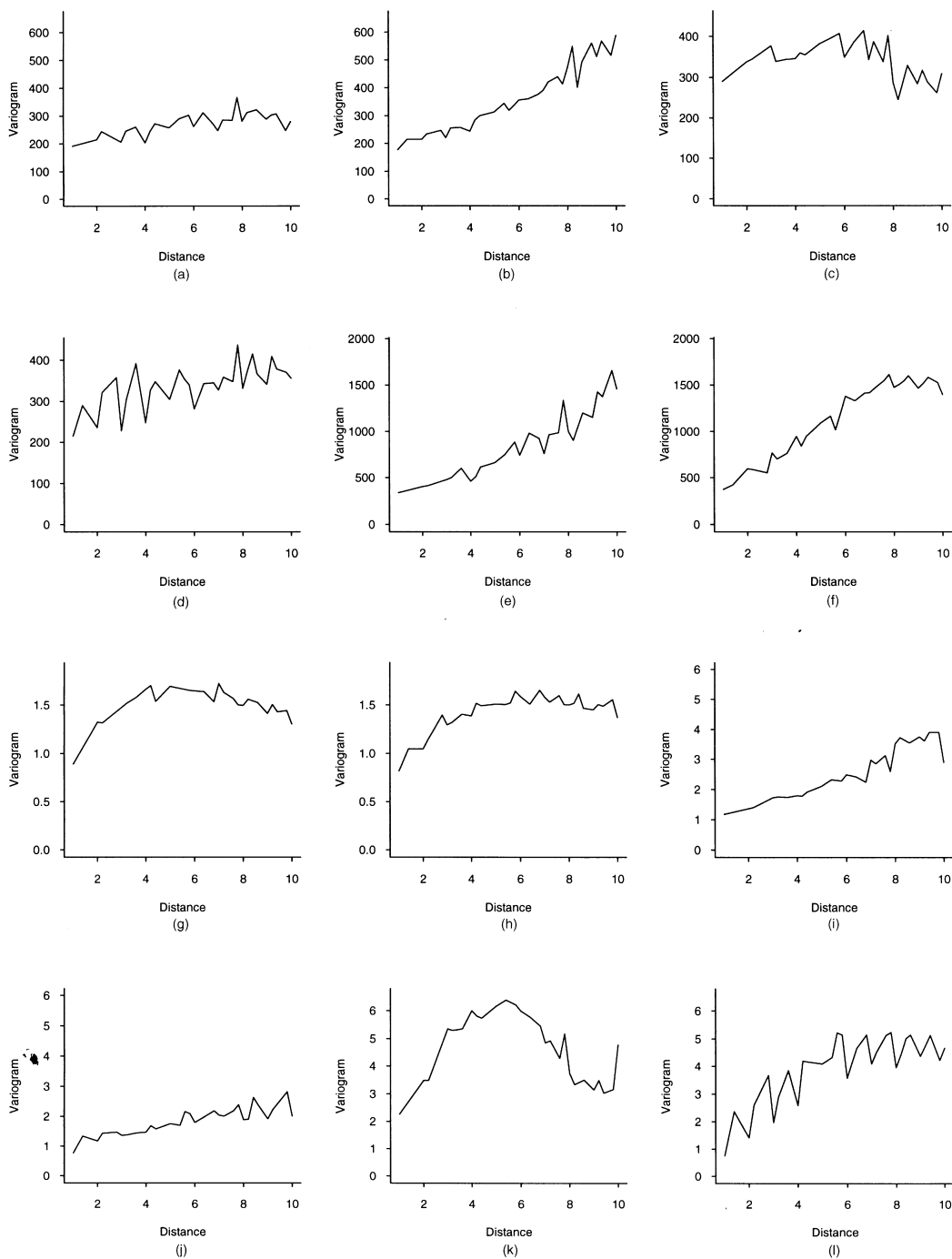


Fig. 13. Variogram plots for estimated fertilities: North Carolina data for (a) county 1, east-west, (b) county 1, north-south, (c) county 4, east-west, (d) county 4, north-south, (e) county 6, east-west and (f) county 6, north-south; simulations of model (2.5)–(2.6) (same lay-out as for the North Carolina data) for (g) $\beta = 0.5$, east-west, (h) $\beta = 0.5$, north-south, (i) $\beta = 0.75$, east-west, (j) $\beta = 0.75$, north-south, (k) $\beta = 0.95$, east-west and (l) $\beta = 0.95$, north-south; the unit of distance is the distance between plots in either direction

evidence is not strong. Are there better diagnostics than the variogram, and what would the authors recommend?

Bernard Silverman (*University of Bristol*)

Fertility is intrinsically an intermittent phenomenon. Therefore it may be helpful to model fertility in a way which allows for occasional large changes as well as smoother locally variability. Multiresolution ideas, e.g. using wavelets, may be useful here. Wavelet methods would allow the priors in Section 2.3 of the paper to be generalized, to allow for such intermittent effects; for some discussion of the range of possible behaviour of functions generated from priors over wavelet coefficients see, for example, Abramovich *et al.* (1998). One approach that suggests itself is to model the fertility down to very small levels and then to integrate over the plots; this may be particularly appropriate for some of the irregular plot patterns that come up in practice.

C. Jennison (*University of Bath*)

In the early 1980s, when there was a flurry of research activity in the spatial analysis of field trials, implementing fully two-dimensional methods and finding an acceptable treatment of outliers and discontinuities in fertility were major challenges. I would like to congratulate the authors on solving these problems elegantly and effectively. Their paper brings together the theoretical study of spatial models and hands-on data analysis in an exemplary manner.

The authors have steered clear of some potential difficulties by avoiding overelaborate models. We should all heed their advice to simplify where possible. In hierarchical modelling, it is tempting to side-step the problem of specifying prior parameter distributions by letting these depend on hyperparameters with dispersed distributions. Unfortunately, it is quite possible that the data will carry little information about such hyperparameters and the usual assumption that the prior input will be swamped by the data likelihood may be unjustified.

In some examples of field trials it is difficult to settle on a single definitive interpretation of the data; for example, a model with high variation in fertility, occasional outliers and a low residual variance may be just as plausible as an alternative model with lesser fertility effects, fewer outliers and a high residual variance. In the Bayesian approach one integrates over these possibilities, weighting each in proportion to its posterior probability: this removes the need to commit to a single interpretation of the data but it may leave an analysis open to doubt if key results are sensitive to *prior* weights and opinion on appropriate prior weights is divided. The prior distributions for the degrees of freedom of *t*-distributions for residual errors and fertility comprise one of the more 'elaborate' features of the authors' models. Supposing other aspects of the model to be correct, I wonder

- (a) how much information are the data likely to provide about these degrees of freedom and
- (b) how sensitive are estimated treatment effects to the prior distributions for these degrees of freedom?

The following contributions were received in writing after the meeting.

George Barnard (*Colchester*)

I regret being unable to be present at the discussion on this most interesting paper but would like to comment that Dr A. E. Jones, who was killed in a lift accident in 1948, studied fertility variations of rows of bean plants at Rothamsted. It was known that the bean yields were affected by worms in the soil whose breeding habits suggested that fertility along a row could reasonably be represented by a Wiener process with positive autocorrelation. On the basis of this assumption Jones could make a considerable reduction in the errors in contrasts. His work was published in *Biometrika*, but I am unable to check the literature to give a more precise reference.

I think that it is true that attention to short-range soil variation has grown considerably since field experimentation has been extended to regions such as Australia and much of North America, where the land involved has been under the plough for much shorter periods than the land at Rothamsted. One might expect that repeated ploughing over a very long period would reduce the local variations in fertility.

M. S. Bartlett (*Exmouth*)

I apologize to the authors for my non-attendance. I fully accept their reasons for not wishing to embark on any general critique of Bayesian *versus* 'frequentist' inference, but I cannot resist including with a few specific comments some more general observations (provoked perhaps by Julian Besag's kind

acknowledgement to me). Otherwise the formidable problems associated with any final resolution of this long-standing controversy might be overlooked.

- (a) Some of my own difficulties with the at times complicated analyses did not seem to agree with the authors' claim (final paragraph) that 'the results are easier to interpret . . . when communicating with non-statisticians'. It is a historical fact that the practical simplicity of Fisher's scheme of design and analysis of agricultural field experiments was quickly appreciated worldwide.

This simplicity was linked with Fisher's identification of the analysis with the design. Such an association seems less compelling for a Bayesian; and Bayesian *design* is not particularly considered by the authors.

- (b) In the paper the special spatial problems with agricultural field trials are stressed, but not much is said about Fisher's use of randomization, which enabled field trials to be classified with other important areas such as medical trials.

Randomization has some problems for frequentists, notably in small samples, but its role for a Bayesian is much less clear. The Fisherian emphasis on tests of significance, and the null hypothesis, also has its limitations. In particular, 'null' results in small samples sometimes misled an experimenter into believing that no real effects existed.

- (c) Any overall claims for Bayesian analysis are unfortunately mixed up with our views on what 'statistics' is all about.

I cannot agree with Stigler (1986) when he calls statistics the 'logic of measurement', a definition linked with his identification of statistics with statistical *inference*, and then the latter with *all* inductive inference! In my view statistics must include somewhere its association with populations or groups of individuals (see Bartlett (1971)). Bayesians who claim that their approach covers *all* inference ignore the rejection of their approach by many scientists and philosophers (see, for example, Mayo (1996)).

- (d) The subjective nature of the Bayesian system, especially the one based on *personal* probability assessments, need not bar its use in some disciplines; but any claim to objectivity has always to cope with the existence of prejudice in some personal assessments (not excluding those of some frequentists).

Nicola G. Best (*Imperial College of Science, Technology and Medicine, London*), **Katja Ickstadt** (*University of North Carolina, Chapel Hill*) and **Robert L. Wolpert** (*Duke University, Durham*)

Continuum models similar to those presented in this paper have recently been developed and presented in Wolpert and Ickstadt (1998) and Best *et al.* (1998), motivated by applications in biodiversity and epidemiology. The differences in application lead to differences in context and detail (our continuous latent two-dimensional random fields are used as mean functions for inhomogeneous point processes, rather than as means for inhomogeneous Gaussian processes, and they are constructed as convolutions of underlying gamma random fields, whereas the authors might prefer convolutions of underlying Gaussian random fields), but the same approach will lead to consistent continuum models for agricultural field experiments.

We agree that continuum models are aesthetically appealing but would argue that they enjoy some practical advantages as well. For example, as the authors observe, continuum models permit coherent inference simultaneously at all levels of spatial aggregation; in contrast, inference based on discrete models must be revised completely whenever a different scale of aggregation (e.g. plot size) is chosen. This has practical implications whenever the data are given continuously, since the use of discrete methods would require that they be aggregated to some arbitrary level, and whenever discrete data must be aggregated further (subplots to plots, plots to larger regions, etc.).

Even for the present agricultural example, in which the aggregation level is inherent in the experimental design, the neighbour-based Markov random field approach is less flexible than a distance-based continuum approach that recognizes explicitly that fertility may not be uniform, even within plots, and that dependence may not be mediated by neighbours, because spatial variation may arise at scales smaller than or larger than the plot size. Models based on an underlying continuum can reveal and exploit such dependence, whereas neighbour-based discrete models cannot.

In an increasingly important class of problems covariate data are available to help to guide inference. In agricultural field experiments we may have measurements of soil conditions, for example, or of aspect or sunlight, that may also affect growth rates, but these measurements may be taken or reported at levels of aggregation incommensurate with the experimental plots. Aggregating such data to the plot level is known to distort or mask true covariate-response relationships and may lead to *ecological*

Table 9. Lags 1 and 5 autocorrelations of the outputs of four different Markov chains with the same limiting distribution†

Sampler	Lag 1 autocorrelation					Lag 5 autocorrelation				
	τ_4	τ_{17}	τ_9	τ_{12}	τ_{20}	τ_4	τ_{17}	τ_9	τ_{12}	τ_{20}
1	0.49	0.45	0.46	0.49	0.66	0.24	0.25	0.31	0.33	0.29
2	0.44	0.40	0.39	0.46	0.63	0.21	0.23	0.27	0.31	0.26
3	0.48	0.43	0.42	0.44	0.63	0.12	0.14	0.13	0.15	0.20
4	0.28	0.22	0.18	0.24	0.50	0.03	0.07	0.06	0.07	0.14

†The autocorrelations from the top five performing treatments are shown.

bias, something widely recognized in the sociological and epidemiological literature (Robinson, 1950; Richardson, 1992). The process of data aggregation will also introduce approximation errors and uncertainty whenever (as usual) covariate data are available at disparate non-nested levels of aggregation. Continuum spatial regression methods such as those introduced in Ickstadt and Wolpert (1998) allow covariate data to be used at their natural levels of aggregation.

Simon Byers (*AT&T Research, Florham Park*)

I would like to express my appreciation to the authors for producing this exposition of a powerful and substantial application of Bayesian methods. I expect to see more difficult problems, such as categorical yields and more complex multiyear multisite trials, tackled soon. I have conducted some experiments with Gibbs samplers for the El Batán posterior distribution to examine the effects of blocking on mobility. Throughout I grouped the degrees of freedom with the auxiliary variables and looked at four different arrangements of the variety and fertility effects. The four blocking schemes were univariate variety and fertility updates, four-variate updates with each variety effect being grouped with its associated fertilities, blocking of all fertilities and then all varieties and finally blocking of all effects together. One advantage of heavier blocking that is not generally considered is its simplicity. An example of this is the last sampler above; it has only one move for the effects where others have more, and the full conditional for this step is simpler to form and calculate because there are more variables in the step, and hence less conditioned on. This must be balanced against the inversion work that is inherent in such large moves. A consideration of classic problematic cases for Gibbs samplers tells us that profound mobility gains can be made from higher levels of blocking. Table 9 shows the effects of increased blocking in terms of autocorrelations in this case and indicates that the heaviest blocking has a clear advantage at these lags. I also note some more subtle effects for the two intermediate samplers.

In this situation, because it is easily possible to implement the full blocking, I favour it because of the above considerations. Where full blocking cannot be implemented owing to the large decomposition a combination of samplers 2 and 3 seems interesting, moving all fertilities together and moving each variety effect with its fertilities. I have also experimented with these types of blocking with the models used for spatial epidemiology in Besag *et al.* (1991) with similar results.

A. C. Davison (*Swiss Federal Institute of Technology, Lausanne*)

This is an impressive paper. My query is not directed at the results, but at the simulation methods used to obtain them.

One obvious concern is whether or not the Markov chain has converged. Much attention has been devoted to diagnostics for convergence, while exact simulation methods offer hope of constructing a chain known to be in equilibrium. My question is about the use of output from such a chain.

It can be dangerous to use improper prior densities when Gibbs sampling. As pointed out by Hobert and Casella (1996), even very experienced statisticians sometimes fail to notice that when improper priors are used the posterior densities can also be improper. Although the simulations may seem perfectly reasonable, they are then actually realizations from a non-existent density. This can be avoided by using proper priors, but the presentation of Monte Carlo output often presupposes that quantities of interest have finite posterior means and variances; see for example Fig. 1 of the present paper. If the model is intractable, using for example *t*-errors with low degrees of freedom, how and when can we be sure of the existence of posterior means and variances? Should we take a pragmatic approach, e.g. estimating tail

indices before reporting posterior averages? Is there a general theory to tell us for what classes of models posterior moments exist? Absent such theory, can we trust averages from *any* Gibbs sampler?

R. N. Edmondson (*Horticulture Research International, Warwick*)

My experience is with small- to medium-sized agricultural and horticultural experiments and I restrict my comments to trials of this type.

I do not regard polynomial trend surfaces as implausible models for positional effects in spatial trials. Often, there will be systematic positional effects due to factors such as shading, temperature and drainage, and for these systematic effects polynomial models can be very plausible. Certainly, for orchard, glasshouse and growth room experiments polynomial trend effects are common and I do not see why they should not be equally common in field trials. I often use polynomial models when checking for interaction effects between designed treatment effects and positional effects.

The strawberry example is a very good example of a bad design. Clearly, as should have been foreseen, there was a trend away from the hedge and the blocks should have been chosen to accommodate this trend. A fairly natural design would have been a $(4 \times 4)/2$ Trojan square design (see Edmondson (1998)) which would have provided a double blocking system for controlling positional effects both parallel to the hedge and at right angles to the hedge.

One difficulty with Trojan square experiments is that treatment information is split between strata and there is a need to combine treatment information across strata by using estimated weights. Conventional combining weights are based on estimated stratum variances and for experiments with few error degrees of freedom the conventional combining weights can be unreliable. The authors mention the combination of treatment information across trials. Can the methods discussed in this paper be used to improve the reliability of combination of treatment information across strata in spatial experiments with few error degrees of freedom? Any method that would strengthen the analysis of this important class of designs would be very valuable.

Walter T. Federer (*Cornell University, Ithaca*)

A field design of an experiment consists of first selecting an experiment design plan (randomized complete-block design (RCBD), incomplete-block design, row-column, etc.) based on knowledge of the suspected experimental variation. The second step arises with the lay-out and conduct of the experiment. Most researchers and statisticians, including the authors, seem to consider only the selected plan and ignore the actual lay-out. The first two designs are row-column designs, not RCBDs. The 50-variety-3-replicate design is a 3-row $\times$ 50-column design. For any field design, there usually are several possible models to consider. Text-books give only one analysis for a particular experiment design plan. Some models to consider for the 3-row $\times$ 50-column design are proposed by Federer (1998), Federer and Wolfinger (1998) and Wolfinger *et al.* (1997). For example, one model would be differential linear, quadratic and perhaps cubic gradients within rows. Another would be common regressions over all rows. Another would involve row and column regressions and some of their interactions. Still another model would be rows, columns and varieties. After exploratory model investigation for an experiment, it is necessary to select a model for analysis. Some guidance in this is provided by Bozivich *et al.* (1956), Box and Cox (1964) and Federer (1994).

The strawberry example appears to be designed as a 4-row $\times$ 8-column design. A regression (trend) analysis for linear, quadratic and perhaps cubic regression should account for column variation and linear and perhaps quadratic regression should account for the row variation. Then, interactions of some of the row and column regressions should account for gradients that are oriented differently from the row-column lay-out.

In the 64-variety-3-replicate example, I do not understand how 24 8-plot incomplete blocks can fit into an 18×11 lattice arrangement. The actual design appears to be 18 rows $\times$ 11 columns for the 64 varieties. A regression, differential gradient, nearest neighbour, autoregressive average, smoothing, kriging or other spatial model could be employed to account for the variation in the experiment. I suspect that a different analysis would be needed for each site.

The parameters of the spatial analysis should be considered as random effects and intereffect information recovered. It appears that the varieties for these examples should be considered as fixed effects (see Federer (1998)).

I would have liked to see a discussion of the situation in which Bayesian and mixed models (restricted maximum likelihood, say) give the same results. Is not mixed model analysis a form of Bayesian analysis? A comparison of different models is not a comparison of different procedures such as

Bayesian, mixed model and Kalman filtering. A good discussion of the relationships between these three procedures is needed.

Arthur Gilmour (*New South Wales Agriculture, Orange*), **Brian Cullis and Alison Smith** (*New South Wales Agriculture, Wagga Wagga*) and **Arūnas Verbyla** (*Biometrics SA, Adelaide*)

Our main objective in Gilmour *et al.* (1997) was to shift the emphasis away from the merits of particular spatial models since no model is universally applicable. Our approach uses data-dependent diagnostics to identify the sources of spatial variation, both natural and extraneous. We then choose a model which reasonably matches the observed spatial variation, not a model which produces the smallest average pairwise variances for variety effects. In contrast Besag and Higdon propose a single spatial model for all situations.

To avoid overfitting, we insist on collaborative modelling and only include fixed effects if there is an identifiable cause. Fitting random row, column and spline effects by using restricted maximum likelihood (REML) protects against overfitting.

The collaboration of statisticians with plant breeders has led to significant improvements in field technique: boustrophedon harvesting is rarely used, multiplot seeders have been redesigned and plot trimming methods have been automated and improved. Statisticians are employed on industry funds to analyse, collate and report results from all major plant improvement programmes in Australia. They use our approach routinely to analyse over 5000 field trials annually under strict protocols.

We now re-analyse the six corn trials presented in Section 6. Our approach is described in Cullis *et al.*

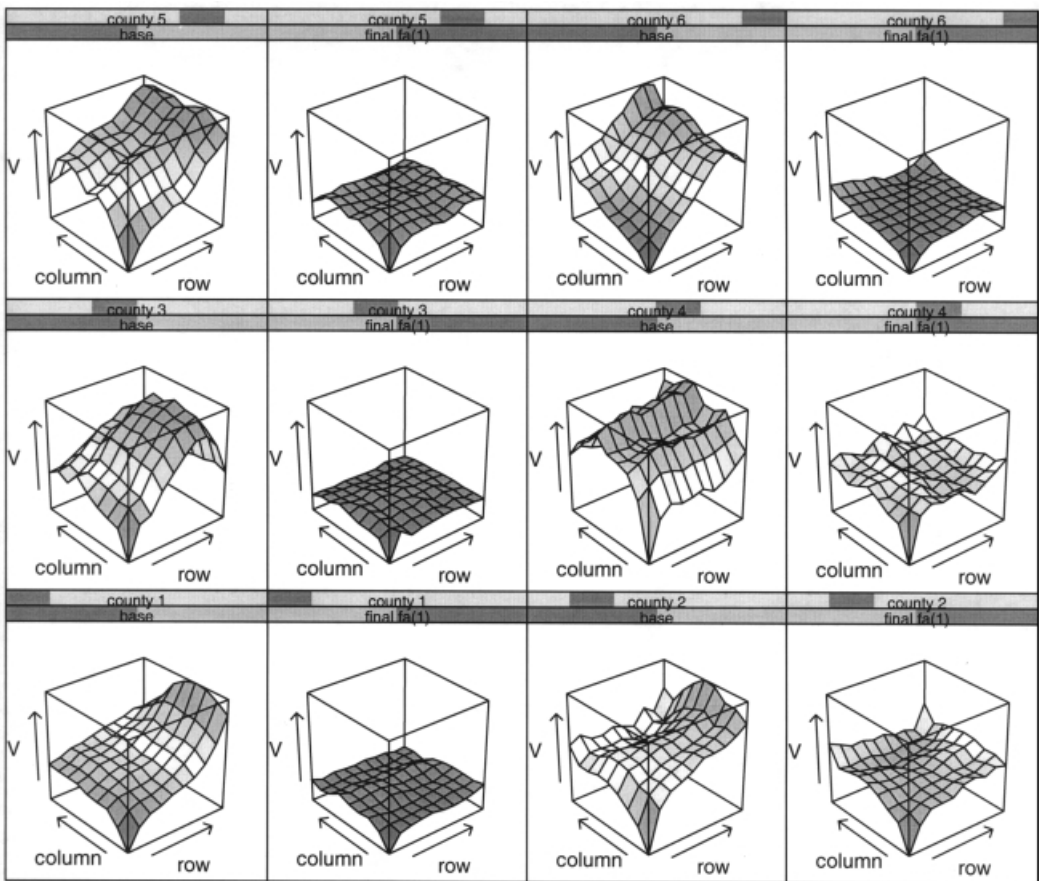


Fig. 14. Wireframe plots of sample variograms for the base and final models for each county

Table 10. Summary of variance models for variety effects and REML log-likelihoods for the corn data of Section 6

Model	Model description	Number of parameters	REML log-likelihood
1	Diagonal	6	−3630.99
2	Uniform	2	−3605.62
3	Common correlation	7	−3597.29
4	Factor analytic	12	−3591.74
5	Unstructured	21	−3586.37

Table 11. REML estimate of the variance matrix for the variety effects†

	County 1	County 2	County 3	County 4	County 5	County 6
County 1	90	0.73	0.42	0.65	0.90	0.44
County 2	0.73	22	0.60	0.62	0.67	0.68
County 3	0.56	0.48	102	0.72	0.63	0.18
County 4	0.70	0.59	0.46	43	0.64	0.44
County 5	0.87	0.74	0.57	0.71	97	0.25
County 6	0.36	0.30	0.23	0.29	0.36	73

†Variances from the FA(1) model are on the diagonal, correlations from the FA(1) model are below the diagonal and correlations from the unstructured model are above the diagonal.

(1998), Gilmour *et al.* (1998) and Smith *et al.* (1998). Fig. 14 presents the sample variograms of the residuals for base and final models for each county. Both models include a separable autoregressive process fitted to the residuals. Varieties were fitted as random effects independently for each county in the base model. The base variograms indicate substantial spatial heterogeneity. Without the opportunity for consultation, we added random effects for rows, columns and spline(rows) as necessary. We included an additional fixed term for county 2, in which the yields of row 1 were extraordinarily high relatively to the other rows. The variograms for the final model show that the spatial heterogeneity has been accommodated.

We then model the variety variance structure across counties as summarized in Table 10. Besag and Higdon effectively use a uniform (i.e. equal variance–equal covariance) model. However, the variety variances differ. The common correlation and factor analytic models with one factor, FA(1) (Smith *et al.*, 1998), are reasonable (Table 11).

The standardized county loadings from the FA(1) model were

$$(0.926, 0.788, 0.610, 0.753, 0.937, 0.385),$$

highlighting the lack of agreement of county 6 with the other five counties. Fig. 15 plots the best linear unbiased predictors of the overall variety effects from the uniform and FA(1) models, showing the degree of reranking.

Marcia Gumpertz (*North Carolina State University, Raleigh*)

Several ideas in this paper will be useful to practitioners fitting models to spatial data. As a fairly traditional data analyst I am particularly interested in understanding

- (a) how Besag and Higdon’s ideas can be applied to classical mixed models and
- (b) which features of their analysis are specific to the Bayesian approach.

Besag and Higdon’s Bayesian model (2.1) is similar to the random-effects model $y = T\tau + F\psi + \epsilon$, where τ is a vector of random variety effects, ψ is a vector of random fertility effects that are spatially correlated, ϵ is a vector of random plot effects, T is a design matrix for varieties and F is a design matrix of factors affecting fertility. In the random-effects model, best linear unbiased prediction provides estimates of individual variety effects.

In addition to the Bayesian framework, two features of Besag and Higdon’s analysis are unusual:

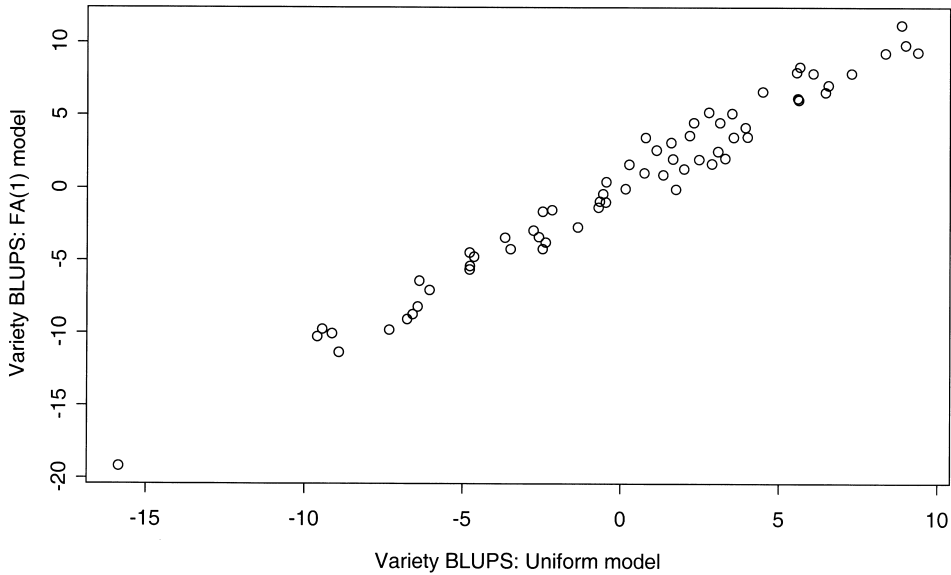


Fig. 15. Plot of the best linear unbiased predictors for the combined variety effects for the FA(1) and uniform models

- (a) the use of the Gaussian intrinsic autoregression model for fertility effects and
- (b) the use of Student's t -distribution for the fertility effects.

The flexibility and non-stationarity of the Gaussian intrinsic autoregression model for fertility effects are attractive. The intrinsic autoregression model also offers a big computational advantage for large spatial data sets, which are often so large that the full $n \times n$ covariance matrix cannot be inverted. I also like the use of Student's t -distribution to account for jumps in fertility, especially with the estimation of the degrees of freedom. An examination of the estimates of the degrees of freedom seems a good way to determine whether the normality assumption is reasonable for the fertility effects. I am not sure whether this type of analysis would be feasible for the frequentist random-effects model. More research in the area of mixed models with non-Gaussian random effects would be welcome.

As the authors mention, Bayesian analysis offers a clear advantage over frequentist analysis in incorporating what is known from previous studies into data analysis. In my view, the barriers to the use of Bayesian models in agricultural research result more from a lack of familiarity than from the philosophy. If practitioners understood the methods and had appropriate software, they would welcome Bayesian methods.

David A. Harville (*IBM T. J. Watson Research Center, Yorktown Heights*) and **Dale L. Zimmerman** (*University of Iowa, Iowa City*)

Bayesian versus frequentist approach

In analysing field plot data, a sensible 'frequentist' and sensible 'Bayesian' are likely to reach very similar conclusions. Both need to account for any substantial variation in fertility. Moreover, the assumptions about the plot effects could just as well be included in the model as in the prior.

It may be sensible (say for publication) for a Bayesian to adopt a 'uniform' prior for τ even in the presence of prior information. However, it may sometimes (e.g. in analysing a screening experiment) be sensible for a frequentist to regard τ as comprising random effects.

Contrary to the authors' assertions, there are frequentist methods for accounting for the uncertainty introduced by the estimation of variances and covariances — refer, for instance, to Harville and Jeske (1992) and Zimmerman and Cressie (1992).

The authors' choice of examples

Although interesting, the authors' examples are not very informative about the relative merits of the

various approaches. We would like to have seen them use data from uniformity trials to subject their approach to the kind of comparative study carried out by Zimmerman and Harville (1991).

The modelling of fertility

The authors' assumptions about the first and second moments of the plot (fertility) effects seem unrealistic. Their assumption of a common mean in combination with their assumption that $W_{ij} = 0$ for $i \neq j$ leaves their approach ill equipped to handle a situation where a number of plots are affected by a common influence (as in the strawberry example where several plots were shaded). We feel that any 'beneficial' effect from taking the distribution of the plot effects to be a t -distribution is more a reflection of the inappropriateness of their assumptions about the first two moments than of any non-normality.

We believe that the so-called geostatistical approach to the modelling of the plot effects (e.g. Zimmerman and Harville (1991)) has many advantages over the authors' approach. Most importantly, it leads to a more sensible (Bayesian or frequentist) analysis. In the geostatistical approach, the spatial dependence is modelled on a continuous scale and in terms of covariances. Consequently, its appropriateness does not depend on the plot size and shape; nor does it require any device for 'edge correction'. Any computational disadvantage of the geostatistical approach can often be overcome by taking advantage of 'separability' or other exploitable structure. We believe that the geostatistical approach is computationally feasible for all except the very largest field plot experiments.

Steven N. MacEachern (*Ohio State University, Columbus*)

It has been a pleasure to read this paper where the case for a Bayesian analysis of field trials has been so strongly presented. The main point of the paper is that the data from such an experiment can be modelled with distributions which allow us to make an inference about the treatment effects (the main quantity of interest) while adjusting for the main sources of heterogeneity among responses — fertility and outliers. The authors implement these strategies, successfully illustrating their use in several examples.

A question which always arises in modelling data is whether the model is adequate, or whether there is a simple extension of the model that produces a better fit to the data. A technique that I have found to be effective is to expand a model by replacing a parametric component with a nonparametric component. As a proponent of nonparametric Bayesian techniques, I see two main uses for these models. The first is to use a nonparametric component where the traditional concept of the random effect comes up, as recommended in Bush and MacEachern (1996). Here, random effects appear when information is combined from a number of trials. The ξ_k^i may be viewed as random effects and the presumed Gaussian distribution replaced by an essentially arbitrary distribution. The simplest model would take the ξ_k^i to be independent and identically distributed draws from some F on which a nonparametric prior distribution is placed. A parsimonious extension of this model would take ξ_k^i from F_{τ_k} , with the F_{τ_k} members of a scale family, defined by $F_{\tau_k} = \lambda_{\tau_k}^{-1/2} F$. Relaxing the Gaussianity assumption allows the data to determine the form of F (or of F_{τ_k}). With the 384 ξ_k^i in Section 6, there should be plenty of information to estimate the form of this distribution.

The second use of nonparametric Bayesian methods appears when the morning-glory data are analysed. A logistic link between effects and p is posited. A more general form of link can be examined with the link taken to follow a nonparametric prior distribution.

Finally, the care given to computational issues, though a secondary point, is a trademark of the authors' work and shines through the paper. A particularly interesting point is the deliberate use of non-identifiable models, even those with improper posteriors! This technique, which the authors have used in previous work, is only now beginning to be appreciated by the general statistics community.

R. Mead (*University of Reading*)

It is good to have a presentation of a Bayesian methodology for the analysis of field experiments, but worrying that all four examples may be considered 'badly designed experiments'. As the authors comment for the El Batán data, it is clear that the standard analysis can be improved on. It would be interesting to compare the Bayesian analyses with sensible traditional statistical analyses for these examples or with standard analyses for less eccentric examples. And a sensible or standard analysis does not, as the authors incorrectly imply, involve the use of multiple comparison.

The El Batán data can be analysed by superimposing incomplete blocks either within the actual blocks or across blocks and this gives an ordering of varieties similar to that in Table 3. I believe that an experimenter would make sensible decisions about which varieties to examine further from a list of

adjusted means (with several alternative analyses leading to similar lists), and that although the information in Table 3 might be diverting it would not help in the making of these decisions.

The strawberry data do indeed appear in Mead (1988) but with the blocks defined (as given to me by the experimenter) as entire horizontal rows in Table 4 and with the superior covariance adjustment using as a single covariate the reciprocal of distance from the hedge. Like the Bayesian analysis, that covariance analysis adjusts P to be higher yielding than E (by about one standard error of difference between adjusted means) and confirms that only variety R1 is markedly different. Note that the design of the trial is actually (and unrecognized by me also) a checker-board of eight half-blocks with two groups of split-plot varieties—hence a sufficiently bad design to justify a Bayesian analysis!

The objectives of a multisite trial such as that for the corn trials must be to assess whether there is a single variety recommendation that can be made for all sites or whether some more specific recommendation is appropriate. In common with various previous methods of analysis, the Bayesian analysis fails to provide an answer and I wonder whether the authors can extend their analysis to answer the real question instead of, like others previously, drawing general conclusions about the relative importance of individual trial results and overall main effects of varieties. I note that a method has been developed to answer the question, in a traditional approach, in a doctoral thesis of Mr Ohemeng-Dapaah.

Jesper Møller (*University of Aalborg*) and **Rasmus Waagepetersen** (*Danish Institute of Agricultural Sciences, Tjele*)

We enjoyed this stimulating and elegant paper. Our comments pertain to the choice of the prior for fertility effects and to some computational aspects.

The choice of the Gaussian pairwise difference prior (2.2) is advantageous mainly computationally and we agree that a more appealing approach is to use a Gaussian process with a continuum index set. It is mentioned in Section 2.3.3 that for moderately large data sets, such as the morning-glory data, the computational task of inverting large non-sparse covariance matrices becomes very demanding routinely, but this is not the experience in Møller *et al.* (1997, 1998) and Brix and Møller (1998), where the fast Fourier transform is used to diagonalize circulant embeddings of covariance matrices (see, for example, Wood and Chan (1994)).

In Møller *et al.* (1997, 1998) and Brix and Møller (1998) a continuum Gaussian field is used to model the log-intensity surface of spatial and spatiotemporal Cox point processes whereby so-called log-Gaussian Cox processes (LGCPs) are obtained. LGCPs satisfy many desirable theoretical properties and the models were applied to forest data and the development of weeds. The techniques developed can easily be adapted to the analysis of spatial count data sets (this is currently being investigated for counts of weed plants) and for spatial continuous data sets. It is, for example, possible to develop diagnostics based on certain theoretical relationships to obtain an idea about an appropriate parametric class of covariance functions (rather than modelling the precision matrix as in Section 2).

The empirical Bayesian prediction procedure in Møller *et al.* (1997, 1998) and Brix and Møller (1998) is based on conditional simulation of the intensity surface by using a so-called Metropolis-adjusted Langevin algorithm (MALA) (Besag, 1994; Roberts and Tweedie, 1997). For the applications we and some of our students (Segel *et al.*, 1998) have considered, the MALA performs much better than usual Metropolis–Hastings algorithms with single-variable updating schemes, and a slightly modified version has the nice theoretical property of geometrical ergodicity. Using a fully Bayesian approach where a hyperprior distribution is introduced for the mean and covariance parameters would only increase the computational burden by a factor of 3.

When simulating from the posterior in this paper it would be interesting to see whether there is any point in replacing the updating steps of τ and ψ by some MALA steps; particularly, for the morning-glory data with the non-Gaussian likelihood we presume that it is advantageous to use the MALA. Incidentally, we believe that an MALA may also be appropriate for certain steps in the Markov chain Monte Carlo calculations in Diggle *et al.* (1998) and for many other applications.

H. D. Patterson (*University of Edinburgh*)

Professor Besag and David Higdon are to be congratulated on their well-written and useful paper. For me, the attractive feature of their method is not that it is Bayesian in origin but that it automatically compensates for some of the selection bias that is inherent in variety testing. Varieties may be successful through a mixture of real achievement and luck in the choice of trial sites, seasons and randomization. Shrinkage of variety means goes a long way towards removing the element of lottery.

Possibly for historical reasons the authors' main concern appears to have been with fertility patterns

in individual trials but I am pleased to see that they also consider combining results across trials (Section 6.2). In general, variety responses vary far more between years and trial locations than between replicate plots. The ‘heavy shrinkage’ they note in their corn varieties example is to my mind completely justified.

The authors used residual maximum likelihood (REML) for the analysis of individual corn trials. REML can also be used for over-trials analysis with variety effects specified as random as well as variety $\times$ years, variety $\times$ locations and the usual error components. Shrinkage, even heavy shrinkage, results when the variety component of variance turns out to be small. It might be interesting to compare the two methods—Bayesian and REML—numerically and algebraically. I should add that I have found it difficult to persuade trials authorities to use REML or indeed to introduce any new statistical adjustment. A breeder whose favourite variety is about to be downgraded is, of course, an even more difficult proposition. I hope that the authors will have better luck with their Bayesian method.

M. Talbot (*Biomathematics and Statistics Scotland, Edinburgh*)

I come to this discussion with a mixture of respect for the authors’ contributions in this area over a long period and some worries about aspects of this paper.

It is surprising that the most appropriate frequentist analysis was not employed when drawing comparisons with the Bayesian methods. For example, in the corn variety trial data in Section 6, an obvious model to recommend is one with a row-plus-column block structure. The residual mean square from such an analysis is 154, when averaged over the six trials. This compares with 351 from an incomplete blocks within replicates analysis and with 371 from a randomized complete-block analysis. Clearly the block structure, and the boustrophedon lay-out, that was chosen for this trial is far from ideal.

The authors touch briefly on the subject of interference between varieties in neighbouring plots. In our experience (Talbot *et al.*, 1995) this is an important practical problem in the analysis of variety trials data, since the potential for bias is more critical than a lack of precision. A partial solution adopted in UK variety trials has been, through design, to guard against interference by means of restricted randomization and, subsequently through analysis, to estimate for any residual effects. An integrated Bayesian design and analysis approach may have advantages, though we have no strong evidence to offer at this stage.

Although it may be natural in a Bayesian setting to produce combined analyses over trials straight from the plot data, I hope that the authors would not recommend this to their students as a routine data analysis procedure. A sequential data summary approach from plot data to trial means and then to over-trials estimates ensures that all relevant information is carried forward. For example, it may not be clear to readers that the yields from county 1 are approximately two and a half times those from county 2. Any variety estimates with such a wide range of yields must be treated with caution.

Bayesian methods certainly have a role to play in the analysis of field experiments but perhaps our efforts as statisticians are best applied at the level of combining data from a series of trials where a decision maker’s priors matter most. It is this feeling which has partly motivated a project that we are currently undertaking. The aim is to help farmers to identify varieties which may be adapted to their specific conditions, using data from trials along with information about the individual farmer’s circumstances. A Bayesian solution seems the most fruitful way to combine the data and the relevant judgments to help in making choices.

Jeremy Tantrum (*University of Washington, Seattle*)

In Section 5 of the paper, there is an analysis of binary data from an agricultural experiment. Such data also arise in studying the spread of a disease in time and space. Thus, one of the examples in Besag (1978) concerns a 179×14 almost square array of endive plants, whose status, healthy (0) or diseased (1), had been recorded on four occasions, $t = 1, 2, 3, 4$, at 4-week intervals. Apart from the fact that diseased plants did not recover, little was known about the disease. As a simple approximation, it was proposed that, conditionally on the states of all other plants at or before time t , any previously healthy plant became diseased at t with probability that depended only on the states of its immediate neighbours at $t - 1$. This assumption was motivated by the apparently slow spread of the disease; see also Bartlett (1975), p. 78. A corresponding two-parameter model led to a straightforward likelihood function and statistical analysis. The simple model seemed to fit the data surprisingly well.

More realistically, if we allow the above conditional probability to depend on the states of immediate neighbours at time t , as well as at $t - 1$, then a more complicated, Markov random field, formulation is required. The simplest specification is a three-parameter autologistic model (Besag, 1974), with parameters

estimated by maximizing the pseudolikelihood (Besag, 1975), which is straightforward, or the likelihood, which requires Markov chain Monte Carlo sampling (Geyer and Thompson, 1992). In this particular case, the respective estimates agree very closely and have the following implication. Consider a plant that is healthy at time $t - 1$ and let m and n denote the number of diseased neighbours at $t - 1$ and t . Then, for fixed n , the plant is markedly *less* likely to become diseased at t as m ($\leq n$) increases. There are at least two plausible explanations. The first is that diseased plants are infective but only briefly; the second that some external agent, such as excess water, causes the disease and is present only at particular times. In the latter case, a reformulation might be preferable, more in keeping with Section 5 of the present paper. In either case, the model in Besag (1978) would seem too simple, despite the apparent slow spread of the disease. Of course, it would be easy to adapt this discussion to the Bayesian paradigm.

The **authors** replied later, in writing, as follows.

We are extremely grateful to all the discussants for their comments and criticisms. We cannot hope to respond to all the interesting points that have been raised but, in the light of the discussion, we begin by making some general remarks and by briefly presenting the analysis of a well-designed experiment. We then address some individual topics in more detail.

We attempted in two previous papers (Besag and Higdon (1993) and Besag *et al.* (1995), section 4) to stimulate interest in the Bayesian analysis of field experiments. Undeterred by our lack of success, we decided to try again. However, it was necessary to restrict the amount of duplication and this may have contributed to some misunderstandings and to a lack of balance (*sic*), in that all three of the yield experiments in the paper were badly designed (not by us!). For comparable Bayesian analyses of well-designed trials, we refer to the factorial experiment and the balanced lattice square in Besag and Higdon (1993) and to the α -designs in Besag *et al.* (1995) and in the example below. We preface the new example by some points that we hope will clarify our views.

General comments

- (a) Our paper suggests that the Bayesian paradigm has an advantage in the general area of ranking and selection, because relevant questions can be posed that have no direct frequentist counterpart. In this sense, we claim that interpretation is simplified. Admittedly, this argument is superficial and certainly, as Bartlett observes, our paper does not attempt to 'embark on a general critique of Bayesian *versus* "frequentist" inference'.
- (b) Many past studies have demonstrated that frequentist methods of spatial analysis are remarkably effective in resurrecting information from badly designed experiments. The argument for their general use is less strong, despite the Australian success described by Cullis, Gilmour, Verbyla and Smith. In a Bayesian framework, there seems no obvious alternative to a spatial formulation. We acknowledge that there are similarities between Bayesian formulations and classical mixed models, as noted by Diggle, Federer and Gumpertz, but of course the interpretations remain distinct. We hope that, in future work, Tantrum will pursue the comparisons between restricted maximum likelihood (REML) and Bayesian methods, suggested by Patterson.
- (c) Despite the examples in the paper, it was never our intention to promote poor design! Nevertheless, badly designed experiments are common in practice and statisticians need to cope with them in the best possible way, not merely to encourage better designs in the future. The comments on the implementation of new methods of design and analysis by Bailey and Patterson are rather depressing.
- (d) There is little experience in representing prior beliefs about treatment and fertility effects. Bayesian analysis of a sequence of experiments seems attractive, at least in principle, because it provides a natural method of updating one's information about treatments from trial to trial. Priors for fertility are likely to reflect previous experience in frequentist modelling. Sensitivity analysis is crucial but was omitted from the present paper. We refer to Besag *et al.* (1995), section 4, for an example in which inference is quite robust to changes in the hyperparameters. Of course, as the prior becomes more intricate, so the sensitivity typically increases.
- (e) Frequentist analyses are often more subjective than is admitted, as can be seen in some parts of the discussion. Even in well-designed experiments, there are often complications and the way in which these are accommodated or indeed ignored becomes partly a matter of individual taste. This also relates to Bartlett's point (d) and to Preece's acknowledgement that decisions are usually made according to several different, possibly conflicting, criteria.
- (f) Bayesian methods in field experiments are in their infancy. Some remaining issues, apart from

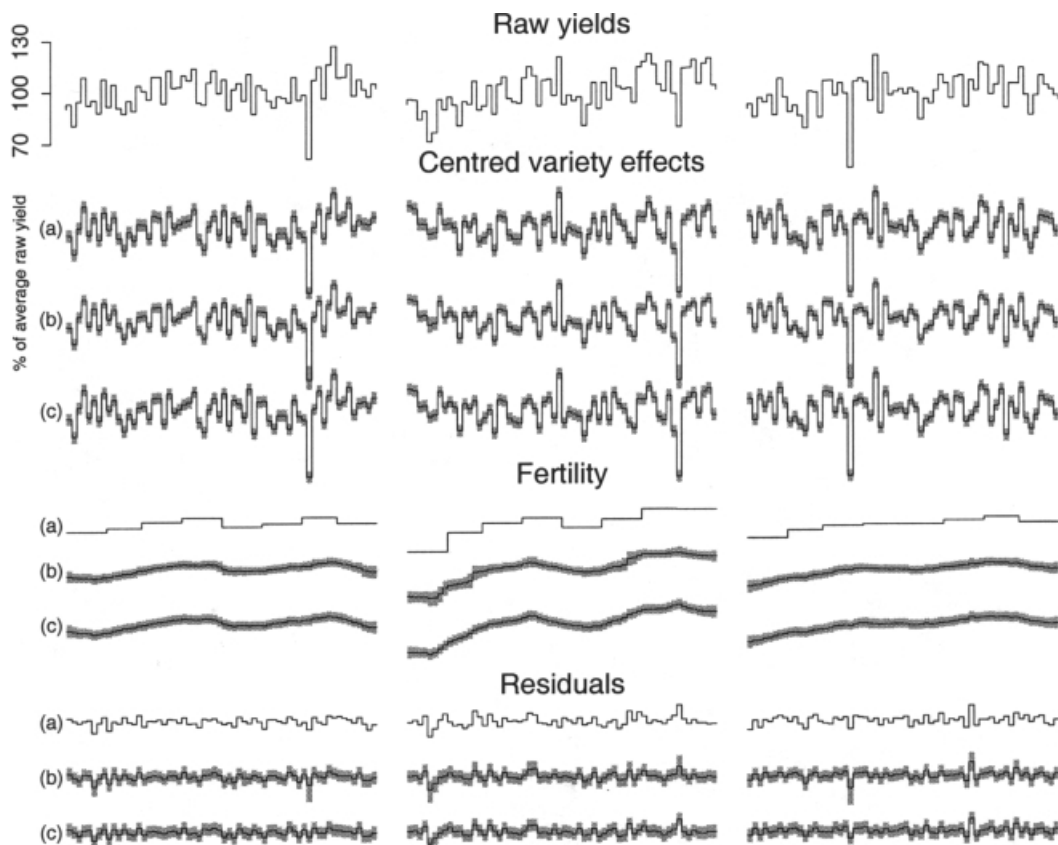


Fig. 16. Yields and additive decompositions under three different analyses (the shaded regions denote pointwise 90% confidence or credible intervals): (a) incomplete block; (b) hierarchical- t with Gaussian variety prior; (c) basic Gaussian with uniform variety prior

those raised in the discussion, are summarized in Section 7 of the paper. We feel that a competition between Bayesian and frequentist analysis is neither warranted nor likely to be productive at this stage. Our paper stops short at a purely numerical comparison, mainly showing similarities under certain assumptions. We continue on this theme after the example.

Analysis of a well-designed experiment

The top part of Fig. 16 displays standardized yields in a trial on 62 varieties of winter wheat, carried out by the Scottish Agricultural College. The data are taken from the same catalogue of experiments, kindly compiled by Tony Hunter at the Agricultural Research Council Unit of Statistics (now Biomathematics and Statistics Scotland), as the example in Besag *et al.* (1995), section 4. The experiment had three replicates in an α -design (Patterson and Williams, 1976). The remaining parts of Fig. 16 display additive decompositions of the data according to three different analyses:

- (a) incomplete block;
- (b) Bayesian hierarchical- t , with a Gaussian prior for variety effects;
- (c) basic Gaussian, with a uniform prior for variety effects.

Also shown are 90% confidence or credible intervals, according to the type of analysis. The similarities between the decompositions are more striking than the differences. Note that the shrinkage of the variety effects in (b) is negligible because there is no support for it in the data. However, the widths of the corresponding credible intervals differ from one another by a factor of 2, whereas those in (c) are

nearly constant. Not surprisingly, the frequentist spatial analysis in Besag and Kempton (1986) produces a decomposition close to (c). We return briefly to the example later in our reply.

Comparisons

Some discussants suggest that the comparisons we make between Bayesian and frequentist analysis are unfair, in optimizing the one but not the other. Several complain that there is no attempt at assessment, which is correct and partly reflects the difficulty in making objective comparisons when the truth is unknown. Thus, in Section 3, for example, we prefer to point out the similarities between numerical results from corresponding spatial analyses, provided that one adopts a uniform prior for the variety effects in the Bayesian procedure, which we would not do in practice. Such agreement, as mentioned above, is to be expected and is also commented on by Harville and Zimmerman, for example. Corresponding remarks apply in Sections 4 and 5, where again we choose the unappealing uniform prior in presenting Bayesian results. Also, although we think it is of interest in Section 4 that our basic spatial formulation produces a plausible analysis without detailed modelling of the effect of the hedge, the point is not exclusively Bayesian. We are disappointed in Pearce's remark that he obtains standard errors of treatment differences that 'lie in the range (0.53, 0.59), which is marginally better than with the Bayesian analysis presented'. In fact, with the preferred Gaussian prior, there is substantial shrinkage, as mentioned in the paper, and the posterior standard deviations for variety differences lie in the range (0.35, 0.51), apart from those involving RL, which are all larger, as seems logical from the lay-out. However, we cannot say which estimates are 'better', because the two methods address different paradigms and, even if they did not, we could not necessarily say which is more correct.

Harville and Zimmerman believe that we should have used uniformity data to make comparisons between analyses, as they do in Zimmerman and Harville (1991), copying the procedure in Besag (1983), which is also used by Besag and Kempton (1984, 1986) and Baird and Mead (1991), for example. This involves the repeated analysis of a single uniformity trial, with dummy treatment lay-outs drawn at random from a particular design, and assumes treatment additivity. It permits fair comparisons, in a randomization framework, between different methods, provided that these also act additively. However, Bayesian analysis, with a Gaussian prior for variety effects, is non-additive and shrinkage will produce results that are close to perfect but that do not generalize. Under a uniform prior, the numerical results should agree closely with the corresponding frequentist spatial formulation, as mentioned above, but it seems unfair to remove the shrinkage. Thus, there seems no alternative to an enormous simulation study, involving the addition of a complete range of treatment effects to the original data. Furthermore, a comparison under the randomization framework is of very limited interest to Bayesians. Of course, we agree that assessment of Bayesian methods is vital but it is not as straightforward as some would have us believe.

Block effects

Several discussants complain about the omission of fixed block effects from our formulation. What we state is that 'any blocking structure that does not have a physical presence is omitted from the model and from the subsequent analysis'. What we mean is that we ignore blocks if their sole purpose is conceptual in defining the lay-out. We support the inclusion of block effects if, for example, they play an operational role in the management of a trial. We would hope that such information is recorded; otherwise we agree that it may be better to adopt a defensive approach and to include them in any case. We definitely do not 'deliberately exclude fixed block effects', as claimed by Bailey, but it would usually be counter-productive in using spatial analysis to include a fixed effect for each of the large number of incomplete blocks that occur in sophisticated designs. Indeed, even the inclusion of complete-block effects, if unwarranted, may break the flow of useful information between adjacent plots, as seems to occur in the balanced lattice square example of Besag and Higdon (1993), section 4.

Plot-based priors for fertility effects

We turn now to 'fertility', which, as we should have made clear, includes, not only the heterogeneity of the soil, but also other sources of spatial variation that are difficult to accommodate in conventional fixed effects. For instance, the influence on yield of some management practices and of topography may be spatial. As a minimum, we intend that the prior distribution for fertility effects has sufficient flexibility to absorb the diverse spatial patterns that are evident in the yields of uniformity trials: see, for example, the Wiebe (1935) data, used later in our reply. In addition, we would like to contain, if not fully to cater for, the variation in moisture content in Preece's example or the effect of the hedge in the strawberry trial or the aberrant row in the North Carolina data, though we agree that in all of these there is a strong case for adding fixed effects. Note the implicit assumption here that treatments are spread out

sufficiently by the design to minimize confounding with spatial effects, and, correspondingly, that the spatial formulation is not so elaborate as to absorb genuine treatment effects. It is such considerations that have led us, rightly or wrongly, to adopt intrinsic autoregressions. These are not only the limiting forms of corresponding stationary models but also have an interpretation as local smoothers, as noted by Fernández and Green. For the precise model, we have favoured a cautious approach in opting for equations (2.5) and (2.6), rather than some version of the more flexible locally quadratic prior in expression (2.7), though extended from eight to 12 neighbours. We may have been too conservative in our choice.

We thank Gilmour, Cullis, Verbyla and Smith for their comments on modelling and their reanalysis of the North Carolina data. Besides frequentist *versus* Bayesian, our respective approaches towards dealing with ‘fertility’ are rather different. Their methods have evolved from Gleeson and Cullis (1987), which addresses one-dimensional adjustment via REML estimation for a small subclass of autoregressive integrated moving average ARIMA($p, d, 0$) models that include those of Besag and Kempton (1984, 1986) and Green *et al.* (1985) as special cases. Subsequently, Cullis, Gleeson and their co-workers developed methods for model selection within the ARIMA class and used the notion of separability (Martin, 1979) to obtain two-dimensional generalizations. The most recent addition to their armoury (e.g. Gilmour *et al.* (1997)) is the introduction of splines and it is at this stage, if not previously, that we wonder whether model selection from very limited data, masked by treatment effects, has got out of control. It is not the splines themselves to which we object but the fact that nothing else is jettisoned along the way. We find it difficult to believe that, among all the terms potentially in the model, there are not some that will tend to detect spatial structure where none exists and hence absorb genuine variety effects. We acknowledge that Gilmour, Cullis, Verbyla and Smith ‘insist on collaborative modelling and only include fixed effects if there is an identifiable cause’ but we question the extent to which REML or any other method can protect against overfitting in such an elaborate formulation. We provide a putative example in Section 7 of the paper, though, as we say there, our results may be fortuitous.

Harville and Zimmerman suggest that it is already overelaborate to drop the Gaussian assumption in one-dimensional fertility adjustment and that this really indicates a flaw in the basic random walk representation. In fact, the switch to t increments retains a useful Gaussian interpretation for any given ν_ψ , as noted in Section 2.5. Harville and Zimmerman also claim incorrectly that intrinsic models imply a common mean for fertility effects but we are grateful to them for pointing out our oversight in the frequentist assessment of prediction errors. As regards Smith’s remark, JB has avoided stationary models for the past 15 years, influenced by empirical findings for uniformity trials and also by the arguments in Wilkinson *et al.* (1983). Thus, random walks for one-dimensional fertility adjustment appear in a frequentist setting in Besag and Kempton (1984, 1986), though they are implicit in the original formulations of J. S. Papadakis in the 1920s to which he eventually returned. Note that, in the traditional analysis of covariance, the regression coefficient must be less than 1 and the corresponding adjustment is not strictly local as claimed by Pearce (1998) and here in the discussion but is attenuated and more akin to that obtained under stationarity. We comment in Section 2.3.2 on the defects of the eight-neighbour locally quadratic model in a Bayesian setting. We do not understand why Pearce regards his particular choice of covariance adjustment as more objective than that of a corresponding prior distribution. Incidentally, Papadakis, in a personal communication in 1984, viewed a regression coefficient that differed much from 1 as evidence of competition and uselessness of the experiment!

We are grateful to Künsch for characterizing intrinsic autoregressions, equations (2.5) and (2.6), on the infinite lattice in terms of constrained increments. This complements the result previously obtained by JB for the corresponding process (2.2) on the vertices of an arbitrary connected finite graph, with $\omega_{ij} = 1$ if (i, j) is an edge of the graph and $\omega_{ij} = 0$ otherwise. Then expression (2.2) is equivalent to a process whose increments along the edges are independent and identically distributed, conditional on their sums being 0 over all circuits of the graph. The proof relies on simple matrix algebra and lemma 3.1 in Besag and Kooperberg (1995). An obvious extension permits the non-zero ω_{ij} s to be unequal. Note that a random walk corresponds to a linear graph and the absence of any constraints.

Continuum priors for fertility effects

As we stated in the paper, we agree that there are advantages in starting from a continuum random field as the fertility prior. However, we are less persuaded that such models are more fundamental to data analysis. Ultimately, all models are wrong and what is important here is to produce a representation that captures the plot-to-plot variability: smaller scales are not relevant! However, as mentioned in the paper, we did experiment with plot splitting and with Brownian motion in one dimension and found the

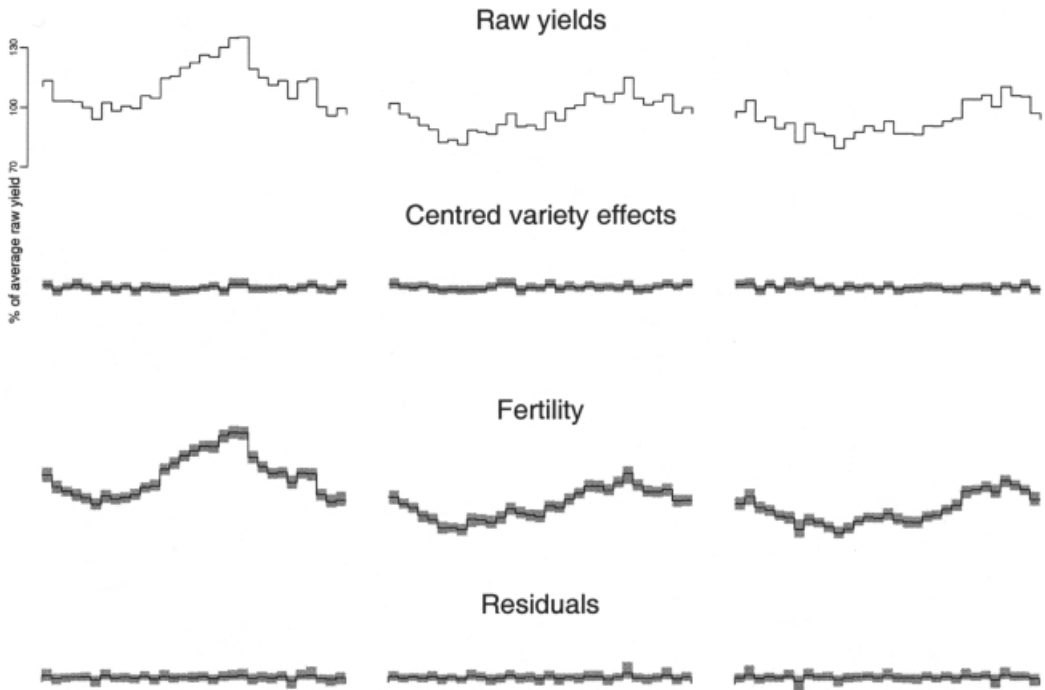


Fig. 17. Yields and additive decompositions under the hierarchical- t formulation for uniformity data (the lines show the posterior mean and the shaded regions show pointwise 90% credible intervals)

conclusions to be quite robust. We thank Barnard for the reference to the interesting related work by Jones (1948), published posthumously with additional comments by Kendall (1948). The paper assumes stationarity and unfortunately does not contain any data, though there is an editorial comment on this omission.

The main disadvantage of the geostatistical approach is its reliance on the variogram. The behaviour of the true variogram close to the origin and at large distances is highly informative but, in practice, there is a lack of data at both scales, as in Smith's figures. Also, conflicts may occur and produce misleading results: for an agricultural example, in which REML is used for estimation, see section 9.7 of Littell *et al.* (1996). In direct fitting, which is often employed for geological data, the class of allowable variogram functions is not known explicitly and sometimes an invalid choice is made, as discussed by Gneiting (1996). Note that, for the intrinsic autoregression (2.5)–(2.6) in Smith's simulations, the true variogram increases approximately as the logarithm of distance (Besag, 1981; Besag and Kooperberg, 1995), which is in close agreement with the empirical results claimed by Fairfield Smith (1938) and Pearce (1976). Note also that the intrinsic model seems to cope adequately with the example in Littell *et al.* (1996).

We look forward to further developments in the continuum models promoted by Best, Ickstadt and Wolpert and by Møller and Waagepetersen, for example. Nevertheless, we repeat that it is not necessary here to model all scales simultaneously; nor is there any requirement to explain, rather than merely to describe, the spatial variation. Best, Ickstadt and Wolpert draw parallels with studies in geographical epidemiology but we contend that the problems there are usually much more complex, particularly in the irregularity of the regions and the desire to draw conclusions about individuals from aggregated data, which cannot be done without additional assumptions. Incidentally, as regards covariates, we are unclear why soil measurements are taken so rarely in field experiments.

Patches, jumps and outliers

Fernández and Green and also Silverman suggest promising approaches to cater for patchiness in a field, via mixture modelling and wavelets respectively. For the much simpler one-dimensional problem, in which we replace Gaussian by t -increments, Jennison questions the extent to which the corresponding

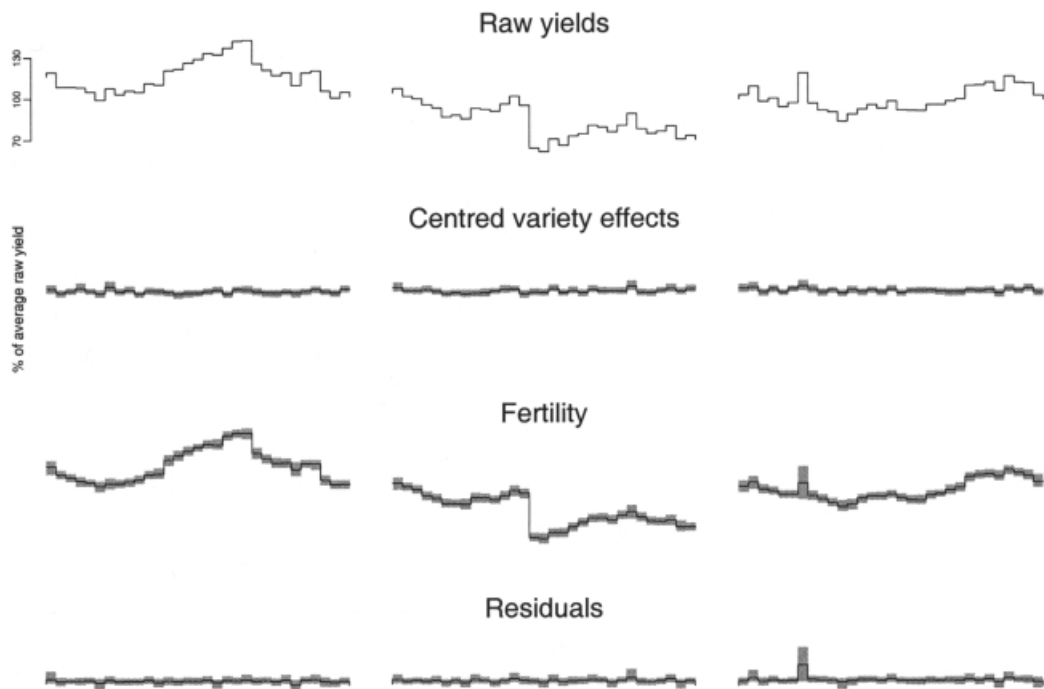


Fig. 18. Yields and additive decompositions under the hierarchical- t formulation for modified uniformity data: a fertility jump was added to the second replicate and an outlier to the third (the lines show the posterior means and the shaded regions show pointwise 90% credible intervals)

degrees of freedom are under the control of the data. We agree that this requires further investigation and perhaps a less elaborate alternative. It might be appropriate here to describe a simulation experiment from several years ago. Fig. 17 shows the additive decomposition for data based on the Wiebe (1935) uniformity trial, using a hierarchical- t formulation, with a Gaussian prior for the superimposed dummy variety effects. Fig. 18 shows the result when the raw data are perturbed by an outlier and a jump in fertility. Such examples were omitted from the paper because they are somewhat ambiguous: the Gaussian prior for variety effects provides desirable shrinkage, also advocated by Patterson in his contribution, but the non-additivity of the analysis implies that the results do not generalize.

Ranking and selection

Preece and Mead rightly criticize us for our reference to multiple-comparison procedures but surely the real issue is that Bayesian methods in ranking and selection do not suffer from similar deficiencies. From a purely pragmatic viewpoint, this is one of the most seductive features of the paradigm. Thus, the hierarchical- t analysis of the El Batán data suggests that the eight varieties 4, 17, 9, 12, 42, 20, 50 and 11 include the most effective one with posterior probability 0.95, which could have been useful in planning a subsequent (better designed!) experiment; and it is irrelevant that we did not name these varieties in advance. In answer to Gilmour, one can calculate $E(\tau_{\max} - \tau_i | y)$ by noting the corresponding empirical quantity in each Markov chain Monte Carlo (MCMC) realization and then averaging over samples. Thus, in the El Batán trial, the posterior expected loss for variety 4 is about 8%, whereas, in the spring barley trial discussed earlier in the reply, it is about 1% for variety 7 and less than about 10% for eight other varieties. In multisite trials, one can, for example, assess the loss that would have been incurred by growing a particular variety at all sites *versus* the unknown best or the known apparent best at each site. We think that this answers the sort of question posed by Mead, subject of course to any defects in the underlying formulation.

Combining information

Glasbey, Edmondson, Gumpertz, Mead, Patterson and Talbot all comment on the role of Bayesian

methods in combining information from different experiments. MacEachern proposes a nonparametric formulation that had not occurred to us. We would have preferred as our example to analyse a sequence of trials but our only data (from Biomathematics and Statistics Scotland) omitted the detailed lay-out in the first of the two experiments. Consequently, we used the North Carolina data, largely to illustrate the type of computations that would be needed for a sequence of experiments: we allude to this in Section 1 of the paper. Our analysis of the corn trials could certainly be improved on and, indeed, it is clear from our statements in the first sentence of Section 6.1 and in the last of Section 6.2 that this is not our preferred Bayesian analysis but one made for comparability with a frequentist analysis. Regarding the comments by Gilmour, Cullis, Verbyla and Smith, we do allow for heterogeneity between counties in the ξ_i s and are puzzled by the lack of shrinkage in their results. We think that Talbot is premature to dismiss hierarchical modelling as a means of combining information from individual trials and do not understand why the standard procedure carries forward 'all relevant information'. Of course, we agree that, in practice, one should always include summaries of the individual trials and, even in the paper, we produce some of these in Figs 6–8, with comments. As regards variation in overall yield between different sites, clearly seen in Fig. 5, counties 2 and 3 were subject to a drought during the spring and to heavy rains in July and August. We do not have an explanation for the disparate row in county 2 but the experimenters doubt that it was caused by a gross error of the type described by Preece. The fertility map in Fig. 5 suggests that the aberration is successfully absorbed and that the estimates of variety effects are not much affected.

Computation

We do not discuss computational issues in much detail in the paper. We agree with Byers that it can be helpful to group variables in the way that he suggests, though in general the statistical gains must be offset against increases in central processor unit time. We thank Künsch for pointing out the analytical solution to the eigenvalue decomposition for W in applying the border method of edge correction. However, we do not agree that Huber functions provide an easy two-dimensional robust analysis unless the asymmetry parameter β is fixed, in which case other models seem equally available. We hope to pursue two-dimensional robustness elsewhere. Møller and Waagepetersen correct our remark about non-sparse matrix inversion, though it is not true that our main reasons for adopting pairwise difference priors (2.2) are computational ones. We are glad they are having success with the Metropolis-adjusted Langevin algorithm (though, more properly, Hastings-adjusted Langevin algorithm!) and surmise that it should be used more widely in MCMC sampling. Incidentally, among many recent advances in spatial simulation, Rue *et al.* (1999) use divide and conquer to simulate conditional autoregressions on large arrays and presumably their algorithm can be modified to cater for intrinsic autoregressions.

Davison raises some interesting issues regarding the validity of MCMC methods. It is of course crucial to detect when a posterior distribution is improper but the condition itself is not necessarily fatal: see, for example, the contribution by MacEachern and its implicit reference to Besag *et al.* (1995). Indeed, the uniform variety prior, which we again emphasize was used purely for numerical comparison, produces a defective overall posterior but in general a proper distribution for variety contrasts. Nevertheless, it is possible to have a proper posterior, with finite means and variances that are poorly estimated by Monte Carlo methods, because of the freak tail behaviour that can occur in pseudo-random samples. We have seen this happen in practice and we sometimes take steps to guard against it. We included posterior means and standard deviations of (centred!) variety effects primarily for comparison with frequentist results, but, in practice, we prefer, both scientifically and numerically, to concentrate on medians, credible regions and posterior probabilities. It is one of the strengths of MCMC methods that such quantities are estimated robustly from the output, without recourse to moment-based asymptotic formulae. As regards convergence, the only foolproof approach is via perfect simulation (e.g. Propp and Wilson (1996), Kendall (1998), Murdoch and Green (1998) and Häggström *et al.* (1999)) but there is now overwhelming evidence that, with reasonable care, MCMC sampling generally produces excellent results. Accuracy can be assessed by various methods; see Geyer (1992), for example.

Other issues

Gilmour and Talbot add competition and interference to the list of possible complications. We agree that design, rather than analysis, should play the key role in tackling these, though additional plots may then be required. One advantage of spatial analysis is that its efficiency is less affected by special purpose design criteria.

There are some disagreements over the lay-out of the strawberry trial, despite which the analyses by

Glasbey, Pearce and Mead seem to reach conclusions that are not materially different from our own, apart from the shrinkage that we would adopt in practice. However, note that our results are from a standard spatial analysis and do not require special covariates, polynomial fits, arbitrary distance functions or omitting certain plots. We discourage fully automatic procedures, except where absolutely necessary, but it seems that (Bayesian) spatial analysis can take some things in its stride, not only in this example. We are grateful to Federer for his comments on all three yield trials. It is plausible that the layout of the El Batán trial was three contiguous columns but a two-dimensional spatial analysis indicates that similarities only occur within replicates. We are surprised at Federer's and Edmondson's reliance on polynomial fits.

Diggle proposes a Poisson-based model that allows for the counts of seeds in the morning-glory example and which agrees closely with the analysis in Higdon (1998). We decided to reduce the data to presence or absence because we have no basis for a Poisson assumption, though the reduction is clearly very wasteful. We would value any further suggestions. In response to Butler, we agree that misspecification can lead to erroneous inference, particularly in the width of credible intervals. For some related work in image analysis, see Tjelmeland and Besag (1998).

References in the discussion

- Abramovich, F., Sapatinas, T. and Silverman, B. W. (1998) Wavelet thresholding via a Bayesian approach. *J. R. Statist. Soc. B*, **60**, 725–749.
- Azaïs, J.-M., Monod, H. and Bailey, R. A. (1998) The influence of design on validity and efficiency of neighbour methods. *Biometrics*, **54**, 1374–1387.
- Bailey, R. A. (1998) Statistics and mathematics: the appropriate use of mathematics within statistics. *Statistician*, **47**, 261–271.
- Baird, D. and Mead, R. (1991) The empirical efficiency and validity of two neighbour models. *Biometrics*, **47**, 1473–1487.
- Bartlett, M. S. (1971) When is inference *statistical* inference? In *Proc. Symp. Foundations of Statistical Inference, Waterloo* (eds V. P. Godambe and D. A. Sprott), pp. 20–31. Toronto: Holt, Rinehart and Winston.
- (1975) *The Statistical Analysis of Spatial Pattern*. London: Chapman and Hall.
- Besag, J. (1974) Spatial interaction and the statistical analysis of lattice systems (with discussion). *J. R. Statist. Soc. B*, **36**, 192–236.
- (1975) Statistical analysis of non-lattice data. *Statistician*, **24**, 179–195.
- (1978) Some methods of statistical analysis for spatial data. *Bull. Int. Statist. Inst.*, **47**, 77–92.
- (1981) On a system of two-dimensional recurrence equations. *J. R. Statist. Soc. B*, **43**, 302–309.
- (1983) Discussion on Nearest-neighbour (NN) analysis of field experiments (by G. Wilkinson, S. Eckert, T. Hancock and O. Mayo). *J. R. Statist. Soc. B*, **45**, 180–183.
- (1992) Discussion on Constrained Monte Carlo maximum likelihood for dependent data (by C. J. Geyer and E. A. Thompson). *J. R. Statist. Soc. B*, **54**, 690.
- (1994) Discussion on Representations of knowledge in complex systems (by U. Grenander and M. I. Miller). *J. R. Statist. Soc. B*, **56**, 591–592.
- Besag, J., Green, P. J., Higdon, D. M. and Mengersen, K. L. (1995) Bayesian computation and stochastic systems (with discussion). *Statist. Sci.*, **10**, 3–66.
- Besag, J. and Higdon, D. M. (1993) Bayesian inference for agricultural field experiments. *Bull. Int. Statist. Inst.*, **55**, book 1, 121–136.
- Besag, J. and Kempton, R. A. (1984) Spatial methods in the analysis of agricultural field trials. In *Proc. 12th Int. Biometric Conf.*, pp. 80–88. Washington DC: Biometric Society.
- (1986) Statistical analysis of field experiments using neighbouring plots. *Biometrics*, **42**, 231–251.
- Besag, J. and Kooperberg, C. L. (1995) On conditional and intrinsic autoregressions. *Biometrika*, **82**, 733–746.
- Besag, J., York, J. and Mollié, A. (1991) Bayesian image restoration, with two applications in spatial statistics (with discussion). *Ann. Inst. Statist. Math.*, **43**, 1–59.
- Best, N. G., Ickstadt, K. and Wolpert, R. L. (1998) Spatial Poisson regression for health and exposure data measured at disparate resolutions. *Discussion Paper 98-36*. Institute of Statistics and Decision Sciences, Duke University, Durham.
- Box, G. E. P. and Cox, D. R. (1964) An analysis of transformations (with discussion). *J. R. Statist. Soc. B*, **26**, 211–252.
- Bozovich, H., Bancroft, T. A. and Hartley, H. O. (1956) Power of analysis of variance test procedures for certain incompletely specified models. *Ann. Math. Statist.*, **27**, 1017–1043.
- Brix, A. and Möller, J. (1998) Space-time multitype log Gaussian Cox processes with a view to modelling weed data. *Research Report R-98-2012*. Department of Mathematics, Aalborg University, Aalborg.
- Bush, C. A. and MacEachern, S. N. (1996) A semiparametric Bayesian model for randomised block designs. *Biometrika*, **83**, 275–285.

- Butler, N. A. (1998) A frequency domain comparison of estimation methods for Gaussian fields. *Communs Statist. Theory Meth.*, **27**, 2325–2342.
- Clatworthy, W. H. (1973) *Tables of Two-associate-class Partially Balanced Designs*. Washington DC: National Bureau of Standards.
- Cochran, W. G. and Cox, G. M. (1957) *Experimental Designs*, 2nd edn. New York: Wiley.
- Cressie, N. A. C. (1993) *Statistics for Spatial Data*. New York: Wiley.
- Cullis, B. R., Gogel, B. J., Verbyla, A. P. and Thompson, R. (1998) Spatial analysis of multi-environment early generation trials. *Biometrics*, **54**, 1–18.
- Diggle, P. and Henderson, R. (1998) A score test for association between longitudinal measurements and recurrent events. *Technical Report ST-98-07*. Lancaster University, Lancaster.
- Diggle, P. J., Tawn, J. A. and Moyeed, R. A. (1998) Model-based geostatistics (with discussion). *Appl. Statist.*, **47**, 299–350.
- Dyke, G. V. (1974) *Comparative Experiments with Field Crops*. London: Butterworth.
- Edmondson, R. N. (1998) Trojan square and incomplete Trojan square designs for crop research. *J. Agric. Sci. Camb.*, **131**, 135–142.
- Fairfield Smith, H. (1938) An empirical law describing heterogeneity in the yields of agricultural crops. *J. Agric. Sci.*, **28**, 1–23.
- Federer, W. T. (1994) Diagnostic procedures for analysis of variance. In *New Progress in Probability and Statistics: Proc. Conf. 'Nicolas de Arriquibar'*, pp. 1–23. Zeist: VSP.
- (1998) Recovering interblock, intergradient, and intervariety information in incomplete block and lattice rectangle designed experiments. *Biometrics*, **54**, 471–481.
- Federer, W. T. and Wolfinger, R. D. (1998) SAS code for recovering intereffect information in experiments with incomplete block and lattice rectangle designs. *Agron. J.*, **90**, 545–551.
- Geyer, C. J. and Thompson, E. A. (1992) Constrained Monte Carlo maximum likelihood for dependent data (with discussion). *J. R. Statist. Soc. B*, **54**, 657–699.
- Gilmour, A. R., Cullis, B. R., Frensham, A. B. and Thompson, R. (1998) (Co)variance structures for linear models in the analysis of plant improvement data. In *COMPSTAT98: Proc. Computational Statistics*, pp. 53–64. Heidelberg: Physica.
- Gilmour, A. R., Cullis, B. R. and Verbyla, A. P. (1997) Accounting for natural and extraneous variation in the analysis of field experiments. *J. Agric. Biol. Environ. Statist.*, **2**, 269–293.
- Gleeson, A. C. and Cullis, B. R. (1987) Residual maximum likelihood (REML) estimation of a neighbor model for field experiments. *Biometrics*, **43**, 277–288.
- Gneiting, T. (1996) Discussion on A simple and efficient space domain implementation of the turning bands method (by C. R. Dietrich). *Wat. Resour. Res.*, **32**, 3391–3396.
- Green, P., Jennison, C. and Seheult, A. (1985) Analysis of field experiments by least squares smoothing. *J. R. Statist. Soc. B*, **47**, 299–315.
- Häggström, O., van Lieshout, M. N. M. and Möller, J. (1999) Characterisation results and Markov chain Monte Carlo algorithms including exact simulation for some spatial point processes. *Bernoulli*, **5**, 641–659.
- Harville, D. A. and Jeske, D. R. (1992) Mean squared error of estimation or prediction under a general linear model. *J. Am. Statist. Ass.*, **87**, 724–731.
- Haywood, J. and Tunnicliffe Wilson, G. (1997) Fitting time series models by minimizing multistep-ahead errors: a frequency domain approach. *J. R. Statist. Soc. B*, **59**, 237–254.
- Higdon, D. (1998) Estimation of genetic effects in the presence of spatial trend and censoring. In *Proc. 13th Int. Wrkshp Statistical Modelling* (eds B. Marx and H. Friedl), pp. 253–260.
- Hobert, J. P. and Casella, G. (1996) The effect of improper priors on Gibbs sampling in hierarchical linear mixed models. *J. Am. Statist. Ass.*, **91**, 1461–1473.
- Ickstadt, K. and Wolpert, R. L. (1999) Spatial regression for marked point processes (with discussion). In *Bayesian Statistics 6* (eds J. M. Bernardo, J. O. Berger, A. P. Dawid and A. F. M. Smith). Oxford: Oxford University Press.
- Jones, A. E. (1948) Systematic sampling of continuous parameter populations. *Biometrika*, **35**, 283–290.
- Kendall, M. G. (1948) Continuation of Dr Jones's paper. *Biometrika*, **35**, 291–296.
- Kendall, W. (1998) Perfect simulation for the area–interaction point processes. In *Probability towards 2000* (eds L. Accardi and C. C. Heyde), pp. 218–234. New York: Springer.
- Künsch, H. R. (1982) Statistical analysis of uniformity trials. *Research Report 32*. Seminar für Statistik, Eidgenössische Technische Hochschule, Zurich.
- (1994) Robust priors for smoothing and image restoration. *Ann. Inst. Statist. Math.*, **46**, 1–19.
- Littell, R. C., Milliken, G. A., Stroup, W. W. and Wolfinger, R. D. (1996) *SAS System for Mixed Models*. Cary: SAS Institute.
- Martin, R. J. (1979) A subclass of lattice processes applied to a problem in planar sampling. *Biometrika*, **66**, 209–217.
- Mayo, D. G. (1996) *Error and the Growth of Experimental Knowledge*. Chicago: University of Chicago Press.
- Mead, R. (1988) *The Design of Experiments*. Cambridge: Cambridge University Press.
- Møller, J., Syversveen, A. R. and Waagepetersen, R. P. (1997) Log Gaussian Cox processes: a statistical model for analyzing stand structural heterogeneity in forestry. In *Proc. 1st Eur. Conf. Information Technology in Agriculture*, pp. 339–342. Copenhagen: Royal Veterinary and Agricultural University.

- (1998) Log Gaussian Cox processes. *Scand. J. Statist.*, **25**, 451–482.
- Murdoch, D. J. and Green, P. J. (1998) Exact sampling from a continuous state space. *Scand. J. Statist.*, **25**, 483–502.
- Patterson, H. D. and Williams, E. R. (1976) A new class of resolvable incomplete block designs. *Biometrika*, **63**, 83–92.
- Pearce, S. C. (1976) An examination of Fairfield Smith's law of environmental variation. *J. Agric. Sci.*, **87**, 21–24.
- (1983) *The Agricultural Field Experiment: a Statistical Examination of Theory and Practice*. Chichester: Wiley.
- (1998) Field experimentation on rough land: the method of Papadakis reconsidered. *J. Agric. Sci. Camb.*, **131**, 1–11.
- Philippeau, G., David, O. and Monod, H. (1996) Interplot competition in cereal variety trials. In *18th Int. Biometric Conf.*, invited papers, pp. 107–116.
- (1997) Importance de la compétition sur céréales à paille et proposition de solutions. In *Proc. Sem. Effets de Compétition dans les Essais Variétaux de Pleinchamp*. Boigneville: Institut Technique des Céréales et des Fourrages.
- Priestley, M. B. (1981) *Spectral Analysis and Time Series*. London: Academic Press.
- Propp, J. G. and Wilson, D. B. (1996) Exact sampling with coupled Markov chains and applications to statistical mechanics. *Rand. Struct. Alg.*, **9**, 223–252.
- Richardson, S. (1992) Statistical methods for geographical correlation studies. In *Geographical and Environmental Epidemiology: Methods for Small-area Studies* (eds P. Elliott, J. Cuzick, D. English and R. Stern), pp. 181–204. Oxford: Oxford University Press.
- Roberts, G. O. and Tweedie, R. L. (1997) Exponential convergence of Langevin diffusions and their discrete approximations. *Bernoulli*, **2**, 341–363.
- Robinson, W. S. (1950) Ecological correlations and the behaviour of individuals. *Am. Sociol. Rev.*, **15**, 351–357.
- Rue, H., Marthinsen, A. and Husby, E. G. (1999) A divide and conquer algorithm for exact simulation of general Gaussian Markov random Fields. *Report 11/99*. Department of Mathematical Sciences, Norwegian University of Science and Technology.
- Segel, S., Sloth, N. and Tøgersen, F. A. (1998) Image analysis: a probabilistic approach using Markov chain Monte Carlo methods. *Diploma Thesis*. Department of Mathematics, Aalborg University, Aalborg.
- Smith, A. B., Cullis, B. R., Gilmour, A. and Thompson, R. (1998) Multiplicative models for the interaction in spatial mixed model analyses of multi-environment trial data. In *Proc. 19th Int. Biometrics Conf., Cape Town*, pp. 91–101.
- Stigler, S. M. (1986) *The History of Statistics*. Cambridge: Harvard University Press.
- Talbot, M., Milner, A. D., Nutkins, A. E. and Law, J. R. (1995) Effect of interference between plots on yield performance in crop variety trials. *J. Agric. Sci. Camb.*, **124**, 335–342.
- Tjelmeland, H. and Besag, J. (1998) Markov random fields with higher-order interactions. *Scand. J. Statist.*, **25**, 415–433.
- Wiebe, G. A. (1935) Variation and correlation in grain yield among 1500 wheat nursery plots. *J. Agric. Res.*, **50**, 331–357.
- Wilkinson, G. N., Eckert, S. R., Hancock, T. W. and Mayo, O. (1983) Nearest neighbour (NN) analysis of field experiments (with discussion). *J. R. Statist. Soc. B*, **45**, 151–211.
- Wolfinger, R. D., Federer, W. T. and Cordero-Brana, O. (1997) Recovering information in augmented designs, using SAS PROC GLM and PROC MIXED. *Agron. J.*, **89**, 856–859.
- Wolpert, R. L. and Ickstadt, K. (1998) Poisson/gamma random field models for spatial statistics. *Biometrika*, **85**, 251–267.
- Wood, A. T. A. and Chan, G. (1994) Simulation of stationary Gaussian processes in $[0, 1]^d$. *J. Comput. Graph. Statist.*, **3**, 409–432.
- Yates, F. (1937) *The Design and Analysis of Factorial Experiments*. Farnham Royal: Commonwealth Agricultural Bureaux.
- (1939) The comparative advantages of systematic and randomized arrangements in the design of agricultural and biological experiments. *Biometrika*, **30**, 440–466.
- Zimmerman, D. L. and Cressie, N. A. C. (1992) Mean squared prediction error in the spatial linear model with estimated covariance parameters. *Ann. Inst. Statist. Math.*, **44**, 27–43.
- Zimmerman, D. L. and Harville, D. A. (1991) A random field approach to the analysis of field-plot experiments and other spatial experiments. *Biometrics*, **47**, 223–239.